



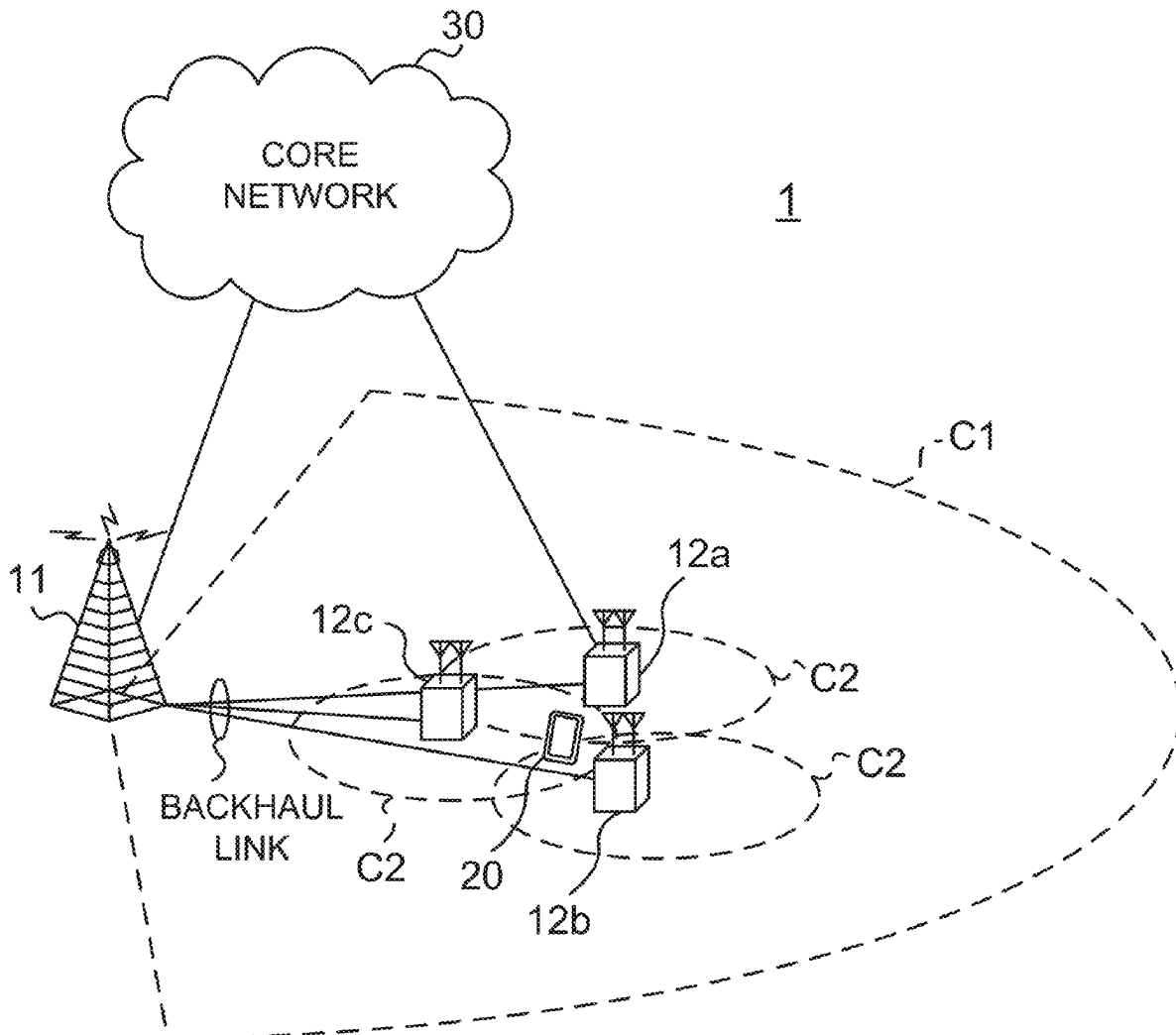
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Matsumura et al.(10) **Pub. No.: US 2025/0286683 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **TERMINAL, RADIO COMMUNICATION
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Tokyo (JP); **Jing Wang**, Beijing (CN)(73) Assignee: **NTT DOCOMO, INC.**, Tokyo (JP)(21) Appl. No.: **18/859,893**(22) PCT Filed: **Apr. 28, 2022**(86) PCT No.: **PCT/JP2022/019322**

§ 371 (c)(1),

(2) Date: **Oct. 24, 2024**(57) **ABSTRACT**

A terminal according to one aspect of the present disclosure includes a receiving section that receives a configuration of a demodulation reference signal (DMRS) of a shared channel, and a control section that applies a frequency domain (FD)-orthogonal cover code (OCC) having a length of 4 or 3 to the DMRS, based on the configuration. According to one aspect of the present disclosure, an appropriate number of DMRS ports can be used.



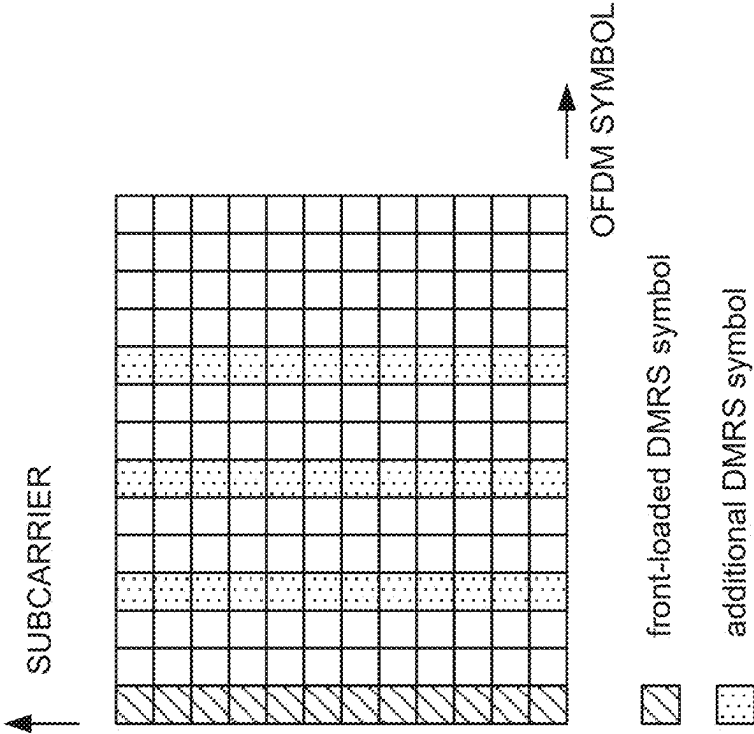


FIG. 1

FIG. 2A

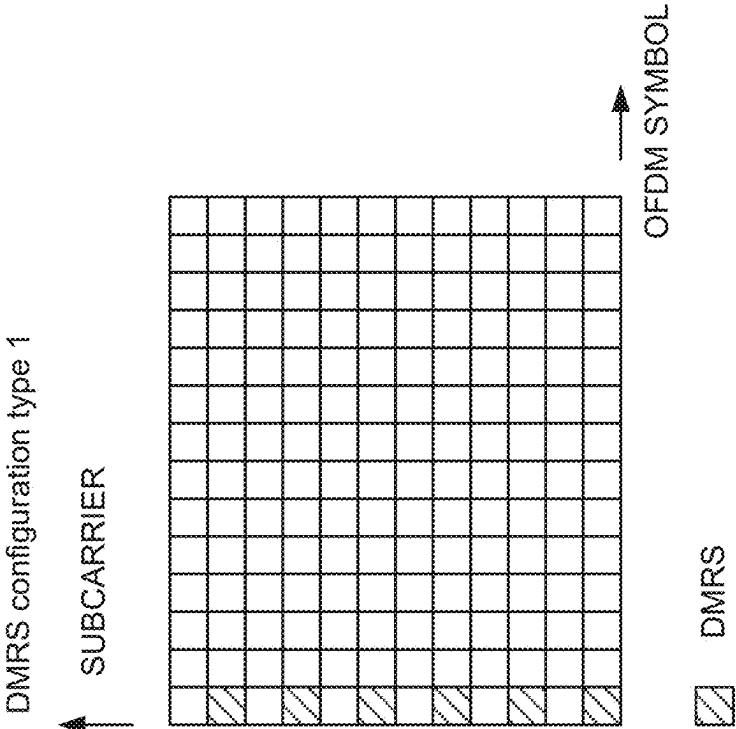


FIG. 2B

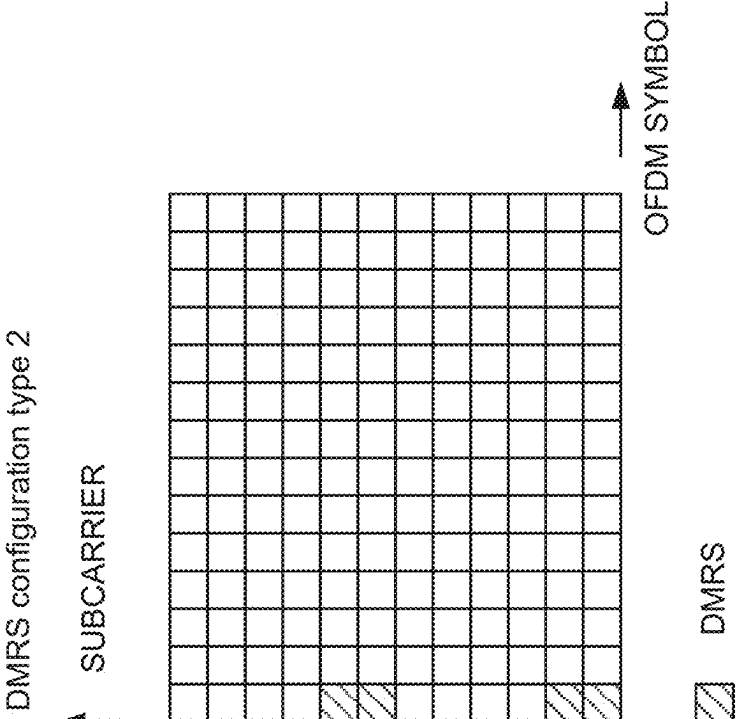


FIG. 3A

Single symbol DMRS

DMRS configuration type 1

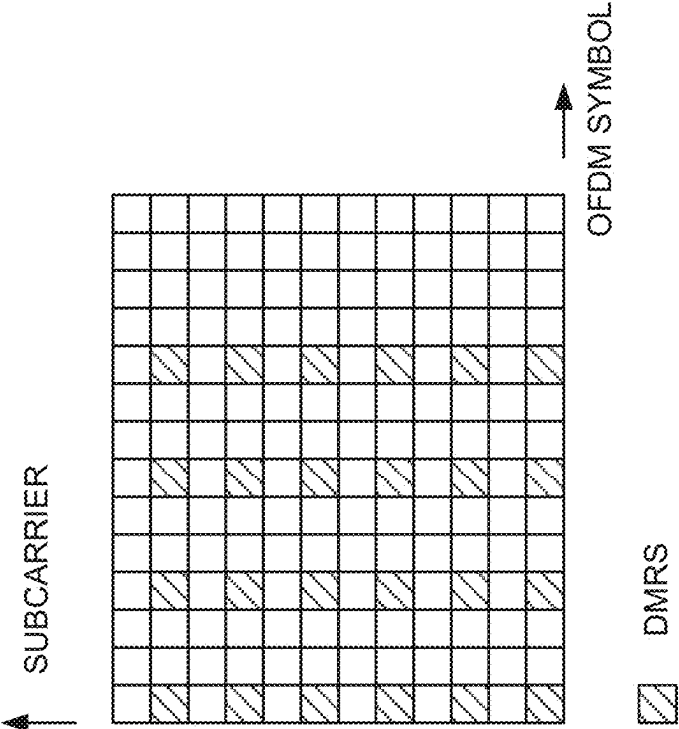


FIG. 3B

Single symbol DMRS

DMRS configuration type 2

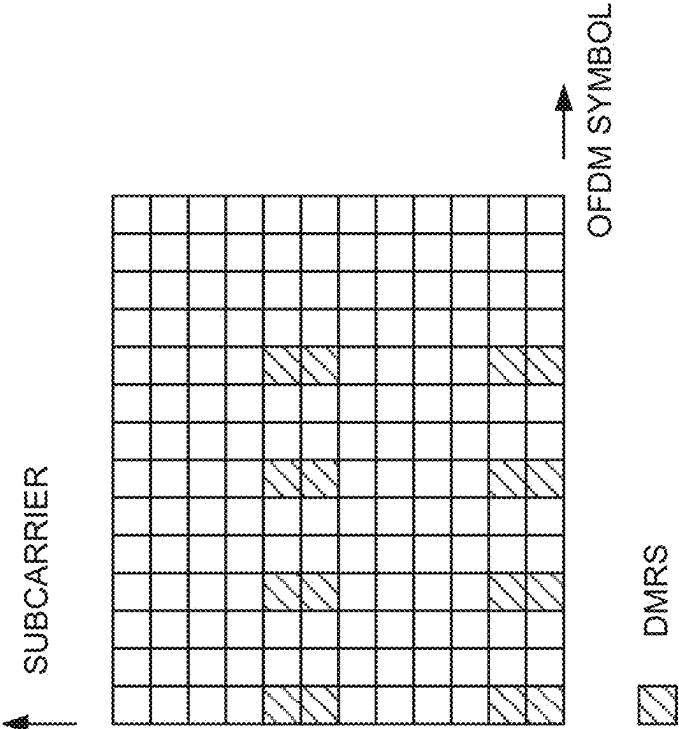


FIG. 4A

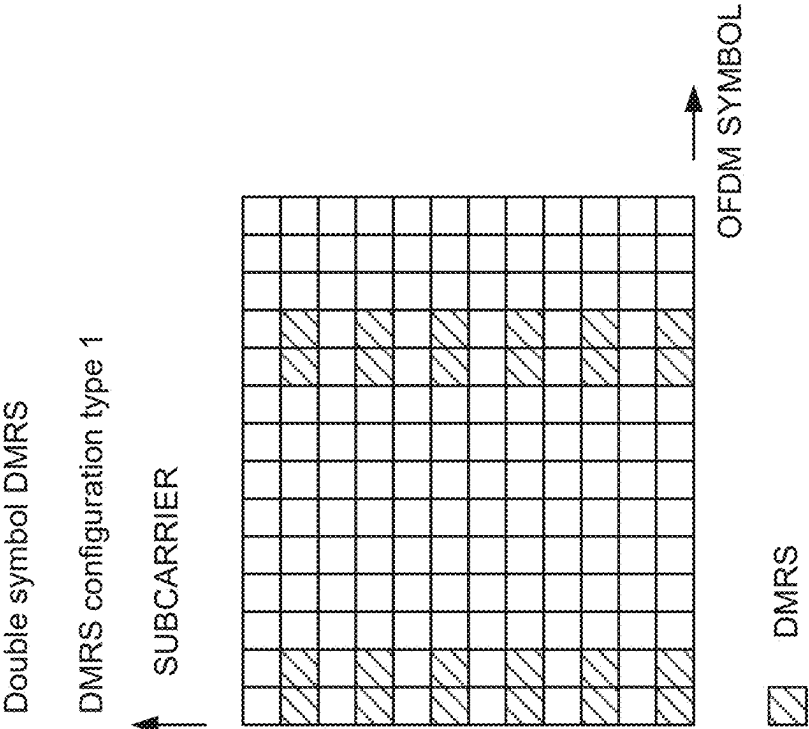
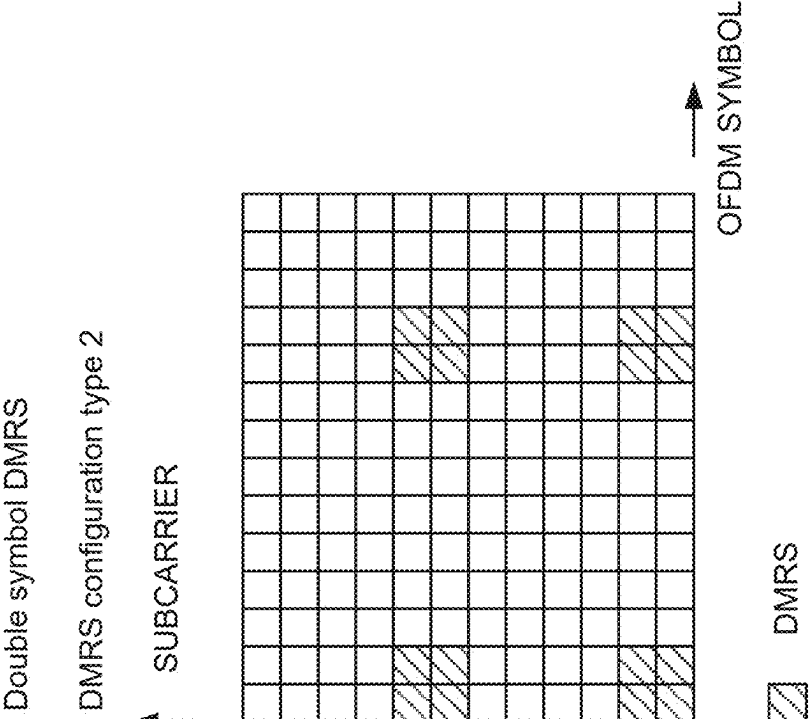


FIG. 4B



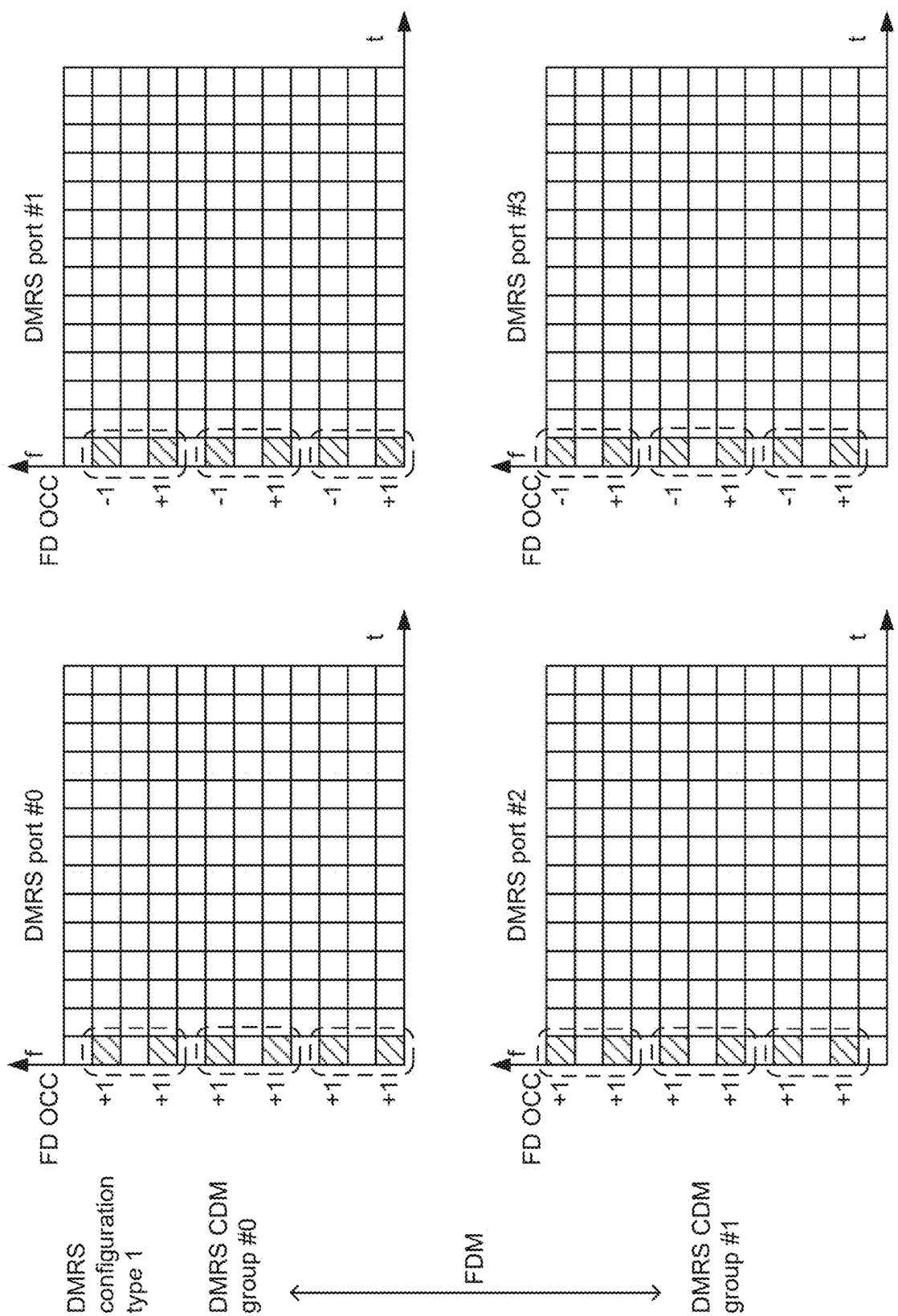


FIG. 5

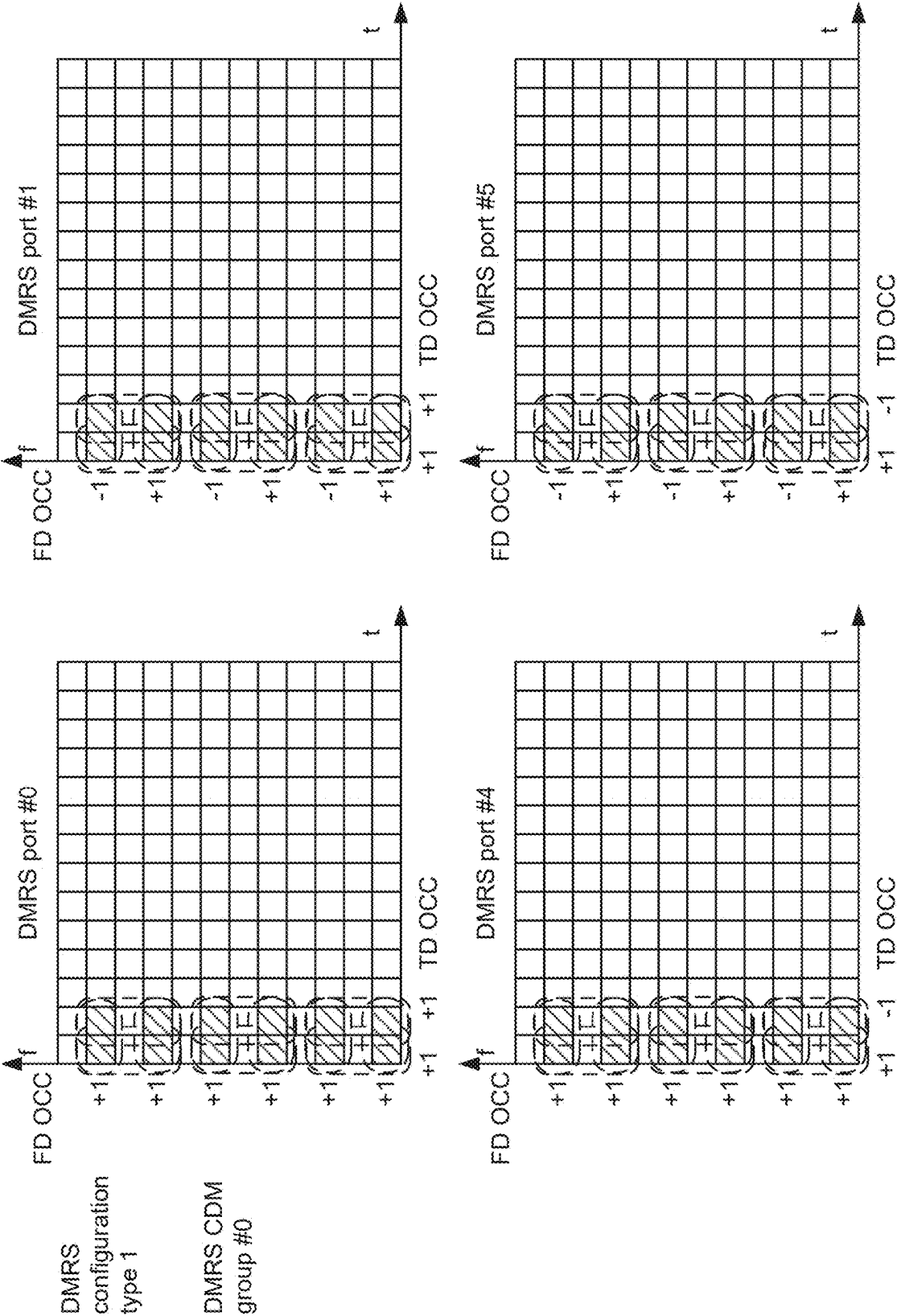
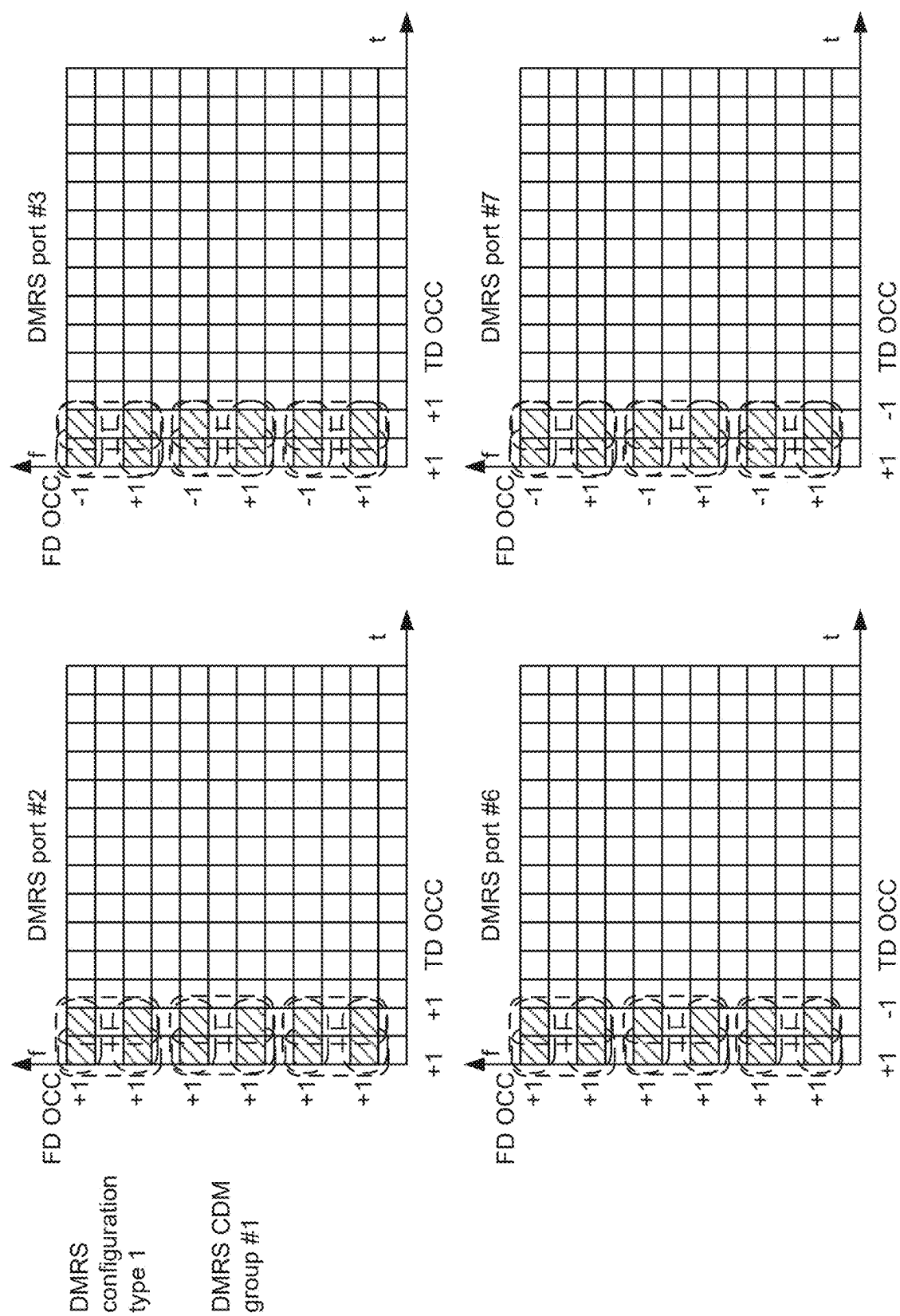


FIG. 6



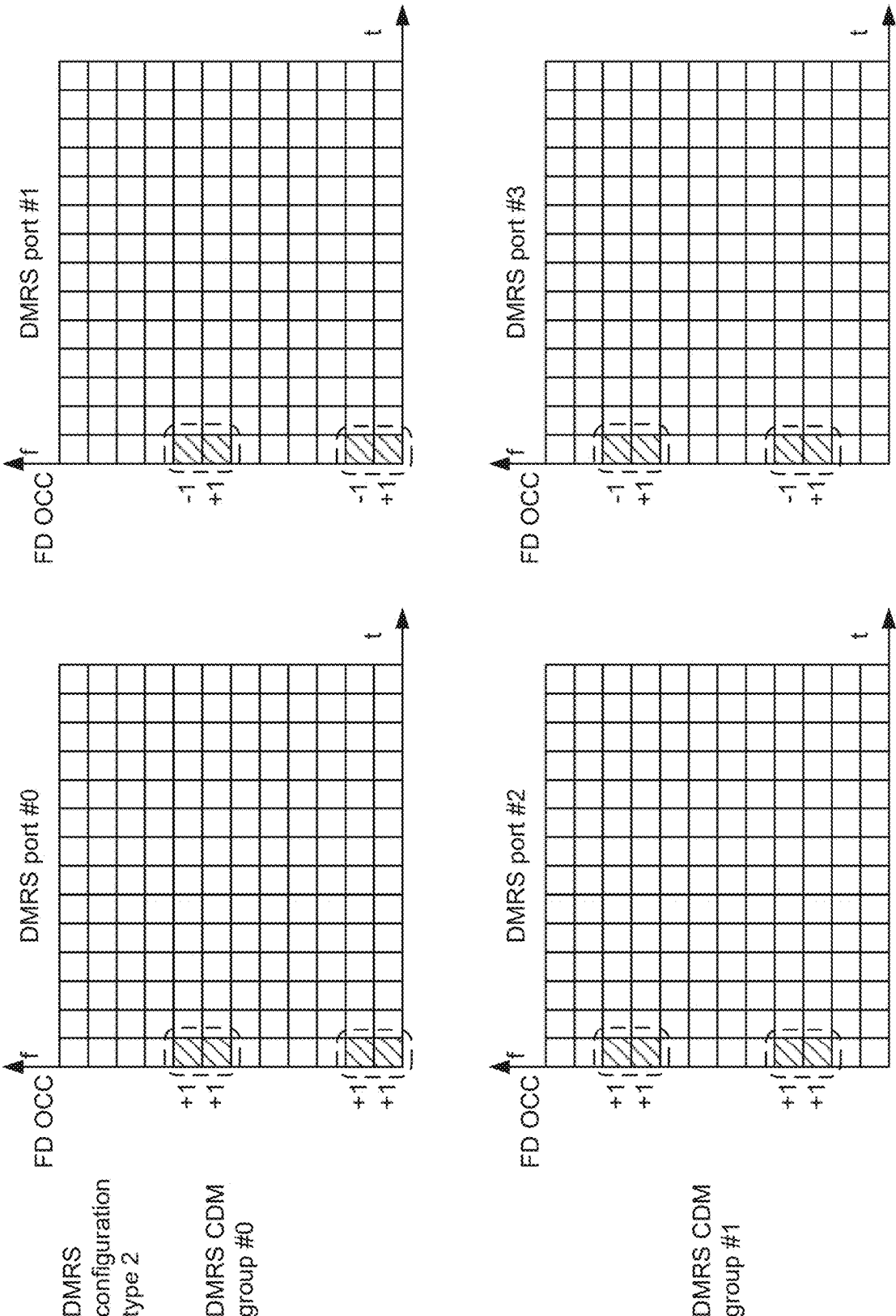


FIG. 8

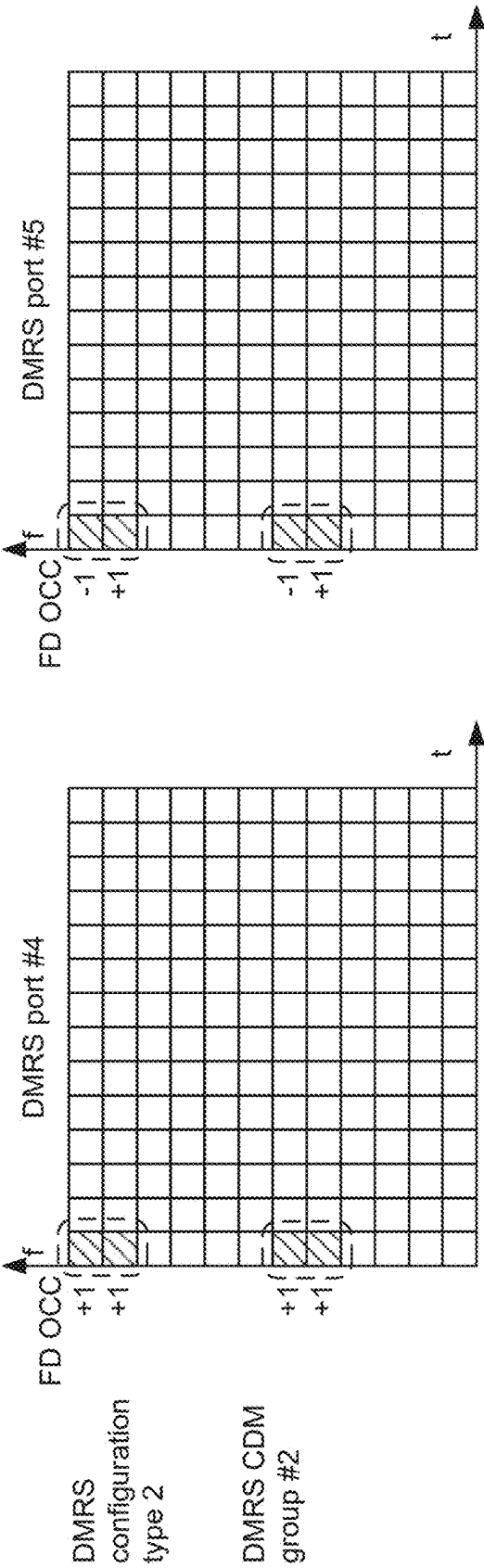
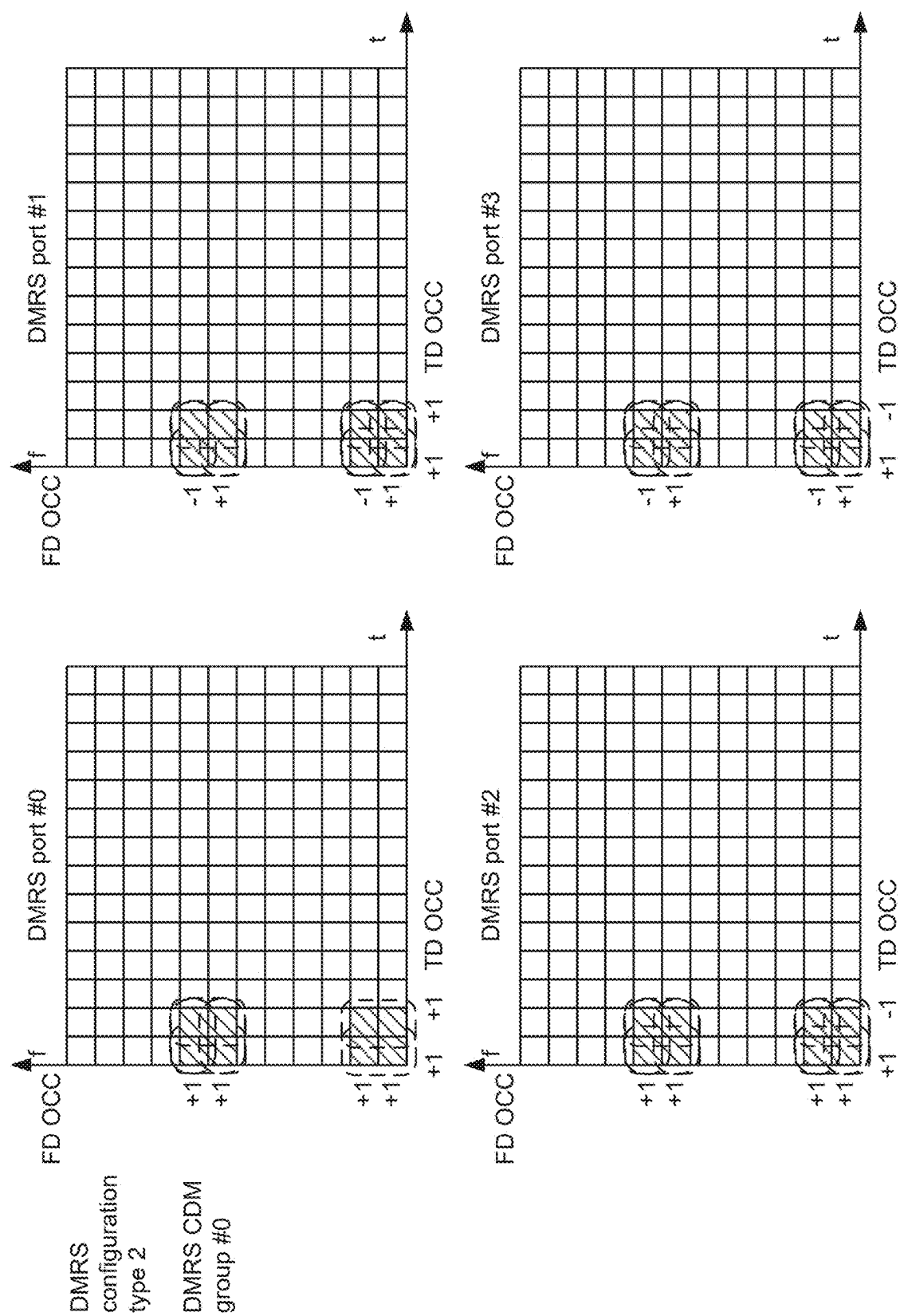
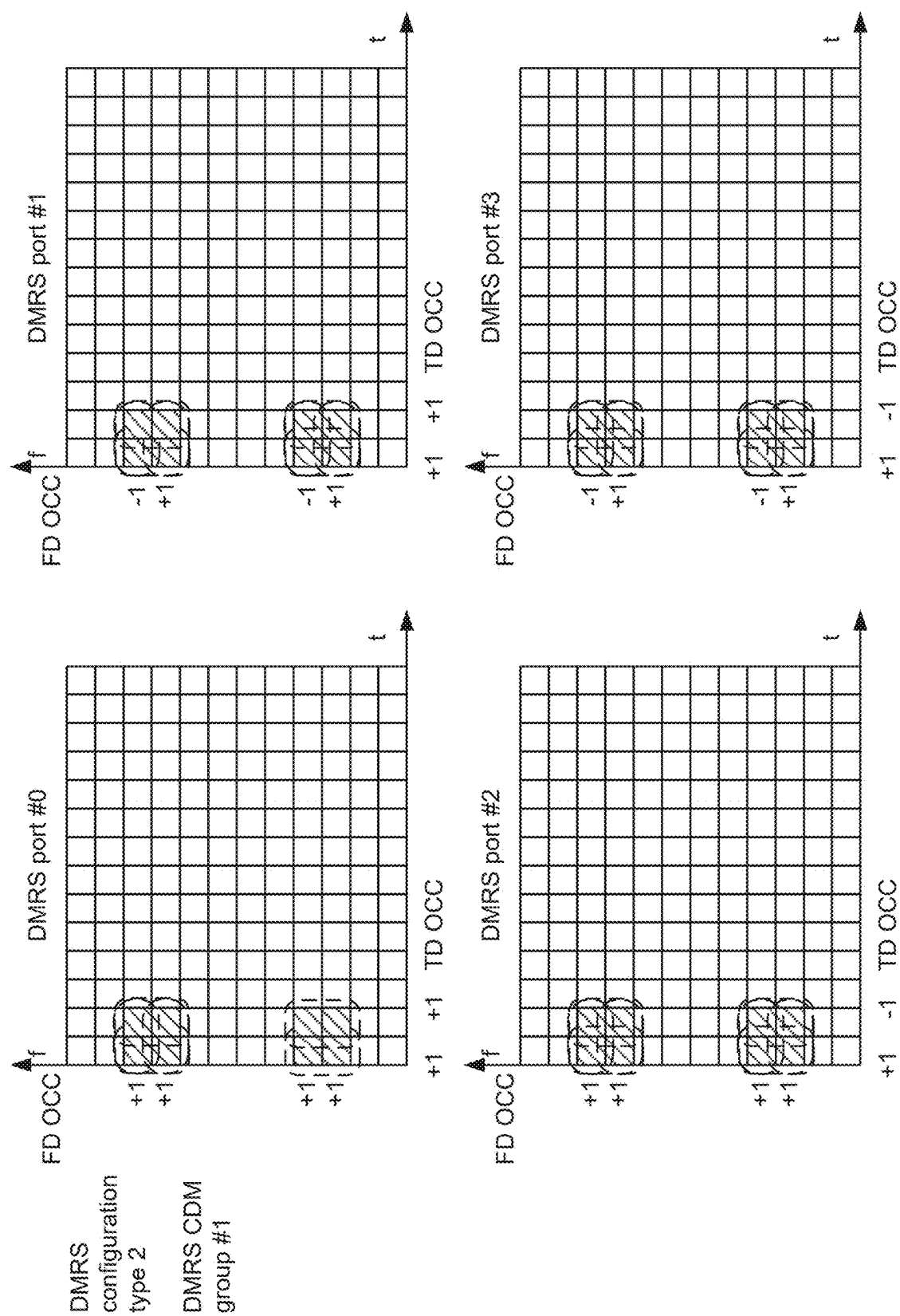


FIG. 9





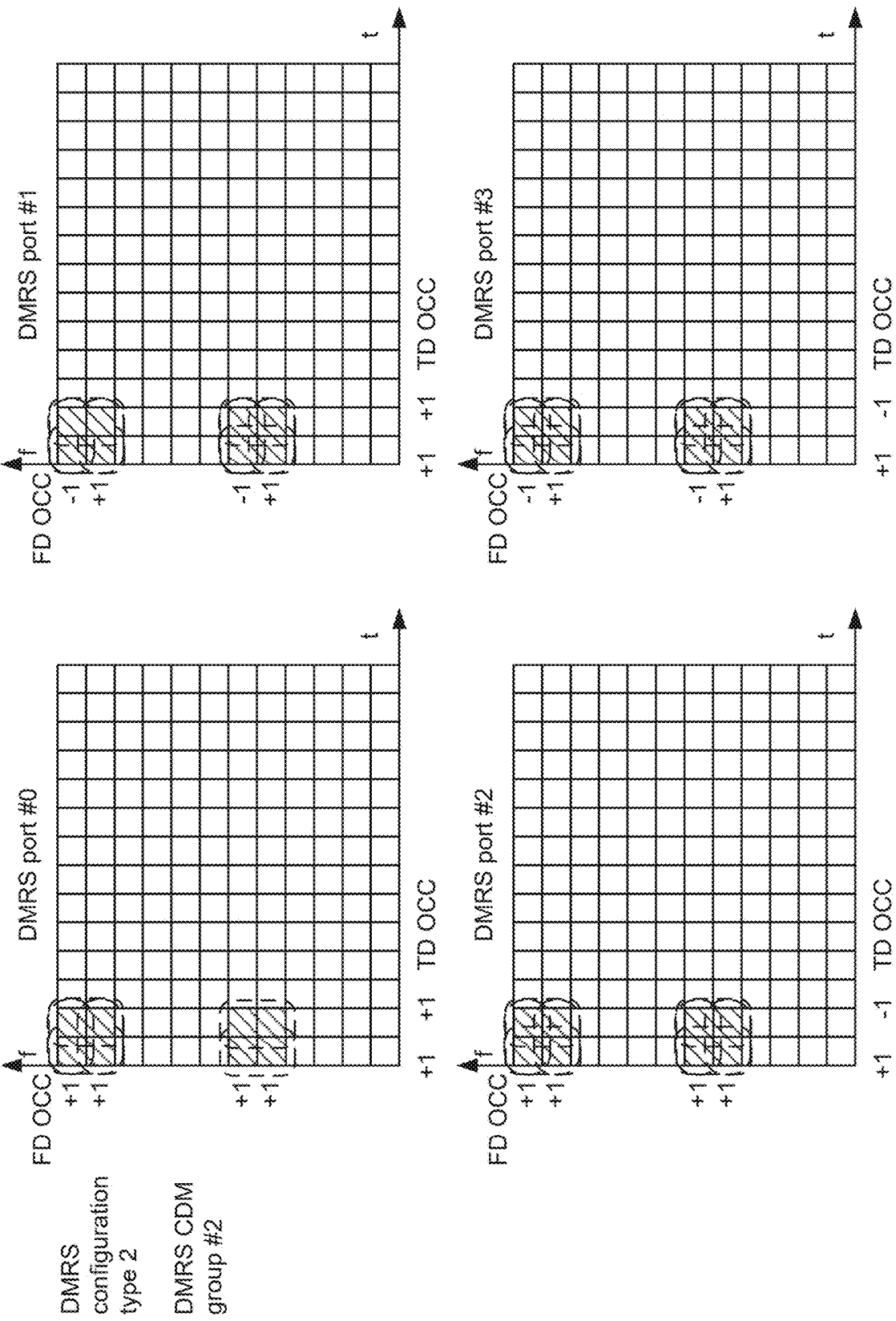


FIG. 12

Parameters for PDSCH DM-RS configuration type 1.

p	CDM group λ	Δ	$w_f(k')$		$w_t(l')$	
			$k' = 0$	$k' = 1$	$l' = 0$	$l' = 1$
1000	0	0	+1	+1	+1	+1
1001	0	0	+1	-1	+1	+1
1002	1	1	+1	+1	+1	+1
1003	1	1	+1	-1	+1	+1
1004	0	0	+1	+1	+1	-1
1005	0	0	+1	-1	+1	-1
1006	1	1	+1	+1	+1	-1
1007	1	1	+1	-1	+1	-1

Parameters for PDSCH DM-RS configuration type 2.

p	CDM group λ	Δ	$w_f(k')$		$w_t(l')$	
			$k' = 0$	$k' = 1$	$l' = 0$	$l' = 1$
1000	0	0	+1	+1	+1	+1
1001	0	0	+1	-1	+1	+1
1002	1	2	+1	+1	+1	+1
1003	1	2	+1	-1	+1	+1
1004	2	4	+1	+1	+1	+1
1005	2	4	+1	-1	+1	+1
1006	0	0	+1	+1	+1	-1
1007	0	0	+1	-1	+1	-1
1008	1	2	+1	+1	+1	-1
1009	1	2	+1	-1	+1	-1
1010	2	4	+1	+1	+1	-1
1011	2	4	+1	-1	+1	-1

FIG. 13

Parameters for PUSCH DM-RS configuration type 1.

$\tilde{p}$	CDM group λ	Δ	$w_f(k')$		$w_t(l')$	
			$k' = 0$	$k' = 1$	$l' = 0$	$l' = 1$
0	0	0	+1	+1	+1	+1
1	0	0	+1	-1	+1	+1
2	1	1	+1	+1	+1	+1
3	1	1	+1	-1	+1	+1
4	0	0	+1	+1	+1	-1
5	0	0	+1	-1	+1	-1
6	1	1	+1	+1	+1	-1
7	1	1	+1	-1	+1	-1

Parameters for PUSCH DM-RS configuration type 2.

$\tilde{p}$	CDM group λ	Δ	$w_f(k')$		$w_t(l')$	
			$k' = 0$	$k' = 1$	$l' = 0$	$l' = 1$
0	0	0	+1	+1	+1	+1
1	0	0	+1	-1	+1	+1
2	1	2	+1	+1	+1	+1
3	1	2	+1	-1	+1	+1
4	2	4	+1	+1	+1	+1
5	2	4	+1	-1	+1	+1
6	0	0	+1	+1	+1	-1
7	0	0	+1	-1	+1	-1
8	1	2	+1	+1	+1	-1
9	1	2	+1	-1	+1	-1
10	2	4	+1	+1	+1	-1
11	2	4	+1	-1	+1	-1

FIG. 14

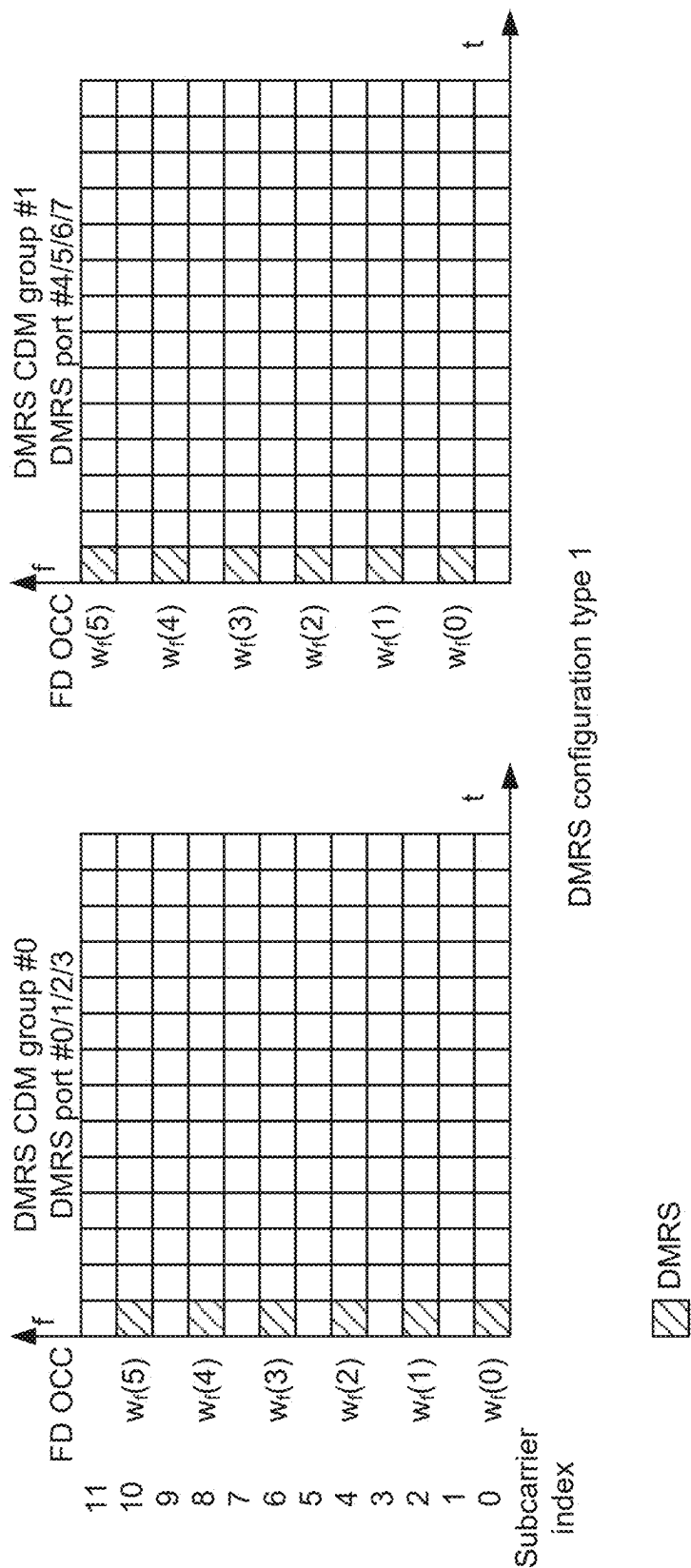


FIG. 15

FIG. 16A

OCC index	$w_i(0)$	$w_i(1)$	$w_i(2)$	$w_i(3)$	$w_i(4)$	$w_i(5)$
0	+1	+1	+1	+1	+1	+1
1	+1	+j	-1	-j	+1	+j
2	+1	-1	+1	-1	+1	-1
3	+1	-j	-1	+j	+1	-j

FIG. 16B

OCC index	$w_i(0)$	$w_i(1)$	$w_i(2)$	$w_i(3)$	$w_i(4)$	$w_i(5)$
0	$\exp(j0*0)$	$\exp(j02*1)$	$\exp(j0*2)$	$\exp(j0*3)$	$\exp(j0*4)$	$\exp(j0*5)$
1	$\exp(j\pi/2*0)$	$\exp(j\pi/2*1)$	$\exp(j\pi/2*2)$	$\exp(j\pi/2*3)$	$\exp(j\pi/2*4)$	$\exp(j\pi/2*5)$
2	$\exp(j\pi*0)$	$\exp(j\pi*1)$	$\exp(j\pi*2)$	$\exp(j\pi*3)$	$\exp(j\pi*4)$	$\exp(j\pi*5)$
3	$\exp(j3\pi/2*0)$	$\exp(j3\pi/2*1)$	$\exp(j3\pi/2*2)$	$\exp(j3\pi/2*3)$	$\exp(j3\pi/2*4)$	$\exp(j3\pi/2*5)$

OCC index	$w_i(0)$	$w_i(1)$	$w_i(2)$	$w_i(3)$	$w_i(4)$	$w_i(5)$
0	+1	+1	+1	+1	+1	+1
1	+1	-1	+1	-1	+1	-1
2	+1	+1	-1	-1	+1	+1
3	+1	-1	-1	+1	+1	-1

FIG. 17

FIG. 18A

OCC1			OCC2		
OCC index	$w_{f1}(0)$	$w_{f1}(1)$	OCC index	$w_{f2}(0)$	$w_{f2}(1)$
0	+1	+1	0	+1	+1
1	+1	-1	1	+1	-1

FIG. 18B

OCC1 index	OCC2 index	$w_f(0)$	$w_f(1)$	$w_f(2)$	$w_f(3)$	$w_f(4)$	$w_f(5)$
0	0	+1	+1	+1	+1	+1	+1
0	1	+1	+1	-1	-1	+1	+1
1	0	+1	-1	+1	-1	+1	-1
1	1	+1	-1	-1	+1	+1	-1

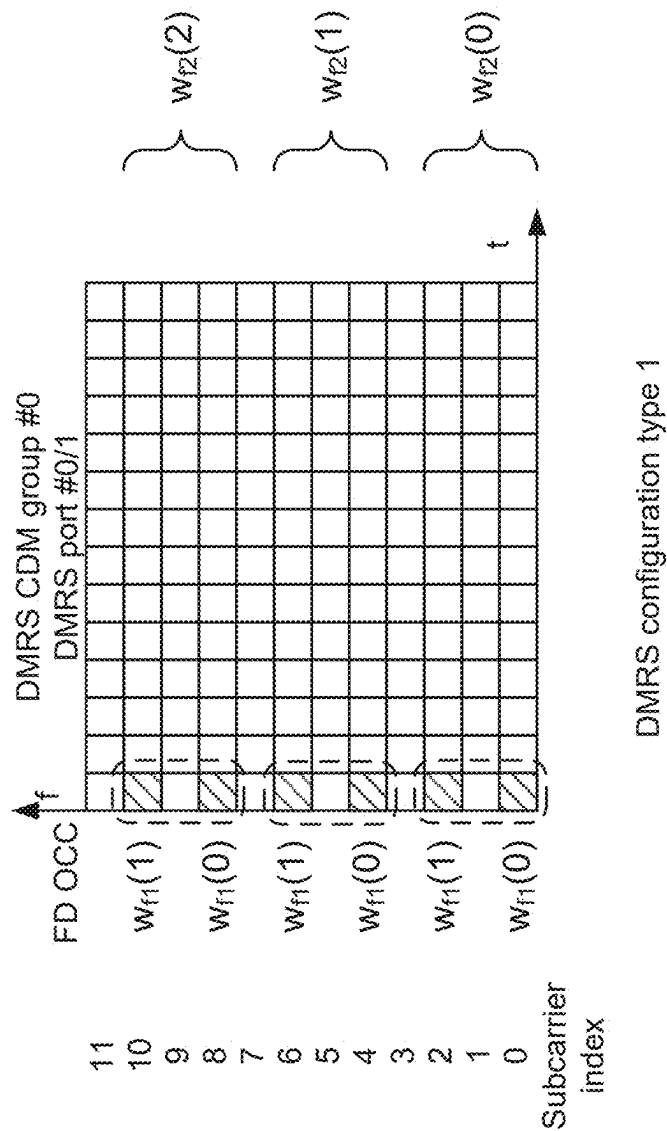


FIG. 19

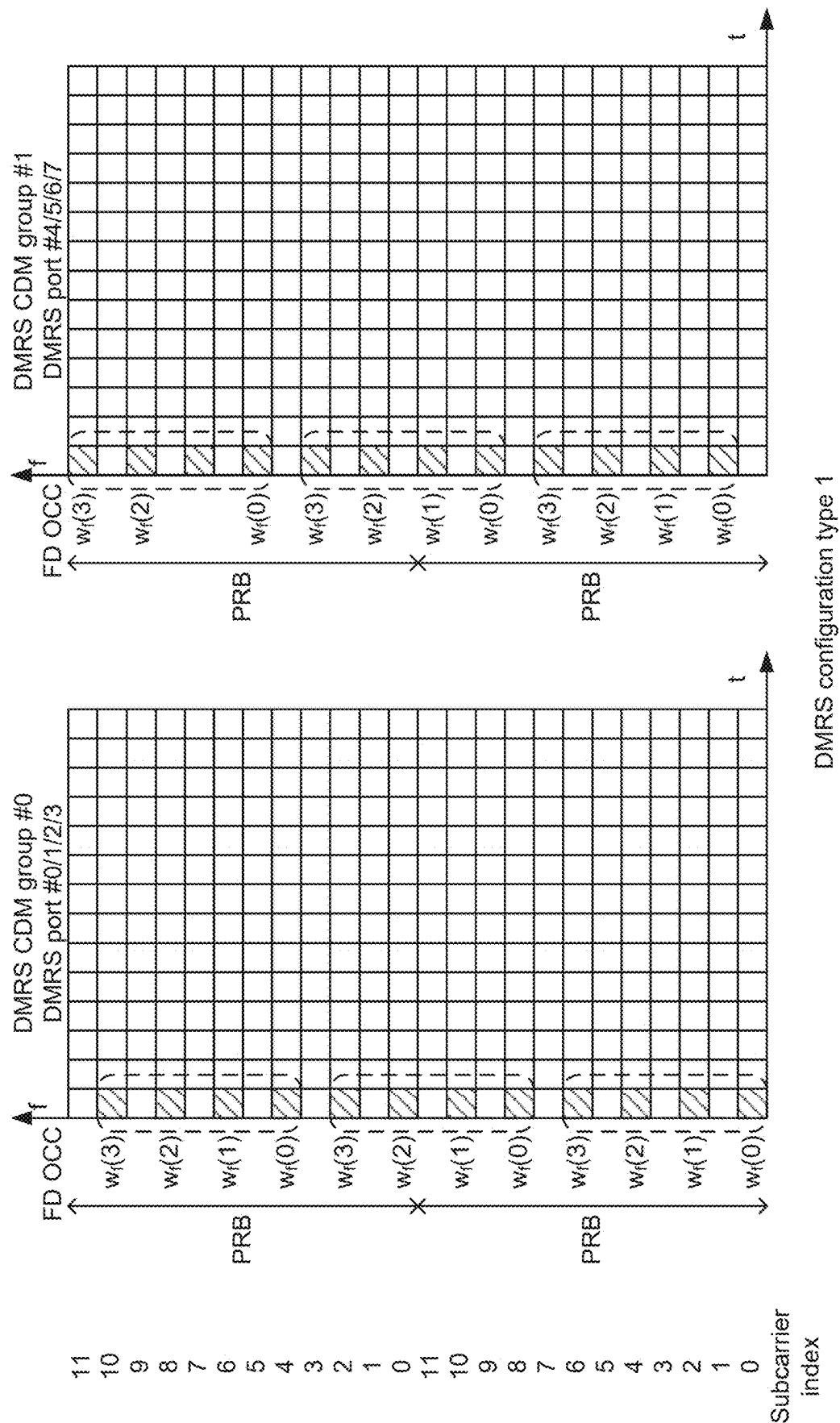


FIG. 20

FIG. 21A

OCC index	$w_i(0)$	$w_i(1)$	$w_i(2)$	$w_i(3)$
0	+1	+1	+1	+1
1	+1	+j	-1	-j
2	+1	-1	+1	-1
3	+1	-j	-1	+j

FIG. 21B

OCC index	$w_i(0)$	$w_i(1)$	$w_i(2)$	$w_i(3)$
0	$\exp(j0*0)$	$\exp(j02*1)$	$\exp(j0*2)$	$\exp(j0*3)$
1	$\exp(j\pi/2*0)$	$\exp(j\pi/2*1)$	$\exp(j\pi/2*2)$	$\exp(j\pi/2*3)$
2	$\exp(j\pi*0)$	$\exp(j\pi*1)$	$\exp(j\pi*2)$	$\exp(j\pi*3)$
3	$\exp(j3\pi/2*0)$	$\exp(j3\pi/2*1)$	$\exp(j3\pi/2*2)$	$\exp(j3\pi/2*3)$

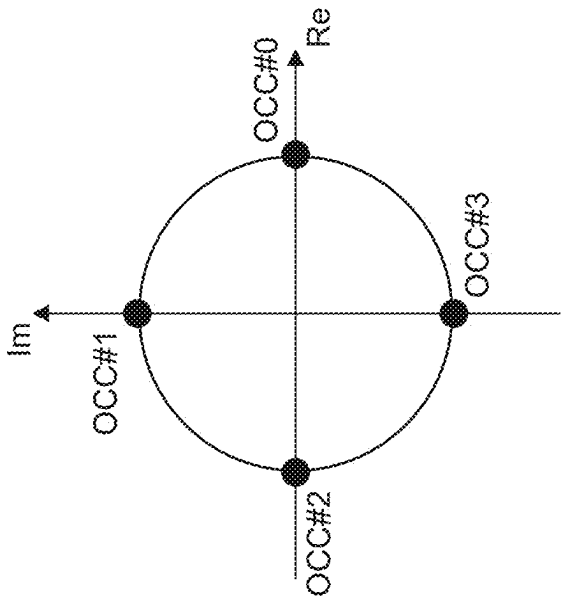


FIG. 22

OCC index	$w_i(0)$	$w_i(1)$	$w_i(2)$	$w_i(3)$
0	+1	+1	+1	+1
1	+1	-1	+1	-1
2	+1	+1	-1	-1
3	+1	-1	-1	+1

FIG. 23

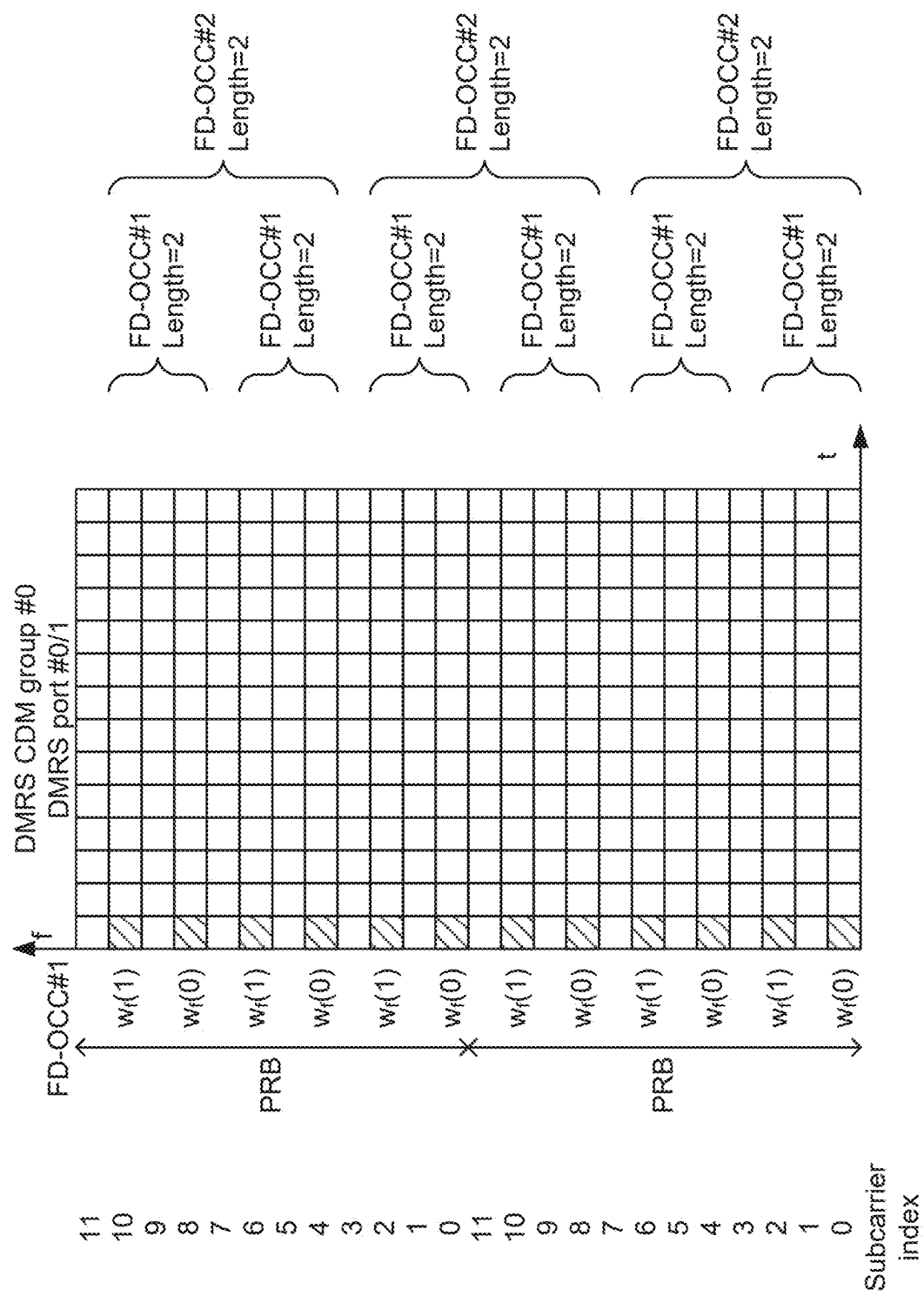
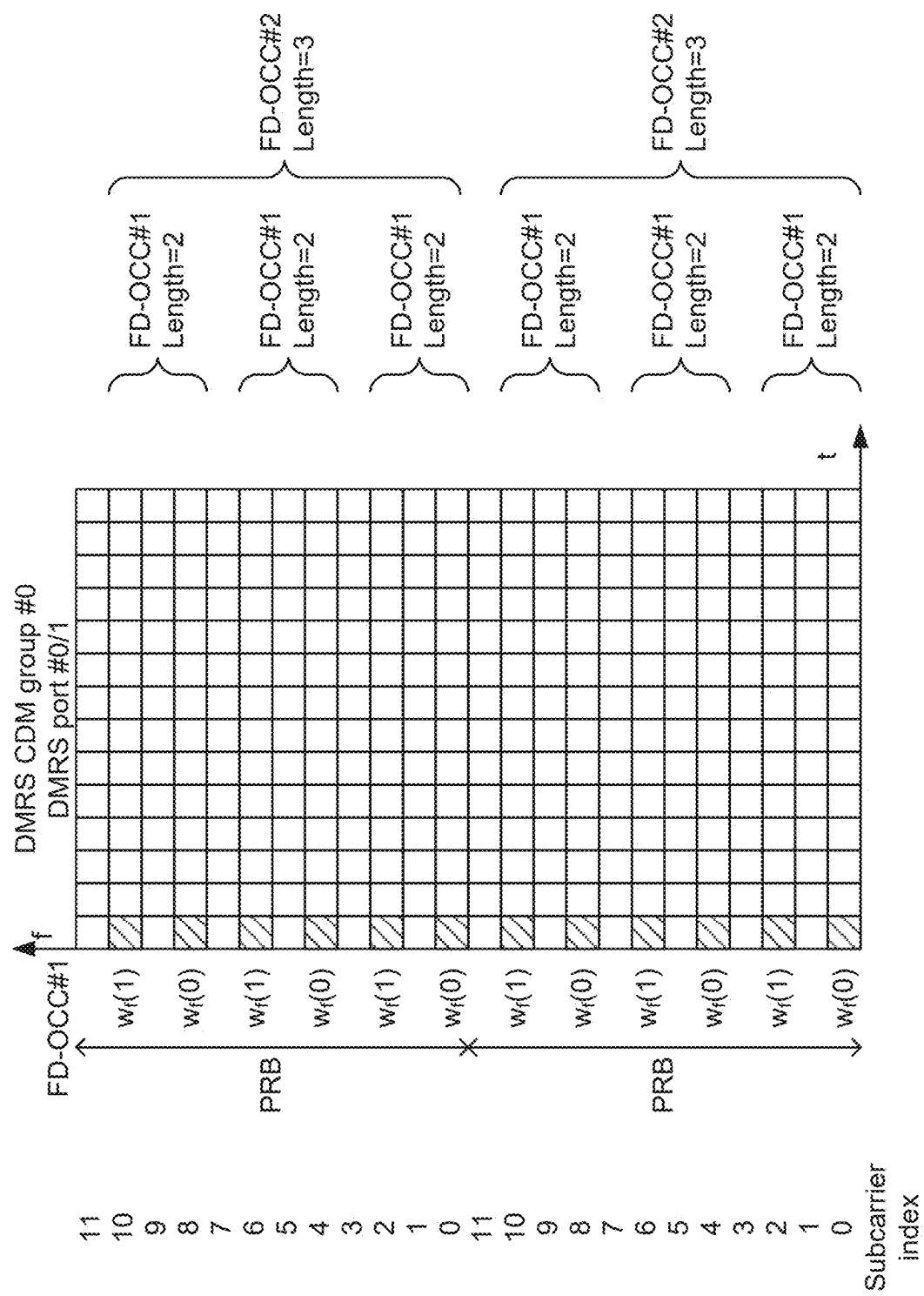


FIG. 24



Ex. 25

FIG.
26B

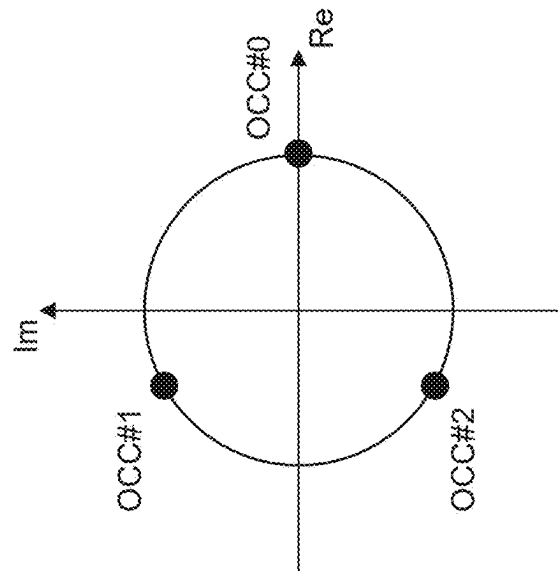


FIG.
26A

OCC index	$w_i(0)$	$w_i(1)$	$w_i(2)$
0	$\exp(j0 \cdot 0)$	$\exp(j02 \cdot 1)$	$\exp(j0 \cdot 2)$
1	$\exp(j2\pi/3 \cdot 0)$	$\exp(j2\pi/3 \cdot 1)$	$\exp(j2\pi/3 \cdot 2)$
2	$\exp(j4\pi/3 \cdot 0)$	$\exp(j4\pi/3 \cdot 1)$	$\exp(j4\pi/3 \cdot 2)$

FIG. 26

OCC index	$w_f(0)$	$w_f(1)$	$w_f(2)$
0	+1	+1	+1
1	+1	-1	+1
2	+1	+1	-1

FIG. 27

FIG. 28B

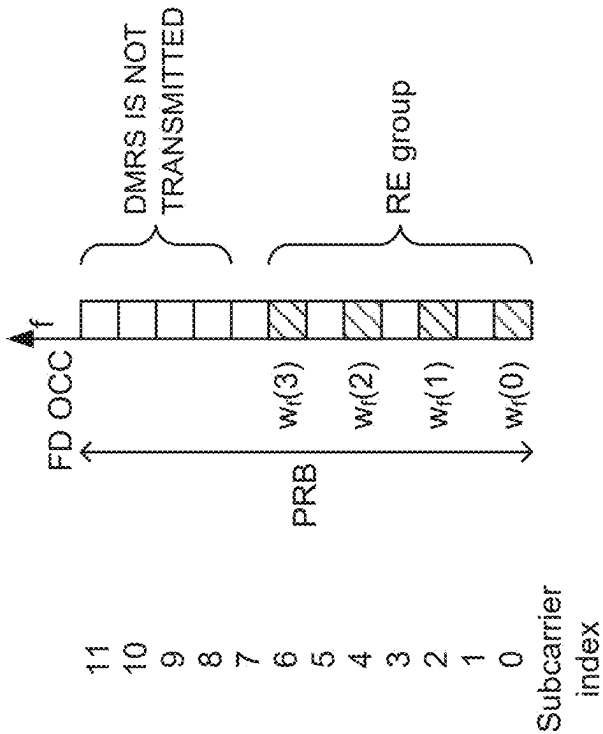
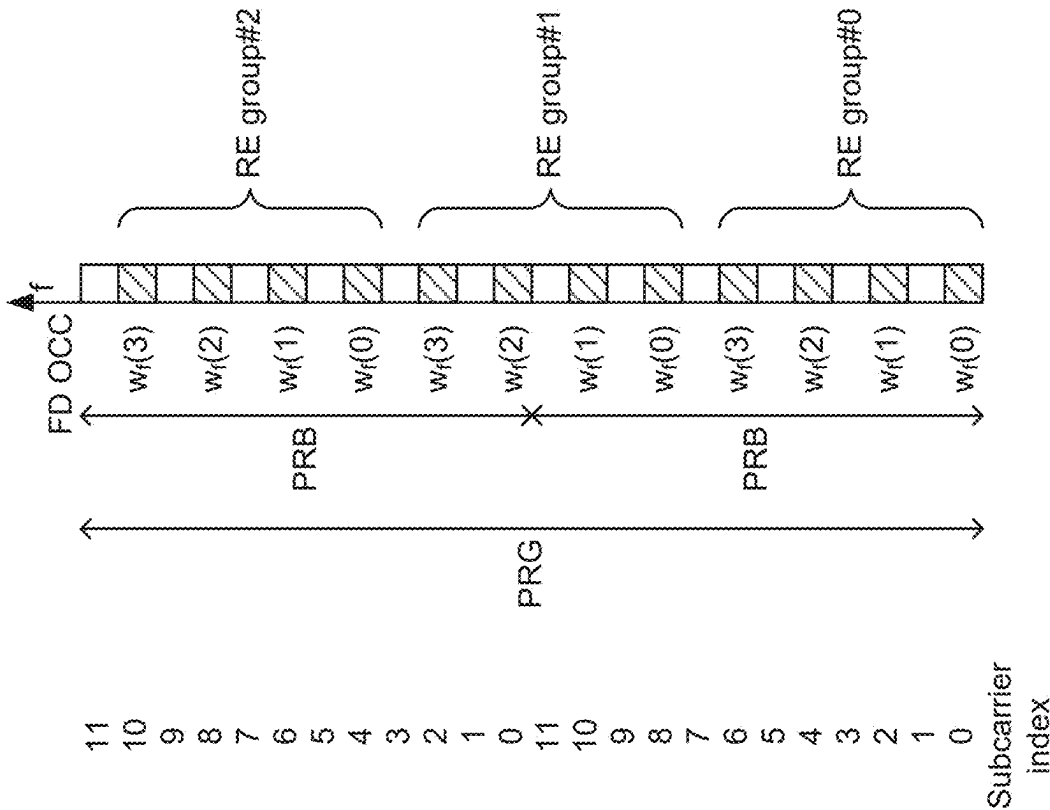


FIG. 28A



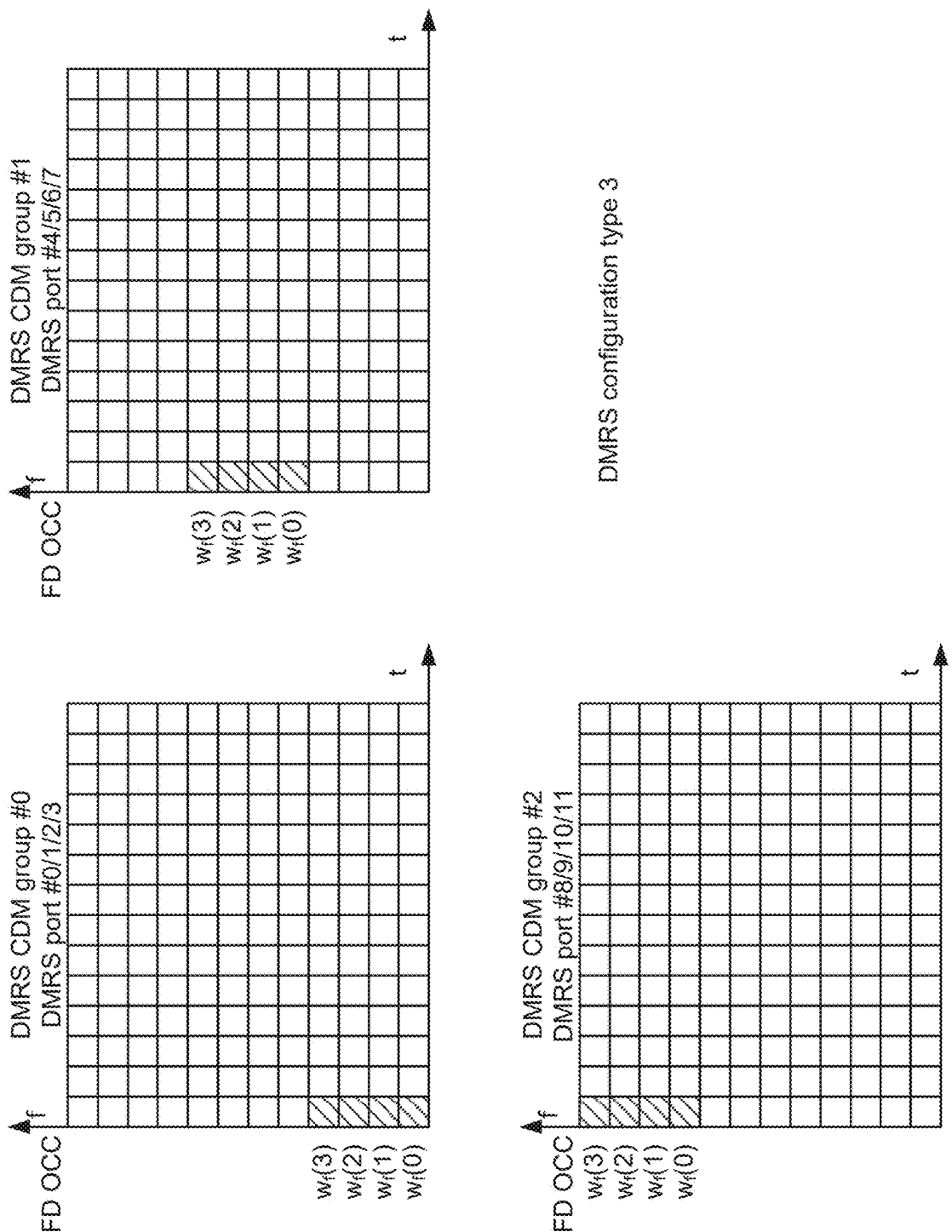


FIG. 29

FIG. 30A

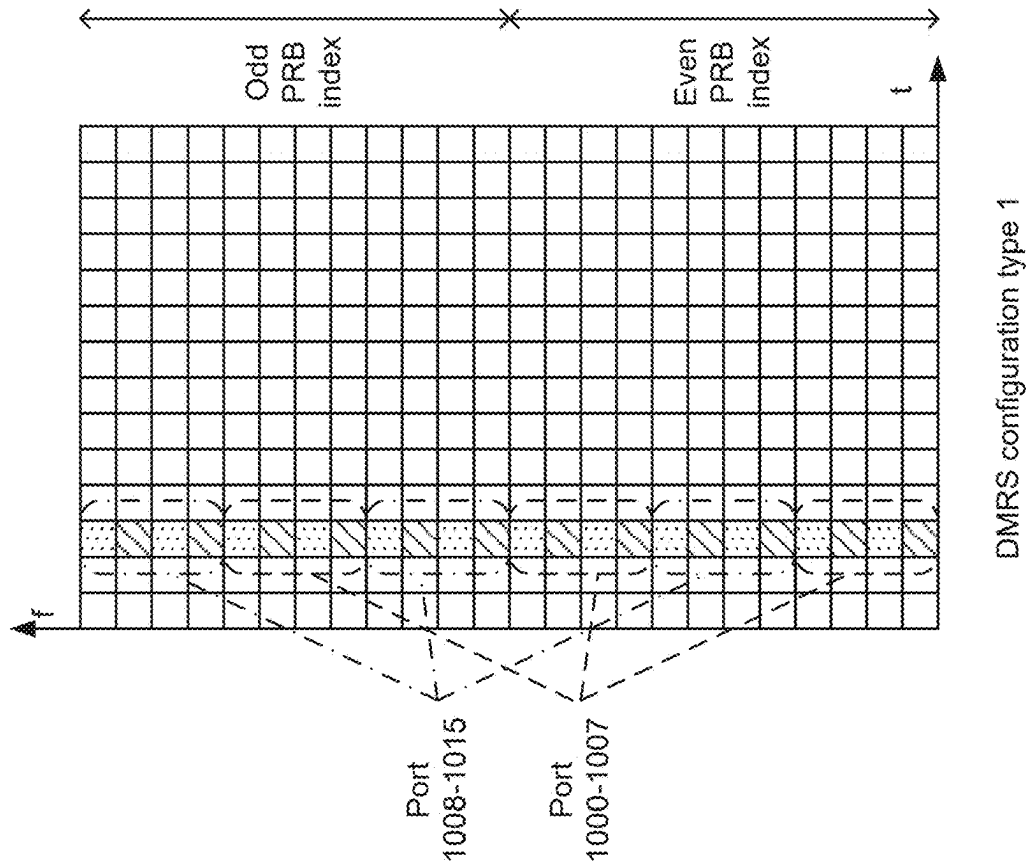
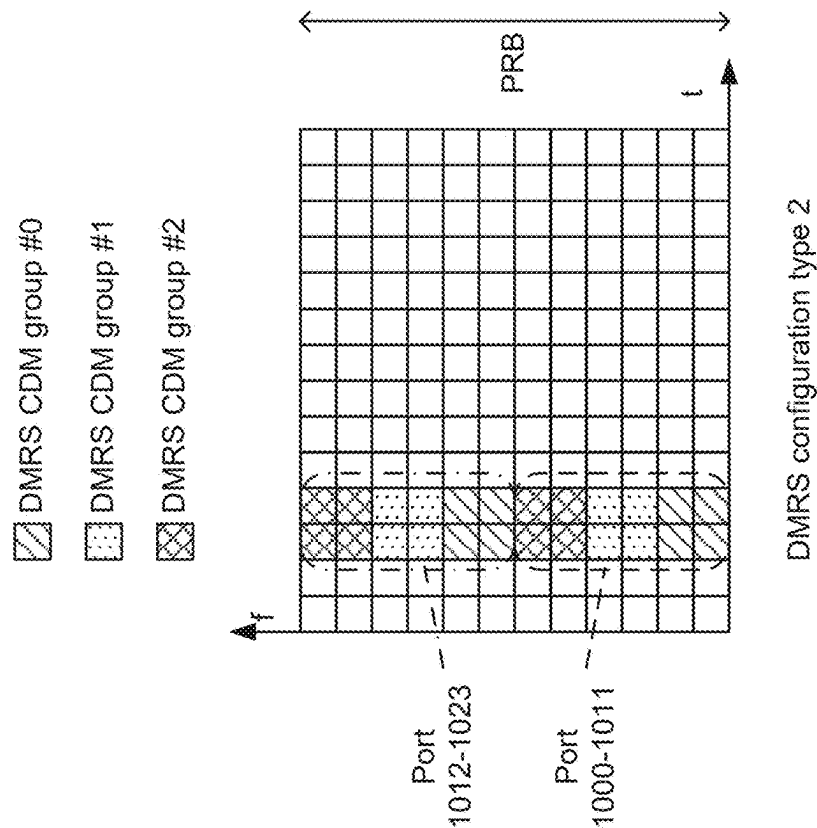


FIG. 30B



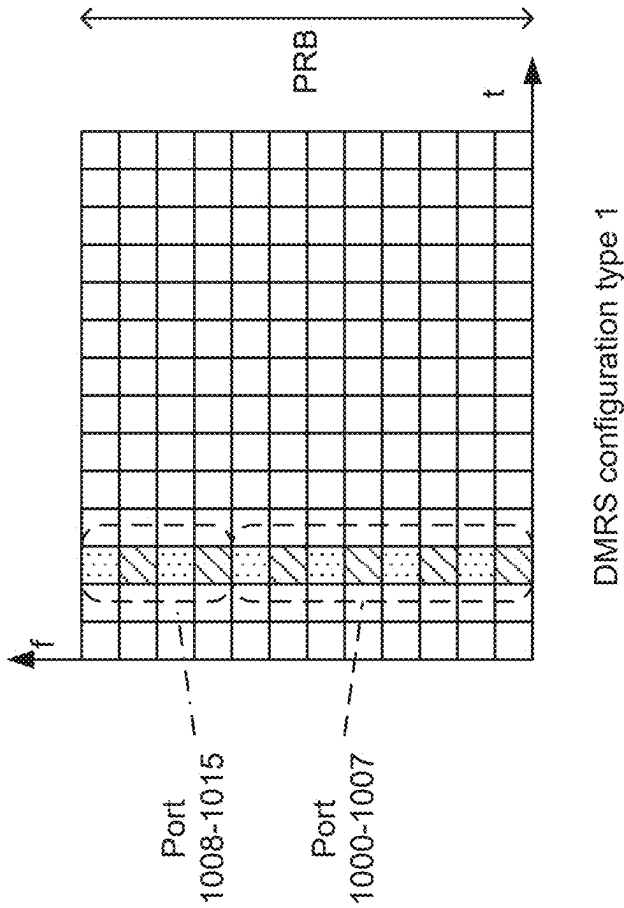


FIG. 31

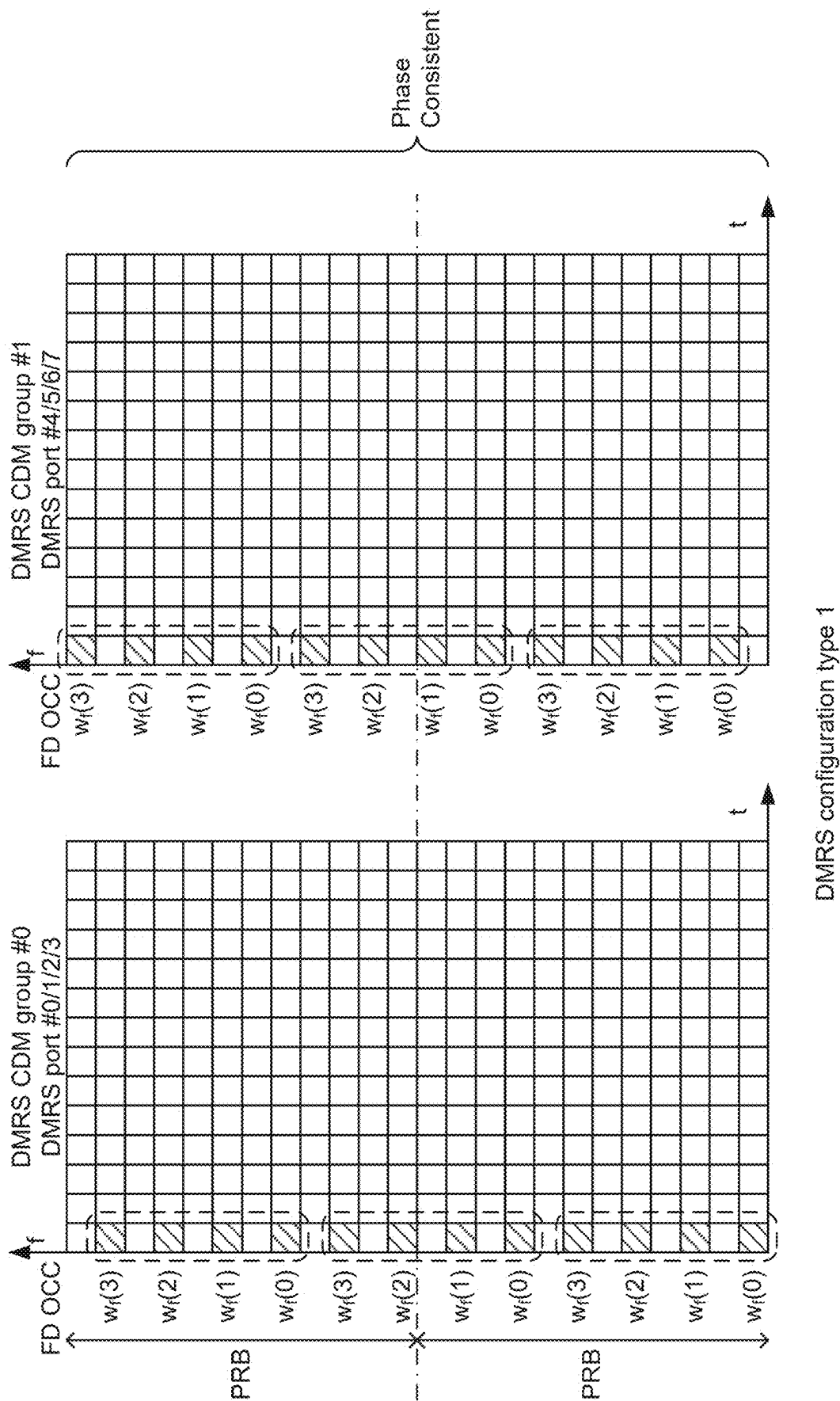


FIG. 32

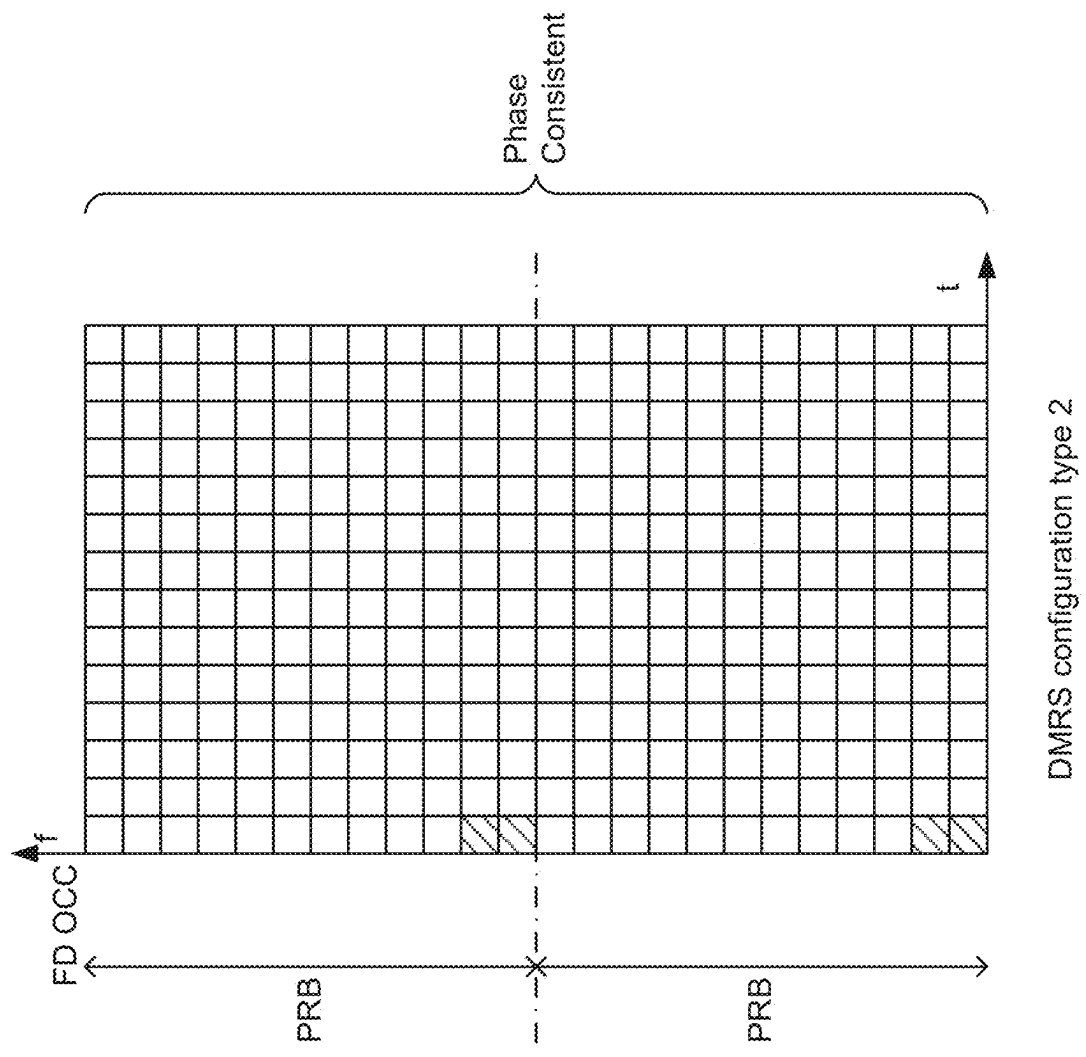


FIG. 33

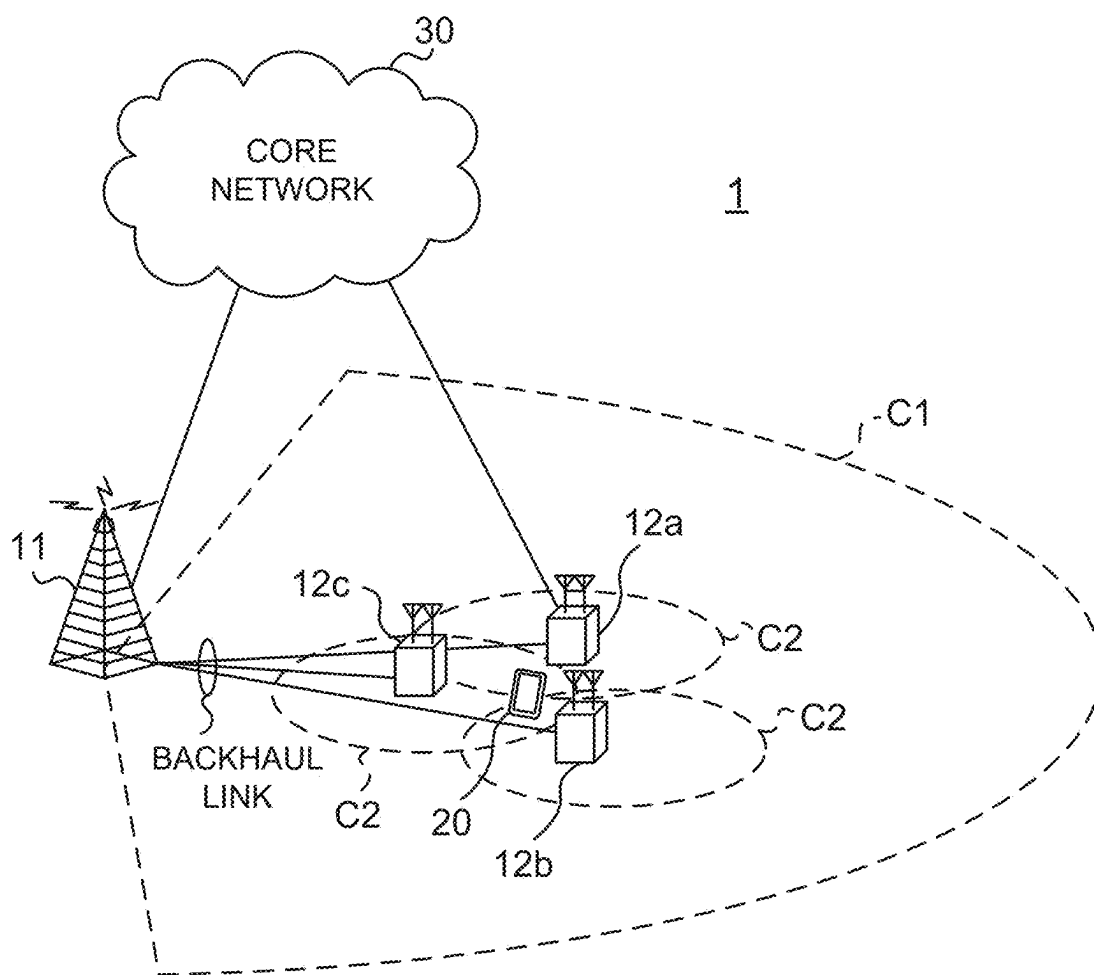


FIG. 34

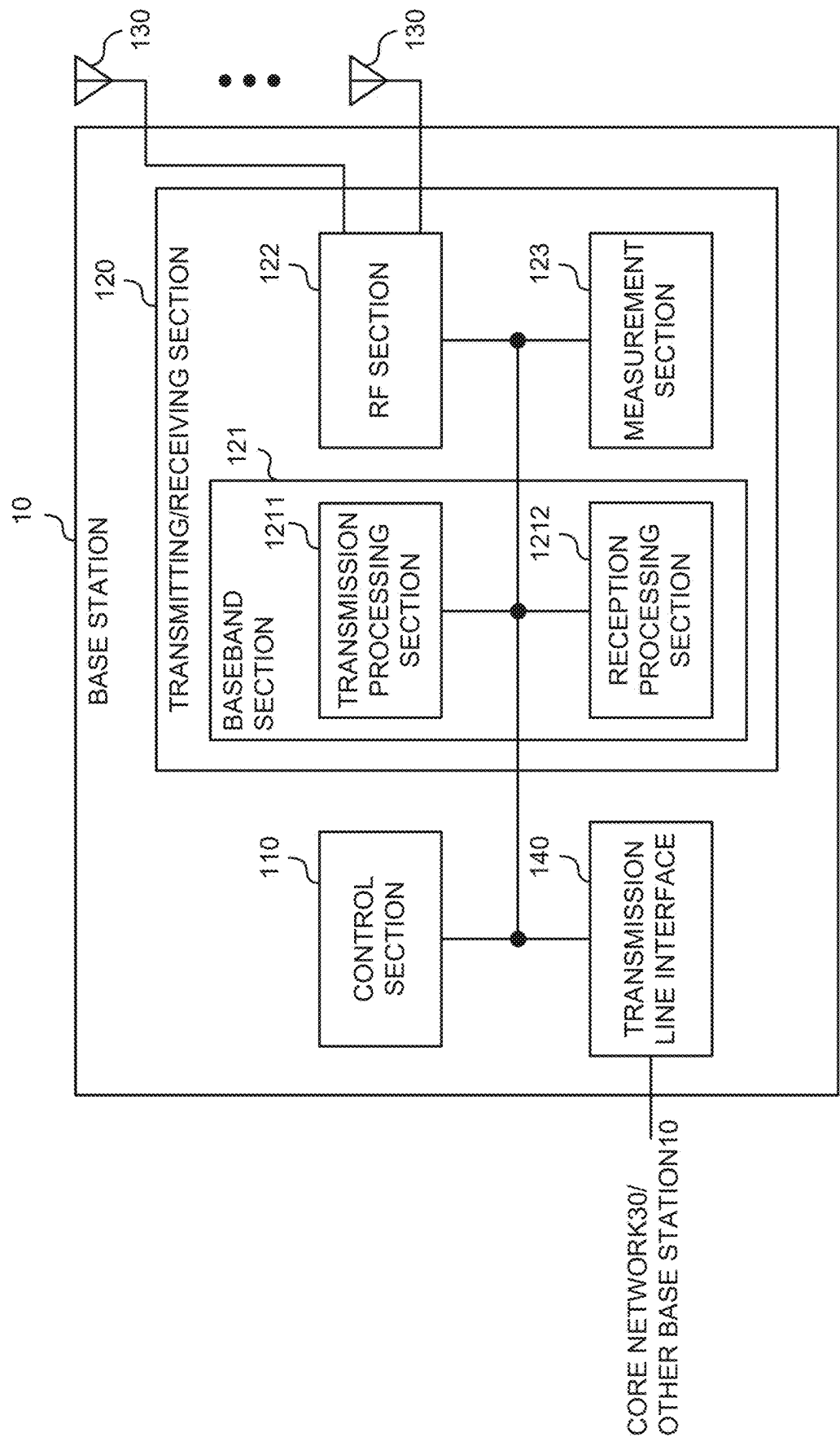


FIG. 35

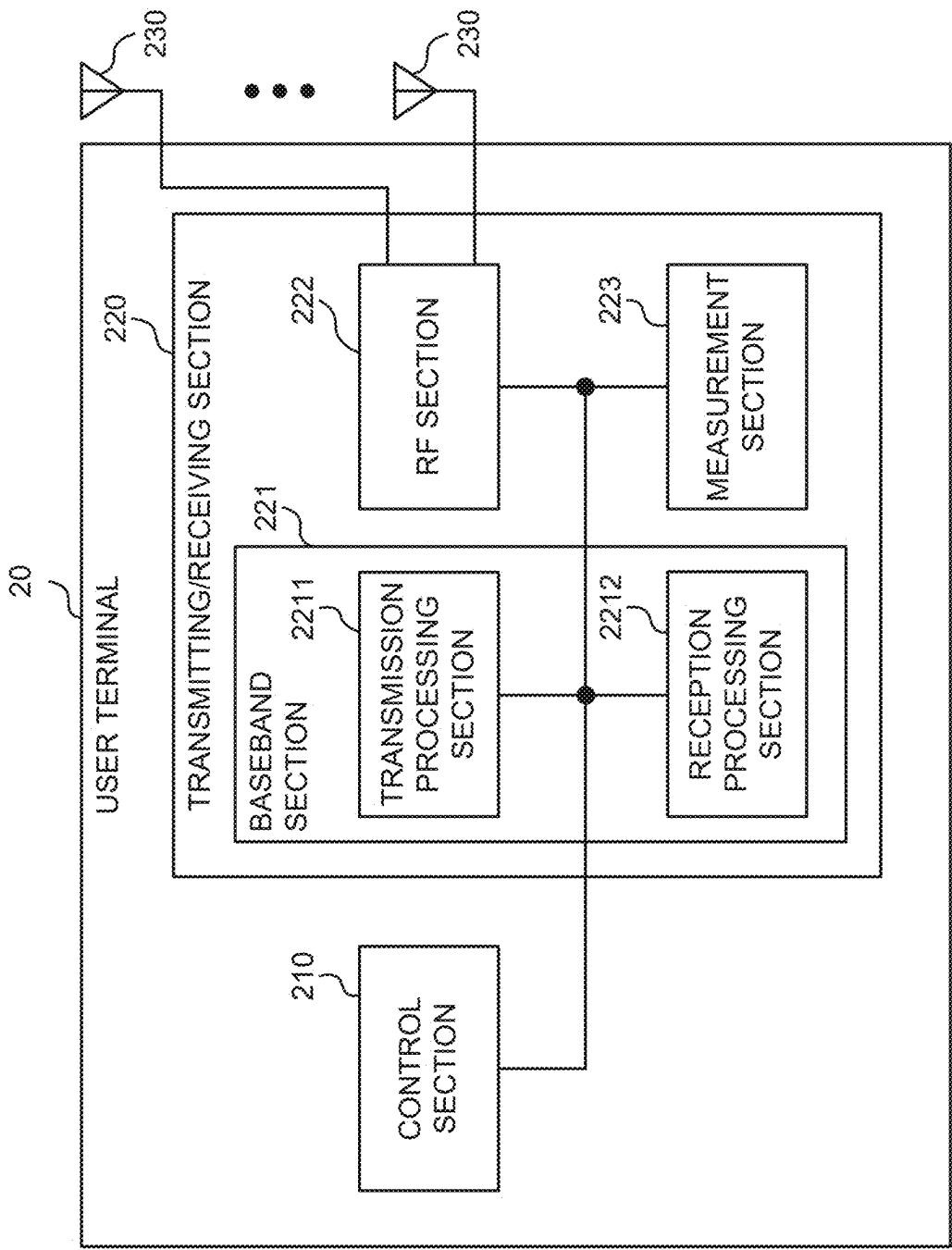


FIG. 36

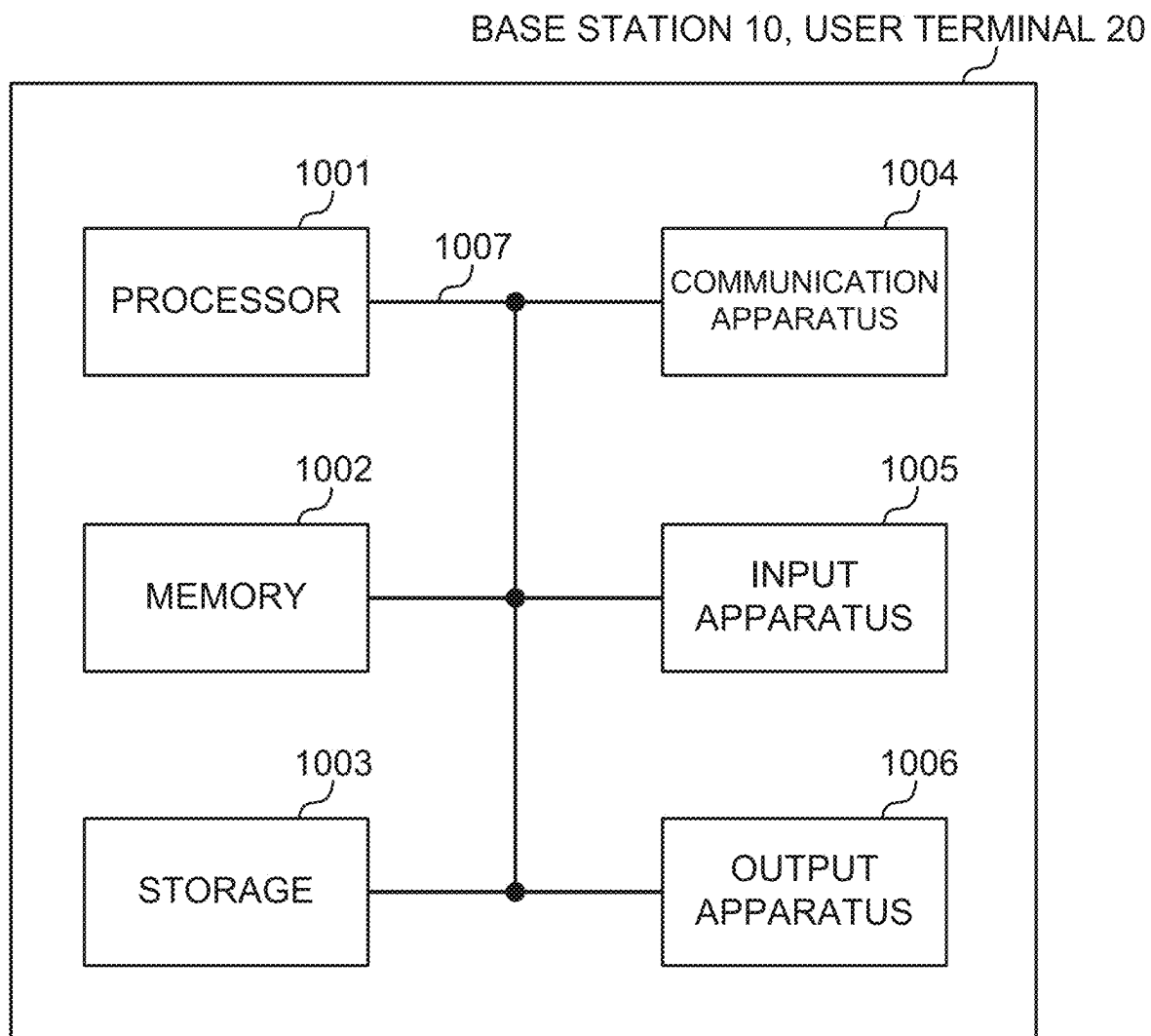


FIG. 37

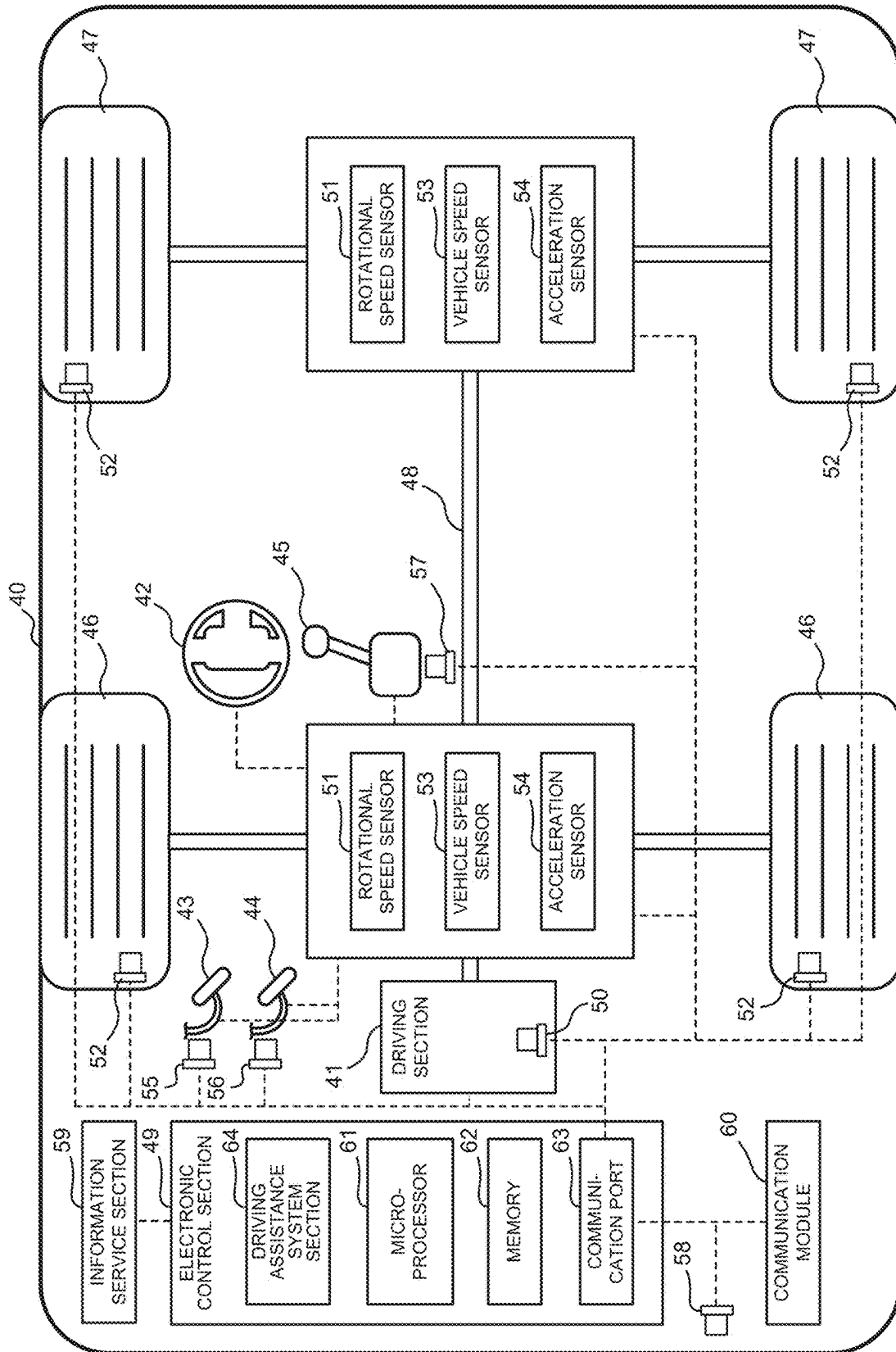


FIG. 38

TERMINAL, RADIO COMMUNICATION METHOD, AND BASE STATION

TECHNICAL FIELD

[0001] The present disclosure relates to a terminal, a radio communication method, and a base station in next-generation mobile communication systems.

BACKGROUND ART

[0002] In a Universal Mobile Telecommunications System (UMTS) network, the specifications of Long-Term Evolution (LTE) have been drafted for the purpose of further increasing high speed data rates, providing lower latency and so on (see Non-Patent Literature 1). In addition, for the purpose of further high capacity, advancement and the like of the LTE (Third Generation Partnership Project (3GPP) Release (Rel.) 8 and Rel. 9), the specifications of LTE-Advanced (3GPP Rel. 10 to Rel. 14) have been drafted.

[0003] Successor systems of LTE (for example, also referred to as “5th generation mobile communication system (5G),” “5G+ (plus),” “6th generation mobile communication system (6G),” “New Radio (NR),” “3GPP Rel. 15 (or later versions),” and so on) are also under study.

CITATION LIST

Non-Patent Literature

[0004] Non-Patent Literature 1: 3GPP TS 36.300 V8.12.0 “Evolved Universal Terrestrial Radio Access (E-UTRA) and Evolved Universal Terrestrial Radio Access Network (E-UTRAN); Overall description; Stage 2 (Release 8),” April, 2010

SUMMARY OF INVENTION

Technical Problem

[0005] In future radio communication systems (for example, NR), a method of beam management is introduced. For example, in NR, forming (or using) beams in at least one of a base station and a user terminal (User Equipment (UE)) has been under study.

[0006] At the same time, for the sake of orthogonalization of layers or the like, a reference signal (for example, a demodulation reference signal (DMRS)) having a plurality of ports is used. In the future radio communication systems, increasing the number of DMRS ports more than that in existing specifications is required. However, how to increase the number of DMRS ports has not yet been actively studied. If an appropriate number of DMRS ports cannot be used, communication throughput/communication quality may be deteriorated.

[0007] In view of this, the present disclosure has one object to provide a terminal, a radio communication method, and a base station that use an appropriate number of DMRS ports.

Solution to Problem

[0008] A terminal according to one aspect of the present disclosure includes a receiving section that receives a configuration of a demodulation reference signal (DMRS) of a shared channel, and a control section that applies a fre-

quency domain (FD)-orthogonal cover code (OCC) having a length of 4 or 3 to the DMRS, based on the configuration.

Advantageous Effects of Invention

[0009] According to one aspect of the present disclosure, an appropriate number of DMRS ports can be used.

BRIEF DESCRIPTION OF DRAWINGS

[0010] FIG. 1 shows an example of mapping of a DMRS.

[0011] FIGS. 2A and 2B show examples of DMRS configuration type 1/2.

[0012] FIGS. 3A and 3B show examples of a single symbol DMRS.

[0013] FIGS. 4A and 4B show examples of a double symbol DMRS.

[0014] FIG. 5 shows an example of DMRS configuration type 1 and the single symbol DMRS.

[0015] FIG. 6 shows a first example of DMRS configuration type 1 and the double symbol DMRS.

[0016] FIG. 7 shows a second example of DMRS configuration type 1 and the double symbol DMRS.

[0017] FIG. 8 shows a first example of DMRS configuration type 2 and the single symbol DMRS.

[0018] FIG. 9 shows a second example of DMRS configuration type 2 and the single symbol DMRS.

[0019] FIG. 10 shows a first example of DMRS configuration type 2 and the double symbol DMRS.

[0020] FIG. 11 shows a second example of DMRS configuration type 2 and the double symbol DMRS.

[0021] FIG. 12 shows a third example of DMRS configuration type 2 and the double symbol DMRS.

[0022] FIG. 13 shows an example of parameters for PDSCH DMRS configuration type 1.

[0023] FIG. 14 shows an example of parameters for PUSCH DMRS configuration type 1.

[0024] FIG. 15 shows an example of application of an FD OCC having a length of 6.

[0025] FIGS. 16A and 16B show examples of the FD OCC having a length of 6.

[0026] FIG. 17 shows another example of the FD OCC having a length of 6.

[0027] FIGS. 18A and 18B show examples of a method of generating the FD OCC having a length of 6.

[0028] FIG. 19 shows an example of a method of applying the FD OCC having a length of 6.

[0029] FIG. 20 shows an example of application of the FD OCC having a length of 4.

[0030] FIGS. 21A and 21B show examples of the FD OCC having a length of 4.

[0031] FIG. 22 shows an example of generation of the FD OCC having a length of 4.

[0032] FIG. 23 shows another example of the FD OCC having a length of 4.

[0033] FIG. 24 shows an example of application of the FD OCC having a length of 2.

[0034] FIG. 25 shows an example of application of the FD OCC having a length of 3.

[0035] FIGS. 26A and 26B show examples of the FD OCC having a length of 3.

[0036] FIG. 27 shows another example of the FD OCC having a length of 3.

[0037] FIGS. 28A and 28B show examples of embodiment #1-4.

[0038] FIG. 29 shows an example of embodiment #2.
 [0039] FIGS. 30A and 30B show examples of embodiment #3.
 [0040] FIG. 31 shows another example of embodiment #3.
 [0041] FIG. 32 shows an example of embodiment #4.
 [0042] FIG. 33 shows another example of embodiment #4.
 [0043] FIG. 34 is a diagram to show an example of a schematic structure of a radio communication system according to one embodiment.
 [0044] FIG. 35 is a diagram to show an example of a structure of a base station according to one embodiment.
 [0045] FIG. 36 is a diagram to show an example of a structure of a user terminal according to one embodiment.
 [0046] FIG. 37 is a diagram to show an example of a hardware structure of the base station and the user terminal according to one embodiment.
 [0047] FIG. 38 is a diagram to show an example of a vehicle according to one embodiment.

DESCRIPTION OF EMBODIMENTS

(Beam Management)

[0048] In NR, a method of beam management is introduced. For example, for NR, forming (or using) beams in at least one of a base station and a UE has been under study.
 [0049] Through application of beam forming (BF), it is expected that difficulty in securing coverage due to increase in carrier frequency be reduced, and radio wave propagation loss be reduced.

[0050] BF is, for example, a technique in which a beam (antenna directivity) is formed by controlling (also referred to as precoding) amplitude/phase of a signal that is transmitted or received from each element by using an ultra multi-element antenna. Note that Multiple Input Multiple Output (MIMO) using such an ultra multi-element antenna is also referred to as massive MIMO.

[0051] Sweeping of beams may be performed in both of transmission and reception, and control may be performed so that an appropriate pair is selected out of candidates of a plurality of patterns of transmit and receive beam pairs. The pair of the transmit beam and the receive beam may be referred to as a beam pair, and may be identified as a beam pair candidate index.

[0052] Note that, in beam management, a single beam is not used, and a plurality of levels of beam control, such as a rough beam and a fine beam, may be performed.

[0053] BF can be categorized into digital BF and analog BF. Digital BF and analog BF may be referred to as digital precoding and analog precoding, respectively.

[0054] Digital BF is, for example, a method in which precoding signal processing is performed on a baseband (for a digital signal). In this case, as many parallel processings, such as inverse fast Fourier transform (IFFT), digital to analog conversion (Digital to Analog Converter (DAC)), and Radio Frequency (RF), as the number of antenna ports (or RF chains) are required. At the same time, as many beams as the number of RF chains can be formed at any timing.

[0055] Analog BF is, for example, a method in which a phase shifter is used in RF. In analog BF, a plurality of beams cannot be formed at the same timing; however, a configuration thereof is easy and can be implemented at a low cost because it is only necessary that phase of RF signals be rotated.

[0056] Note that a hybrid BF configuration, which is a combination of digital BF and analog BF, can be implemented as well. In NR, introduction of massive MIMO has been under study. When forming of a great number of beams is intended to be performed by means of only digital BF, however, a circuit configuration costs much. Thus, use of the hybrid BF configuration is also assumed.

(TCI, Spatial Relation, QCL)

[0057] For NR, control of reception processing (for example, at least one of reception, demapping, demodulation, and decoding) and transmission processing (for example, at least one of transmission, mapping, precoding, modulation, and coding) of at least one of a signal and a channel (which may be referred to as a signal/channel; hereinafter, in a similar manner, "A/B" may be interpreted as "at least one of A and B") based on a transmission configuration indication state (TCI state) has been under study.

[0058] The TCI state may be a state applied to a downlink signal/channel. A state that corresponds to the TCI state applied to an uplink signal/channel may be expressed as spatial relation.

[0059] The TCI state is information related to quasi-collocation (QCL) of the signal/channel, and may be referred to as a spatial reception parameter, spatial relation information (SRI), or the like. The TCI state may be configured for the UE for each channel or for each signal.

[0060] QCL is an indicator indicating statistical properties of the signal/channel. For example, when a given signal/channel and another signal/channel are in a relationship of QCL, it may be indicated that it is assumable that at least one of Doppler shift, a Doppler spread, an average delay, a delay spread, and a spatial parameter (for example, a spatial reception parameter (spatial Rx parameter)) is the same (the relationship of QCL is satisfied in at least one of these) between such a plurality of different signals/channels.

[0061] Note that the spatial reception parameter may correspond to a receive beam of the UE (for example, a receive analog beam), and the beam may be identified based on spatial QCL. The QCL (or at least one element in the relationship of QCL) in the present disclosure may be interpreted as sQCL (spatial QCL).

[0062] For the QCL, a plurality of types (QCL types) may be defined. For example, four QCL types A to D may be provided, which have different parameter(s) (or parameter set(s)) that can be assumed to be the same, and such parameter(s) (which may be referred to as QCL parameter(s)) are described below:

[0063] QCL type A: Doppler shift, Doppler spread, average delay, and delay spread

[0064] QCL type B: Doppler shift and Doppler spread

[0065] QCL type C: Doppler shift and Average delay

[0066] QCL type D: Spatial reception parameter

[0067] Types A to C may correspond to QCL information related to synchronization processing of at least one of time and frequency, and type D may correspond to QCL information related to beam control.

[0068] A case that the UE assumes that a given control resource set (CORESET), channel, or reference signal is in a relationship of specific QCL (for example, QCL type D) with another CORESET, channel, or reference signal may be referred to as QCL assumption.

[0069] The UE may determine at least one of a transmit beam (Tx beam) and a receive beam (Rx beam) of the signal/channel, based on the TCI state or the QCL assumption of the signal/channel.

[0070] The TCI state may be, for example, information related to QCL between a channel as a target (or a reference signal (RS) for the channel) and another signal (for example, another downlink reference signal (DL-RS)). The TCI state may be configured (indicated) by higher layer signaling or physical layer signaling, or a combination of these.

[0071] In the present disclosure, the higher layer signaling may be, for example, any one or combinations of Radio Resource Control (RRC) signaling, Medium Access Control (MAC) signaling, broadcast information, and the like.

[0072] The MAC signaling may use, for example, a MAC control element (MAC CE), a MAC Protocol Data Unit (PDU), or the like. The broadcast information may be, for example, a master information block (MIB), a system information block (SIB), minimum system information (Remaining Minimum System Information (RMSI)), other system information (OSI), or the like.

[0073] The physical layer signaling may be, for example, downlink control information (DCI).

[0074] A channel for which the TCI state is configured (indicated) may be, for example, at least one of a downlink shared channel (Physical Downlink Shared Channel (PDSCH)), a downlink control channel (Physical Downlink Control Channel (PDCCH)), an uplink shared channel (Physical Uplink Shared Channel (PUSCH)), and an uplink control channel (Physical Uplink Control Channel (PUCCH)).

[0075] The RS (DL-RS) to have a QCL relationship with the channel may be, for example, at least one of a synchronization signal block (SSB), a channel state information reference signal (CSI-RS), and a reference signal for measurement (Sounding Reference Signal (SRS)). Alternatively, the DL-RS may be a CSI-RS used for tracking (also referred to as a Tracking Reference Signal (TRS)), or a reference signal used for QCL detection (also referred to as a QRS).

[0076] The SSB is a signal block including at least one of a primary synchronization signal (PSS), a secondary synchronization signal (SSS), and a broadcast channel (Physical Broadcast Channel (PBCH)). The SSB may be referred to as an SS/PBCH block.

[0077] An information element of the TCI state ("TCI-state IE" of RRC) configured using higher layer signaling may include one or a plurality of pieces of QCL information ("QCL-Info"). The QCL information may include at least one of information related to the DL-RS to have a QCL relationship (DL-RS relation information) and information indicating a QCL type (QCL type information). The DL-RS relation information may include information such as an index of the DL-RS (for example, an SSB index, or a non-zero power CSI-RS (NZP CSI-RS) resource ID (Identifier)), an index of a cell in which the RS is located, and an index of a Bandwidth Part (BWP) in which the RS is located.

(Development of MIMO Technology and Beam)

[0078] Incidentally, the MIMO technology has hitherto been used in a frequency band (or a frequency band) lower than 6 GHz. However, further application to a frequency band higher than 6 GHz in future has been under study.

[0079] Note that the frequency band lower than 6 GHz may be referred to as sub-6, frequency range (FR) 1, or the

like. The frequency band higher than 6 GHz may be referred to as above-6, FR2, a millimeter wave (mmW), FR4, or the like.

[0080] It is assumed that the maximum number of MIMO layers is limited by an antenna size.

[0081] It is expected that, even with mmW, through use of MIMO of a high degree as well as cooperation of a plurality of UEs, the degree of freedom and diversity of MIMO multiplexing are enhanced, which further leads to enhancement of throughput.

[0082] In this manner, it is assumed that, in future radio communication systems (for example, NR of Rel. 17 or later versions), even in a high frequency (for example, FR2), operation of only a digital beam without use of an analog beam (which may be referred to as full digital operation) is used and operation implementing dominant use of the digital beam is used.

[0083] For example, in a case of the full digital operation, by simultaneously applying orthogonal precoding (or an orthogonal beam, a digital beam) to a plurality of UEs, improvement of spectral efficiency can be expected. When the digital beam cannot be appropriately applied, interference between the UEs is increased, which leads to deterioration of communication quality (or reduction of cell capacity). Note that orthogonality according to the present disclosure may be interpreted as semi-orthogonality.

[0084] When the base station (which may be interpreted as a transmission/reception point (TRP), a panel, or the like) can transmit only one beam at given time, the base station performs transmission and reception by switching beams for the UE. When the base station can transmit a plurality of beams at given time, the base station can simultaneously perform transmission and reception to and from a plurality of UEs by using different beams.

[0085] Even if the base station adopts the full digital operation, the Rel-15 UE ought to be supported as far as the Rel-15 UE is present.

(DMRS)

[0086] A front-loaded DMRS is a first (first symbol or a symbol near the first symbol) DMRS for earlier demodulation. An additional DMRS can be configured by the RRC, for the sake of a high-speed moving UE or a high modulation and coding scheme (MCS)/rank (FIG. 1). A frequency position of the additional DMRS is the same as that of the front-loaded DMRS.

[0087] For the time domain, DMRS mapping type A or B is configured. In DMRS mapping type A, DMRS position 1₀ is counted with a symbol index in the slot. 1₀ is configured by an MIB or a parameter (dmrs-TypeA-Position) in a common serving cell configuration (ServingCell-ConfigCommon). DMRS position 0 (reference point 1) means a first symbol of a slot or each frequency hop. In DMRS mapping type B, DMRS position 1₀ is counted with a symbol index in a PDSCH/PUSCH. 1₀ is invariably 0. DMRS position 0 (reference point 1) means a first symbol of a PDSCH/PUSCH or each frequency hop.

[0088] The DMRS position is defined by a table in a specification, and is dependent upon duration of the PDSCH/PUSCH. The position of the additional DMRS is fixed.

[0089] For the frequency domain, DMRS configuration type 1 or 2 is configured. DMRS configuration type 1 has a comb structure, and can be applied to both of CP-OFDM

(transport precoding=disabled) and DFT-S-OFDM (transport precoding=enabled). DMRS configuration type 2 can be applied only to CP-OFDM. FIG. 2A shows an example of DMRS configuration type 1. FIG. 2B shows an example of DMRS configuration type 2.

[0090] A single symbol DMRS or a double symbol DMRS is configured.

[0091] The single symbol DMRS is normally used (mandatory in Rel. 15). In the single symbol DMRS, the number of additional DMRSs (symbols) is {0, 1, 2, 3}. The single symbol DMRS supports both of a case in which frequency hopping is enabled and a case in which frequency hopping is disabled. If a maximum number (maxLength) in an uplink DMRS configuration (DMRS-UplinkConfig) is not configured, the single symbol DMRS is used. FIG. 3A shows an example of the single symbol DMRS (number of additional DMRSs=3) of DMRS configuration type 1. FIG. 3B shows an example of the single symbol DMRS (number of additional DMRSs=3) of DMRS configuration type 2.

[0092] The double symbol DMRS is used for a larger number of DMRS ports (in particular, MU-MIMO). In the double symbol DMRS, the number of additional DMRSs (symbols) is {0, 1}. The double symbol DMRS supports a case in which frequency hopping is disabled. If a maximum number (maxLength) in the uplink DMRS configuration (DMRS-UplinkConfig) is 2 (len2), whether the single symbol DMRS is used or the double symbol DMRS is used is determined by DCI or a configured grant. FIG. 4A shows an example of the double symbol DMRS (number of additional DMRSs=1) of DMRS configuration type 1. FIG. 4B shows an example of the double symbol DMRS (number of additional DMRSs=1) of DMRS configuration type 2.

[0093] From the above, as configuration patterns allowing the DMRS, the following combinations are conceivable.

[0094] DMRS configuration type 1, DMRS mapping type A, single symbol DMRS

[0095] DMRS configuration type 1, DMRS mapping type A, double symbol DMRS

[0096] DMRS configuration type 1, DMRS mapping type B, single symbol DMRS

[0097] DMRS configuration type 1, DMRS mapping type B, double symbol DMRS

[0098] DMRS configuration type 2, DMRS mapping type A, single symbol DMRS

[0099] DMRS configuration type 2, DMRS mapping type A, double symbol DMRS

[0100] DMRS configuration type 2, DMRS mapping type B, single symbol DMRS

[0101] DMRS configuration type 2, DMRS mapping type B, double symbol DMRS

[0102] A plurality of DMRS ports mapped to the same REs (time and frequency resources) are referred to as a DMRS CDM group.

[0103] For DMRS configuration type 1 and the single symbol DMRS, four DMRS ports can be used. In each DMRS CDM group, two DMRS ports are multiplexed using an FD OCC having a length of 2. Among a plurality of DMRS CDM groups (two DMRS CDM groups), two DMRS ports are multiplexed using FDM (FIG. 5).

[0104] For DMRS configuration type 1 and the double symbol DMRS, eight DMRS ports can be used. In each DMRS CDM group, two DMRS ports are multiplexed using an FD OCC having a length of 2, and two DMRS ports are multiplexed using a TD OCC. Among a plurality of DMRS

CDM groups (two DMRS CDM groups), two DMRS ports are multiplexed using FDM (FIG. 6 and FIG. 7).

[0105] For DMRS configuration type 2 and the single symbol DMRS, six DMRS ports can be used. In each DMRS CDM group, two DMRS ports are multiplexed using an FD OCC having a length of 2. Among a plurality of DMRS CDM groups (three DMRS CDM groups), three DMRS ports are multiplexed using FDM (FIG. 8 and FIG. 9).

[0106] For DMRS configuration type 2 and the double symbol DMRS, 12 DMRS ports can be used. In each DMRS CDM group, two DMRS ports are multiplexed using an FD OCC having a length of 2, and two DMRS ports are multiplexed using a TD OCC. Among a plurality of DMRS CDM groups (three DMRS CDM groups), three DMRS ports are multiplexed using FDM (FIG. 10, FIG. 11, and FIG. 12).

[0107] Here, although an example of DMRS mapping type B is shown above, the same holds true for DMRS mapping type A.

[0108] In parameters for a PDSCH DMRS (FIG. 13), DMRS ports 1000 to 1007 can be used for DMRS configuration type 1, and DMRS ports 1000 to 1011 can be used for DMRS configuration type 2.

[0109] In parameters for a PUSCH DMRS (FIG. 14), DMRS ports 0 to 7 can be used for DMRS configuration type 1, and DMRS ports 0 to 11 can be used for DMRS configuration type 2.

(Joint Channel Estimation)

[0110] When joint channel estimation (coverage enhancement scheme) for a plurality of slots/sub-slots is configured, the TD OCC may be applied to the plurality of slots. When joint channel estimation (coverage enhancement scheme) for a plurality of slots/sub-slots is configured, the phase of a signal across the plurality of slots/sub-slots may be assumed to be continuous/coherent. Configuration of joint channel estimation may be equivalent to configuration of DMRS bundling (for example, PUSCH-DMRS-Bundling for the PUSCH, PUCCH-DMRS-Bundling for the PUCCH).

[0111] For the UL/DL, a configured time domain window may be configured for the UE. For example, the configuration may include at least two of an index of a starting slot/sub-slot, an index of an ending slot/sub-slot, and duration of the window. The TD OCC across the plurality of slots/sub-slots may be applied within the window.

(Ports of Reference Signal)

[0112] For the sake of orthogonalization of the MIMO layers or the like, a reference signal (for example, a demodulation reference signal (DMRS) or a CSI-RS) having a plurality of ports is used.

[0113] For example, regarding single user MIMO (SU-MIMO), a different DMRS port/CSI-RS port may be configured for each layer. Regarding multi user MIMO (MU-MIMO), a different DMRS port/CSI-RS port may be configured for each layer in one UE and for each UE.

[0114] Note that, when the number of CSI-RS ports having a value larger than the number of layers used for data is used, it is expected that more accurate channel state measurement can be performed based on the CSI-RS, which contributes to improvement of throughput.

[0115] In Rel-15 NR, regarding the DMRS having a plurality of ports, through the use of frequency division

multiplexing (FDM), a frequency domain orthogonal cover code (FD-OCC), a time domain OCC (TD-OCC), or the like, a maximum of eight ports are supported for a type 1 DMRS (in other words, DMRS configuration type 1), and a maximum of 12 ports are supported for a type 2 DMRS (in other words, DMRS configuration type 2).

[0116] In Rel-15 NR, as the FDM mentioned above, a pattern of transmission frequency having a shape of teeth of a comb (comb-like resource set) is used. As the FD-OCC mentioned above, a cyclic shift (CS) is used. The TD-OCC mentioned above can only be applied to a double symbol DMRS.

[0117] The OCC according to the present disclosure may be interchangeably interpreted as an orthogonal code, orthogonalization, a cyclic shift, or the like.

[0118] Note that the type of the DMRS may be referred to as a DMRS configuration type.

[0119] Among the DMRSs, the DMRS that is subjected to resource mapping in the unit of consecutive (adjacent) two symbols may be referred to as a double symbol DMRS, and the DMRS that is subjected to resource mapping in the unit of one symbol may be referred to as a single symbol DMRS.

[0120] Both of the DMRSs may be mapped to one or more symbols per slot, according to the length of a data channel. The DMRS that is mapped to a start position of a data symbol may be referred to as a front-loaded DMRS, and the DMRS that is additionally mapped to a position other than the start position may be referred to as an additional DMRS.

[0121] In a case of DMRS configuration type 1 and the single symbol DMRS, the Comb and the CS may be used for orthogonalization. For example, up to 4 antenna ports (APs) may be supported by using two types of Combs and two types of CSs (Comb2+2CS).

[0122] In a case of DMRS configuration type 1 and the double symbol DMRS, the Comb, the CS, and the TD-OCC may be used for orthogonalization. For example, up to 8 APs may be supported by using two types of Combs, two types of CSs, and TD-OCCs ($\{1, 1\}$ and $\{1, -1\}$).

[0123] In a case of DMRS configuration type 2 and the single symbol DMRS, the FD-OCC may be used for orthogonalization. For example, up to 6 APs may be supported by applying the orthogonal code (2-FD-OCC) to two resource elements (REs) adjacent to each other in the frequency direction.

[0124] In a case of DMRS configuration type 2 and the double symbol DMRS, the FD-OCC and the TD-OCC may be used for orthogonalization. For example, up to 12 APs may be supported by applying the orthogonal code (2-FD-OCC) to two REs adjacent in the frequency direction and applying the TD-OCC ($\{1, 1\}$ and $\{1, -1\}$) to two REs adjacent in the time direction.

[0125] In Rel-15 NR, regarding the CSI-RS having a plurality of ports, through the use of FDM, time division multiplexing (TDM), a frequency domain OCC, a time domain OCC, or the like, a maximum of 32 ports are supported. Regarding orthogonalization of the CSI-RS as well, a method similar to that for the DMRS described above may be applied.

[0126] Incidentally, a group of DMRS ports subjected to orthogonalization by means of the FD-OCC/TD-OCC as described above is also referred to as a code division multiplexing (CDM) group.

[0127] Different CDM groups are subjected to FDM, and are thus orthogonal to each other. In contrast, in the same

CDM group, orthogonality of the applied OCC may be collapsed due to channel change or the like. In this case, when a signal in the same CDM group is received with different received power, the near-far problem is caused, which may lead to a failure in securing orthogonality.

[0128] Here, the TD-OCC/FD-OCC of the DMRS according to Rel-15 NR will be described. The DMRS mapped to the resource element (RE) may correspond to a sequence, which is obtained by multiplying a parameter (which may be referred to as a sequence element or the like) $w_f(k')$ of the FD-OCC and a parameter (which may be referred to as a sequence element or the like) $w_t(l')$ of the TD-OCC on a DMRS sequence.

[0129] The TD-OCC and the FD-OCC of the DMRS according to Rel-15 NR both correspond to the OCC having a sequence length (which may be referred to as an OCC length) of 2. Thus, a possible value of each of k' and l' in the above is 0 or 1. By multiplying the FD-OCC by the unit of the RE, the DMRS having two ports can be multiplexed by using the same time and frequency resources (two REs). When both of the FD-OCC and the TD-OCC are applied, the DMRS having four ports can be multiplexed by using the same time and frequency resources (four REs).

[0130] The two tables for the parameters for the PDSCH DMRS described above respectively correspond to DMRS configuration type 1 and type 2. Note that p represents a number of the antenna port, and A represents a parameter for shifting (offsetting) the frequency resource.

[0131] For example, to antenna ports 1000 and 1001, $\{w_f(0), w_f(1)\}=\{+1, +1\}$ and $\{w_t(0), w_t(1)\}=\{+1, -1\}$ are respectively applied, so as to be subjected to orthogonalization by using the FD-OCC.

[0132] To antenna ports 1000 to 1001 and antenna ports 1002 to 1003 (in a case of type 2, antenna ports 1004 to 1005 as well), A of different values is applied, so that FDM is applied. Thus, antenna ports 1000 to 1003 (or 1000 to 1005) corresponding to the single symbol DMRS are subjected to orthogonalization by using the FD-OCC and FDM.

[0133] To antenna ports 1000 to 1003 and antenna ports 1004 to 1007 of type 1, $\{w_f(0), w_f(1)\}=\{+1, +1\}$ and $\{w_t(0), w_t(1)\}=\{+1, -1\}$ are respectively applied, so as to be subjected to orthogonalization by using the TD-OCC. Thus, antenna ports 1000 to 1007 (or 1000 to 1011) corresponding to the double symbol DMRS are subjected to orthogonalization by using the FD-OCC, the TD-OCC, and FDM.

[0134] In Rel. 15, a total number of DMRS ports in the single symbol DMRS of (PDSCH DMRS) configuration type 1 is 2 (achieved owing to comb/FDM) $\times$ 2 (achieved owing to FD OCC)=4 ports. In Rel. 15, a total number of DMRS ports in the double symbol DMRS of (PDSCH DMRS) configuration type 1 is 2 (achieved owing to comb/FDM) $\times$ 2 (achieved owing to TD OCC)=8 ports.

[0135] In Rel. 15, a total number of DMRS ports in the single symbol DMRS of (PDSCH DMRS) configuration type 2 is 3 (achieved owing to FDM) $\times$ 2 (achieved owing to FD OCC)=6 ports. In Rel. 15, a total number of DMRS ports in the double symbol DMRS of (PDSCH DMRS) configuration type 2 is 3 (achieved owing to comb) $\times$ 2 (achieved owing to FD OCC) $\times$ 2 (achieved owing to TD OCC)=12 ports.

[0136] Studies have been carried out on defining a larger number of orthogonal DMRS ports for DL and UL MU-MIMO and aiming for a common design for DL and UL

DMRSs without increasing overhead of the DMRSs for CP-OFDM and doubling of a maximum number of orthogonal DMRS ports in both of the single symbol DMRS and the double symbol DMRS to support up to 24 orthogonal DMRS ports for each applicable DMRS type.

[0137] The TD OCC and the FD OCC to be applied to the DMRS have been under study. However, a method of multiplexing a plurality of DMRS ports has not been fully studied yet. For example, performance of the TD OCC deteriorates in a case with a high Doppler shift. Unless the method of multiplexing a plurality of DMRS ports is fully studied, reduction in communication throughput or the like may be caused.

[0138] In view of this, the inventors of the present invention came up with the idea of a method of multiplexing a plurality of DMRS ports.

[0139] Embodiments according to the present disclosure will be described in detail with reference to the drawings as follows. Note that the embodiments (for example, the cases) described below may each be used individually, or at least two of them may be applied in combination.

[0140] In the present disclosure, “A/B” and “at least one of A and B” may be interchangeably interpreted. In the present disclosure, “A/B/C” may mean “at least one of A, B, and C”.

[0141] In the present disclosure, activate, deactivate, indicate, select, configure, update, determine, and the like may be interchangeably interpreted. In the present disclosure, “support,” “control,” “controllable,” “operate,” “operable,” and the like may be interchangeably interpreted.

[0142] In the present disclosure, radio resource control (RRC), an RRC parameter, an RRC message, a higher layer parameter, an information element (IE), a configuration, and the like may be interchangeably interpreted. In the present disclosure, a Medium Access Control control element (MAC Control Element (CE)), an update command, an activation/deactivation command, and the like may be interchangeably interpreted.

[0143] In the present disclosure, the higher layer signaling may be, for example, any one or combinations of Radio Resource Control (RRC) signaling, Medium Access Control (MAC) signaling, broadcast information, and the like.

[0144] In the present disclosure, the MAC signaling may use, for example, a MAC control element (MAC CE), a MAC Protocol Data Unit (PDU), or the like. The broadcast information may be, for example, a master information block (MIB), a system information block (SIB), minimum system information (Remaining Minimum System Information (RMSI)), other system information (OSI), or the like.

[0145] In the present disclosure, the physical layer signaling may be, for example, downlink control information (DCI), uplink control information (UCI), or the like.

[0146] In the present disclosure, an index, an identifier (ID), an indicator, a resource ID, and the like may be interchangeably interpreted. In the present disclosure, a sequence, a list, a set, a group, a cluster, a subset, and the like may be interchangeably interpreted.

[0147] In the present disclosure, a panel, a panel group, a beam, a beam group, a precoder, an Uplink (UL) transmission entity, a transmission/reception point (TRP), a base station, spatial relation information (SRI), a spatial relation, an SRS resource indicator (SRI), a control resource set (CORESET), a Physical Downlink Shared Channel (PDSCH), a codeword (CW), a transport block (TB), a

reference signal (RS), an antenna port (for example, a demodulation reference signal (DMRS) port), an antenna port group (for example, a DMRS port group), a group (for example, a spatial relation group, a code division multiplexing (CDM) group, a reference signal group, a CORESET group, a Physical Uplink Control Channel (PUCCH) group, a PUCCH resource group), a resource (for example, a reference signal resource, an SRS resource), a resource set (for example, a reference signal resource set), a CORESET pool, a downlink Transmission Configuration Indication state (TCI state) (DL TCI state), an uplink TCI state (UL TCI state), a unified TCI state, a common TCI state, quasi-collocation (QCL), QCL assumption, and the like may be interchangeably interpreted.

[0148] In the present disclosure, a frequency domain index, a subcarrier index, an RE index, a PRB index, and an RB index may be interchangeably interpreted.

[0149] In the present disclosure, applying the OCC to a signal/resources and multiplying an element of each RE of a signal/resources by a corresponding element of the OCC may be interchangeably interpreted.

(Radio Communication Method)

[0150] In each embodiment, a DMRS, a DL DMRS, a UL DMRS, a PDSCH DMRS, and a PUSCH DMRS may be interchangeably interpreted.

[0151] In each embodiment, an orthogonal sequence, an OCC, an FD OCC, and a TD OCC may be interchangeably interpreted.

[0152] In each embodiment, a DMRS port, an antenna port, and a port may be interchangeably interpreted. In each embodiment, a port index and a port number may be interchangeably interpreted. In each embodiment, a DMRS CDM group and a CDM group may be interchangeably interpreted. In each embodiment, an antenna port indication and an antenna port field may be interchangeably interpreted.

[0153] In each embodiment, joint channel estimation and DMRS bundling may be interchangeably interpreted.

[0154] In each embodiment, a DMRS for the PDSCH (DMRS port number $p=1000, 1001, \dots, 10xx$) and a DMRS for the PUSCH (DMRS port number $p=0, 1, \dots, xx$) may be interchangeably interpreted.

[0155] In each embodiment, OCC index $m=\{0, 1, 2, \dots\}$, DMRS port number $p=(1000, 1001, \dots)$, and DMRS port number $p=\{0, 1, \dots\}$ may be interchangeably interpreted.

[0156] Although in each embodiment, an example in which the FD OCC is applied to the single symbol DMRS is used, in each embodiment, the FD OCC may be similarly applied to the double symbol DMRS.

[0157] The UE may receive a configuration (for example, an RRC IE) of a PDSCH/PUSCH DMRS. The UE may apply the FD-OCC having a length of $6/4/3/2$ to the PDSCH/PUSCH DMRS, based on the configuration.

Embodiment #1

[0158] The present embodiment relates to the FD OCC for DMRS configuration type 1.

[0159] For DMRS configuration type 1, the FD OCC having a length of 6 may be used.

[0160] FIG. 15 shows an example in which one FD OCC is applied in one PRB. CDM group #0 corresponds to DMRS port #0/#1/#2/#3, and is mapped to the same REs

having even-numbered frequency domain indices (0, 2, . . . , offset $\Delta=0$). CDM group #1 corresponds to DMRS port #4/#5/#6/#7, and is mapped to the same REs (REs different from the REs of CDM group #1) having odd-numbered frequency domain indices (1, 3, . . . , offset $\Delta=1$).

[0161] The FD OCC having a length of 6 may be one of some of the following codes, or may be a sequence different from those.

[Code 1-1]

[0162] The FD OCC may be six complex sequences as in the example of FIG. 16A. As in the example of FIG. 16B, the FD OCC $w_f(k')$ corresponding to $k'=\{0, 1, 2, 3, 4, 5\}$ may be generated using cyclic shifts $\alpha=\{0, \pi/2, \pi, 3\pi/2, 2\pi, 5\pi/2\}$ for $w_f(0)$, respectively. $k'=\{0, 1, 2, 3\}$ may correspond to cyclic shifts at regular intervals (for example, $\alpha=\{0, -\pi/2, -\pi, -3\pi/2, -2\pi, -5\pi/2\}$, $\alpha=\{0, \pi, \pi/2, 3\pi/2, 0, \pi\}$), or may correspond to cyclic shifts at irregular intervals (for example, $\alpha=\{0, 2\pi/6, 2*2\pi/6, 3*2\pi/6, 4*2\pi/6, 5*2\pi/6\}$).

[Code 1-2]

[0163] As in the example of FIG. 17, the FD OCC may be generated based on a Walsh sequence having a length of 4. $w_f(k')$ corresponding to $k'=\{4, 5\}$ may be $w_f(k')$ corresponding to $k'=\{0, 11\}$.

[Code 1-3]

[0164] The FD OCC may be generated using multiplication of two OCCs. For example, the FD OCC may be generated using multiplication (FIG. 18B) of OCC 1 and OCC 2 (FIG. 18A). As in the example of FIG. 19, two subcarriers may be regarded as one subcarrier group, six subcarriers may be divided into three subcarrier groups, OCC 1 may be applied to two subcarriers in each subcarrier group, and OCC 2 may be applied to three subcarrier groups. At least one of OCC 1 and OCC 2 may be an orthogonal sequence. OCC 2 may be generated using a cyclic shift for a given sequence. The cyclic shift may be $\alpha=\{0, \pi, 2\pi\}$, $\alpha=\{0, 2\pi/3, 4\pi/3\}$, or the like.

[0165] A longer FD OCC length (a wider bandwidth, for example, an FD OCC having a length of 6 or more) leads to deterioration of robustness against a larger delay spread (frequency domain channel selectivity). In view of this, the OCC having higher robustness against a larger delay spread is used.

Embodiment #1-1

[0166] For DMRS configuration type 1, the FD OCC having a length of 4 may be applied across one or more PRBs. In this case, it may be defined that the UE does not assume to be scheduled with the PDSCH/PUSCH with an odd number of PRBs. It is preferable that the same precoder be applied to the FD OCC. One FD OCC may be applied to one precoding resource block group (PRG). According to such an FD OCC, in comparison to the FD OCC having a length of 6, higher performance can be achieved in a large delay spread.

[0167] FIG. 20 shows an example in which three FD OCCs are applied across two PRBs. CDM group #0 corresponds to DMRS port #0/#1/#2/#3, and is mapped to the same REs having even-numbered frequency domain indices (0, 2, . . . , offset $\Delta=0$). CDM group #1 corresponds to DMRS

port #4/#5/#6/#7, and is mapped to the same REs (REs different from the REs of CDM group #1) having odd-numbered frequency domain indices (1, 3, . . . , offset $\Delta=1$).

[0168] The FD OCC may be applied from the lowest RE index (lowest subcarrier index) in a set of PRBs. A plurality of PRBs in the set may be ordered (indexed) in accordance with at least one order out of some of the following orders.

[0169] [[Order 1]] To be ordered from the lowest PRB index in the PRBs scheduled for the PDSCH/PUSCH.

[0170] [[Order 2]] To be ordered from the lowest PRB index in the PRG.

[0171] [[Order 3]] To be ordered from the lowest PRB index in the BWP.

[0172] Order 2/3 is common to a plurality of different UEs and is thus preferable.

[0173] The FD OCC having a length of 4 may be one of some of the following codes, or may be a sequence different from those.

[Code 2-1]

[0174] As in the example of FIG. 21A, the FD OCC may be four complex sequences. As in the example of FIG. 21B, the FD OCC $w_f(k')$ corresponding to $k'=\{0, 1, 2, 3\}$ may be generated using cyclic shifts $\alpha=\{0, \pi/2, \pi, 3\pi/2\}$ (FIG. 22) for $w_f(0)$, respectively. $k'=\{0, 1, 2, 3\}$ may correspond to cyclic shifts at regular intervals (for example, $\alpha=\{0, -\pi/2, -\pi, -3\pi/2\}$, $\alpha=\{0, \pi, \pi/2, 3\pi/2\}$), or may correspond to cyclic shifts at irregular intervals (for example, $\alpha=\{0, 2\pi/6, 2*2\pi/6, 3*2\pi/6\}$).

[Code 2-2]

[0175] As in the example of FIG. 23, the FD OCC may be four Walsh sequences. The FD OCC may be the same as the TD OCC for the CSI-RS.

Embodiment #1-2

[0176] Other FD-OCC #2 may be applied to a plurality of existing FD OCCs #1 each having a length of 2 (the plurality of existing FD OCCs #1 may be multiplied by other FD-OCC #2).

[Length of 2]

[0177] FD OCC #2 may be an OCC having a length of 2. As in the example of FIG. 24, two elements of one FD OCC #2 may be respectively applied to two FD OCCs #1 subjected to FDM (two FD OCCs #1 may be respectively multiplied by the two elements). FD OCC #1/#2 having a length of 2 may be defined similarly to existing specifications.

[Length of 3]

[0178] FD OCC #2 may be an FD OCC having a length of 3. As in the example of FIG. 25, three elements of one FD OCC #2 may be respectively applied to three FD OCCs #1 subjected to FDM (three FD OCCs #1 may be respectively multiplied by the three elements). FD OCC #2 having a length of 3 may be an OCC having a length of 3 in embodiment #1-3 to be described later.

[0179] The length of FD OCC #1 may be a number (for example, 3/4/6/8) different from these examples. The length of FD OCC #2 may be a number (for example, 4/6/8) different from these examples. A combination of the length

of FD OCC #1 and the length of FD OCC #2 may be a number (for example, 2 and 4, 3 and 2, 4 and 2, 3 and 3, or the like) different from these examples.

Embodiment #1-3

[0180] For DMRS configuration type 1, the FD OCC having a length of 3 may be applied across one or more PRBs. The OCC having a length of 3 may be one of some of the following codes, or may be a sequence different from those.

[Code 3-1]

[0181] As in the examples of FIG. 26A and FIG. 26B, regarding the OCC having a length of 3, a plurality of OCCs may be generated by applying a plurality of different cyclic shifts to a base sequence.

[Code 3-2]

[0182] As in the example of FIG. 27, regarding the OCC having a length of 3, three sequences may be defined in a specification.

Embodiment #1-4

[0183] As described above, when the FD-OCC having a sequence length of $M (>2)$ is applied, there may be an excess of REs in a physical resource block (PRB) (that is, there may be REs to which the whole FD-OCC is not applied in one PRB). Such a case in which there is an excess of REs in a specific bandwidth (for example, a PRB) may be addressed using the methods shown in FIGS. 28A and 28B.

[0184] Note that, in other words, the case corresponds to a case in which the number of REs (subcarriers) of DMRS resources in a specific bandwidth cannot be divided by M without a remainder.

[0185] An RE group (RE set) includes M non-consecutive REs, and for each RE group, M elements of the FD-OCC are respectively applied to M REs in the RE group. FIGS. 28A and 28B are diagrams to show examples of mapping of the FD-OCC in embodiment #1-4. In the present example, four elements of the FD-OCC are respectively applied to four REs in the RE group.

[0186] When the FD-OCC cannot be applied with the REs in only one PRB, as shown in FIG. 28A, the UE may assume that a plurality of RE groups to which the FD-OCC is applied are present across a plurality of PRBs. In FIG. 28A, the FD-OCC may be respectively applied to a total of four REs, i.e., REs with $k=8$ and 10 of PRB 0 and REs with $k=0$ and 2 of PRB 1, using $w_f(0)$, $w_f(1)$, $w_f(2)$, and $w_f(3)$. In FIG. 28A, three RE groups (groups 1 to 3) may be used in two PRBs.

[0187] The PRBs to which the RE group of the FD-OCC belongs may be consecutive PRBs, or may be non-consecutive PRBs. As shown in FIG. 28A, the PRBs to which the RE group of the FD-OCC belongs may be included in the same precoding resource block group (PRG).

[0188] One PRG may include one or more PRBs to which the same precoding is applied. This is because orthogonalization can be appropriately performed even when the FD-OCC is applied across a plurality of PRBs in the PRG, as it is assumed that precoding is applied for each PRG. One PRB may be interpreted as a plurality of PRBs.

[0189] The RE group to which the FD-OCC is applied may be configured by combining the REs in the PRG in

ascending order or descending order of frequency (in FIG. 28A, the REs are combined in ascending order).

[0190] Note that the same FD-OCC may be applied to different RE groups, or different FD-OCCs may be applied to different RE groups (for example, the index of the FD-OCC may be incremented to be applied for each group).

[0191] For example, the FD-OCC of one OCC index (a set of $w_f(k')$) may be applied to all of groups 1, 2, and 3 of FIG. 28A. The FD-OCC of the OCC index $\{0, 1, 2\}$ having a length of 4 described above may each be applied to groups 1, 2, and 3 of FIG. 28A. Note that the OCC index p may be interpreted as $\text{mod}(p, M)$ or the like. In other words, the OCC index may be incremented among the RE groups to be repeatedly used.

[0192] Note that, for example, when one PRG=odd-numbered PRBs, it may be assumed that there is an excess of REs in one PRB even when the FD-OCC is applied to the RE group across a plurality of PRBs as shown in FIG. 28A.

[0193] When the FD-OCC cannot be applied to all of the REs in one PRB or one PRG (there is an excess of REs of the DMRS), as shown in FIG. 28B, the UE may assume that the DMRS is not transmitted (which may be referred to as being dropped, muted, or the like), or that the DMRS is transmitted without application of the FD-OCC in the excess of REs in the one PRB or in the one PRG. According to such a configuration, compatibility with a Rel-15 DMRS can be suitably maintained.

[0194] In the case of FIG. 28B, the FD-OCC may be applied to a total of four REs with $k=0, 2, 4$, and 6 of the PRB/PRG shown in the figure, using $w_f(0)$, $w_f(1)$, $w_f(2)$, and $w_f(3)$, respectively. In contrast, in the REs with $k=8$ and 10 of the PRB/PRG, the DMRS need not be transmitted, or the DMRS to which the FD-OCC is not applied may be transmitted.

[0195] Note that it may be assumed that the TD-OCC to be described later is applied to the excess of REs in the PRB/PRG (or the REs to which the FD-OCC is not applied).

[0196] The UE may assume that a specific FD-OCC is applied to the excess of REs in the one PRB or in the one PRG. The specific FD-OCC may be the FD-OCC of Rel-15 NR (the OCC having a sequence length of 2), or may correspond to multiplication of a part of $w_f(k')$ included in the set of $w_f(k')$ corresponding to the index of the OCC having a sequence length of M described above. For example, in the case of FIG. 28B, the OCC indices $w_f(0)$ and $w_f(1)$ may be respectively applied to the REs with $k=8$ and 10 of the PRB/PRG.

[0197] Information as to whether or not the DMRS is transmitted using the excess of REs in the PRB/PRG, whether or not the OCC (FD-OCC/TD-OCC) is applied to the excess of REs in the PRB/PRG, or the like may be configured (indicated) to the UE, using higher layer signaling, physical layer signaling, or a combination of these. Based on the information, the UE may determine whether given REs are to be used for transmission and reception of the DMRS, or may determine transmission and reception processing of the REs of the DMRS.

[0198] Note that the PRG of the present disclosure may be interpreted as a width of PRB bundling (frequency resources to which PRB bundling is applied), or may be interpreted as a width of the PDSCH (a transmission bandwidth of the PDSCH).

Embodiment #1-5

[0199] For DMRS configuration type 1, the FD OCC having a length of 6 and the FD OCC having a length of 4 are compared. Using the FD OCC having a length of 6, the PDSCH/PUSCH with any number of PRBs can be scheduled. Using the FD OCC having a length of 4, the PDSCH/PUSCH with only even-numbered PRBs can be scheduled. Using the FD OCC having a length of 4, higher performance can be achieved in a large delay spread.

[0200] Two alternatives of the FD OCC having a length of 6 and the FD OCC having a length of 4 may be defined in a specification. The UE may report support of one of those alternatives. The base station may configure one of those alternatives, based on the report.

[0201] A plurality of alternatives of the TD OCC, the FD OCC, and FDM may be defined in a specification. FDM may refer to mapping of a plurality of/different DMRSs (a plurality of/different DMRS ports) to different frequency domain resources (subcarriers). The UE may report support of one of those alternatives. The base station may configure one of those alternatives for the PDSCH/PUSCH DMRS, based on the report. With the TD OCC, deterioration of performance due to a high delay spread can be reduced. With the FD OCC, deterioration of performance due to a high Doppler spread can be reduced. FDM is preferable for multiplexing with an existing (Rel-15/16) UE.

[0202] Switching of DMRS multiplexing schemes may be configured/indicated by an RRC IE/MAC CE, or may be indicated by DCI. The indication by DCI may be a new DCI field, or may be an existing DCI field. Switching of the DMRS multiplexing schemes may include an indication of one of the TD OCC, the FD OCC, and FDM, or may include an indication of the length of the OCC.

[0203] For example, when the number of PRBs scheduled for the PDSCH/PUSCH is an even number, the FD OCC having a length (for example, $4/3/2$) equal to or less than L (for example, $L=4$) may be applied, otherwise the FD OCC having a length (for example, 6) greater than L may be applied.

[0204] For example, when the number of PRBs scheduled for the PDSCH/PUSCH is greater than X , the FD OCC having a length (for example, 6) greater than L (for example, $L=4$) may be applied, otherwise the FD OCC having a length (for example, $4/3/2$) equal to or less than L may be applied or the FD OCC of Rel. 15 (that is, the FD OCC having a length of 2) may be applied.

[0205] For example, when the number of layers scheduled for the PDSCH/PUSCH is greater than Y , the FD OCC having a length (for example, 6) greater than L (for example, $L=4$) may be applied, otherwise the FD OCC having a length (for example, $4/3/2$) equal to or less than L may be applied or the FD OCC of Rel. 15 (that is, the FD OCC having a length of 2) may be applied.

[0206] At least one of X , Y , and L may be defined in a specification, or may be configured by a higher layer.

[0207] According to the present embodiment, the number of DMRS ports can be increased, and the DMRS can be appropriately multiplexed.

Embodiment #2

[0208] The present embodiment relates to a DMRS mapping method.

[0209] A new DMRS mapping method in the frequency domain may be introduced, or an existing DMRS mapping method may be updated.

[0210] The FD OCC having a length of M may be applied to M consecutive REs. For example, M may be 4, may be 3, or may be another number. The DMRS mapping method may be referred to as DMRS configuration type 3.

[0211] In a case of application to M consecutive REs, in comparison to a case of application to M non-consecutive REs as in embodiment #1, higher performance can be achieved in a large delay spread. Because stay of channels on eight consecutive REs is difficult, multiplexing with the DMRS ports of the existing (Rel-15/16) UE is difficult.

[0212] In the example of FIG. 29, DMRS port #0/#1/#2/#3 is associated with CDM group #0, and $W_{\lambda}(0)$ to $W_{\lambda}(3)$ of the FD OCC are respectively applied to subcarrier indices #0 to #3 of the DMRS. DMRS port #4/#5/#6/#7 is associated with CDM group #1, and $W_{\lambda}(0)$ to $W_{\lambda}(3)$ of the FD OCC are respectively applied to subcarrier indices #4 to #7 of the DMRS. DMRS port #8/#9/#10/#11 is associated with CDM group #0, and $W_{\lambda}(0)$ to $W_{\lambda}(3)$ of the FD OCC are respectively applied to subcarrier indices #8 to #11 of the DMRS. The DMRS of CDM group #0/#1/#2 may be subjected to FDM.

[0213] According to the present embodiment, the DMRS and the FD OCC can be appropriately mapped to frequency domain resources.

Embodiment #3

[0214] The present embodiment relates to FDM.

[0215] In FDM, the FD OCC having a length of 2 may be applied. The FD OCC may be the FD OCC of Rel. 15. Two adjacent or consecutive REs may be grouped to form an RE group.

[0216] For DMRS configuration type 1, mapping from the DMRS ports to the REs may be different depending upon whether the PRB index is an even number or an odd number. For example, for an odd number of PRBs, the UE may conform to one of some of the following schedulings.

[Scheduling 1]

[0217] It may be defined that the UE does not assume to be scheduled with the PDSCH/PUSCH with an odd number of PRBs.

[Scheduling 2]

[0218] The UE can be scheduled with the PDSCH/PUSCH with an odd number of PRBs, and in this case, the number of REs for smaller port numbers and the number of REs for larger port numbers may be different.

[0219] Mapping from the DMRS ports to the REs may be determined depending upon whether the RE (subcarrier) having the lowest index to which a given DMRS port is mapped is present in the PRB having an even-numbered PRB index or is present in the PRB having an odd-numbered PRB index. The PRB index may be one of some of the following indices.

[0220] [[Index 1]] A PRB index in the scheduled PDSCH/PUSCH (a local PRB index in one or more PRBs allocated to the scheduled PDSCH/PUSCH).

[0221] [[Index 2]] A PRB index in one PRG.

[0222] [[Index 3]] An actual PRB index (a PRB index in the BWP).

[0223] Index 2/3 is common to a plurality of different UEs and is thus preferable.

[0224] In the example of FIG. 30A, two REs having even-numbered adjacent RE indices from RE index 0 of PRB index 0 may form an RE group, and the FD OCC having a length of 2 may be applied. Two REs having odd-numbered adjacent RE indices from RE index 1 of PRB index 0 may form an RE group, and the FD OCC having a length of 2 may be applied.

[0225] Furthermore, RE group-level comb2 (FDM) may be applied. To the RE group of RE indices 0 and 2 of PRB index 0, the RE group of RE indices 8 and 10, and the RE group of RE indices 4 and 6 of PRB index 1 and the RE group of RE indices 1 and 3 of PRB index 0, RE indices 9 and 11, and the RE group of RE indices 5 and 7 of PRB index 1, RE-level comb2 may be applied, and DMRS ports 1000 to 1007 may be mapped. To the RE group of RE indices 4 and 6 of PRB index 0, the RE group of RE indices 0 and 2 of PRB index 1, and the RE group of RE indices 8 and 10 and the RE group of RE indices 5 and 7 of PRB index 0, the RE group of RE indices 1 and 3 of PRB index 1, and the RE group of RE indices 9 and 11, RE-level comb2 may be applied, and DMRS ports 1008 to 1015 may be mapped. To DMRS ports 1000 to 1007 and DMRS ports 1008 to 1015, RE group-level comb2 (FDM) may be applied.

[0226] For DMRS configuration type 2, mapping from the DMRS ports to the REs may be common, regardless of the PRB index.

[0227] In the example of FIG. 30B, two REs having even-numbered adjacent RE indices may form an RE group, and the FD OCC having a length of 2 may be applied. The RE group of RE indices 0 and 1, the RE group of RE indices 2 and 3, and the RE group of RE indices 4 and 5 may be subjected to FDM, and DMRS ports 1000 to 1011 may be mapped to those RE groups. The RE group of RE indices 6 and 7, the RE group of RE indices 8 and 9, and the RE group of RE indices 10 and 11 may be subjected to FDM, and DMRS ports 1012 to 1023 may be mapped to those RE groups. DMRS ports 1000 to 1011 and DMRS ports 1012 to 1024 may be subjected to FDM.

[0228] For DMRS configuration type 1, mapping from the DMRS ports to the REs may be common, regardless of the PRB index. Channel estimation accuracy of two DMRS ports subjected to FDM may be different.

[0229] In the example of FIG. 31, two REs having even-numbered adjacent RE indices from RE index 0 may form an RE group, and the FD OCC having a length of 2 may be applied. Two REs having odd-numbered adjacent RE indices from RE index 1 may form an RE group, and the FD OCC having a length of 2 may be applied.

[0230] To the RE group of RE indices 0 and 2, the RE group of RE indices 4 and 6, the RE group of RE indices 1 and 3, and the RE group of RE indices 5 and 7, DMRS ports 1000 to 1007 may be mapped. To the RE group of RE indices 8 and 10 and the RE group of RE indices 9 and 11, DMRS ports 1008 to 1015 may be mapped. In the present example, although more REs are mapped to the DMRS ports corresponding to lower RE indices, more REs may be mapped to the DMRS ports corresponding to higher RE indices.

[0231] According to the present embodiment, a plurality of DMRSs can be appropriately subjected to FDM.

Embodiment #4

[0232] The present embodiment relates to operation across a plurality of PRBs.

[0233] The FD OCC across a plurality of PRBs or joint channel estimation across a plurality of PRBs requires additional UE complexity/functions. The UE may report support of the FD OCC across a plurality of PRBs, or may report support of joint channel estimation across a plurality of PRBs. The UE/base station may report/configure whether or not phases are consistent (continuous) across a plurality of PRBs, may report/configure whether or not the FD OCC across a plurality of PRBs operates, or may report/configure that joint channel estimation across a plurality of PRBs operates.

[0234] In order to enable the FD OCC across a plurality of PRBs or joint channel estimation across a plurality of PRBs, both of the transmitter and the receiver may recognize consistency (continuity) of phases between a plurality of consecutive PRBs (for example, two PRBs). Phase consistency may be maintained across a plurality of PRBs in one PRG in the DL. Phase consistency may be maintained across all of the PRBs in the UL.

[0235] FIG. 32 shows an example of DMRS configuration type 1. In the present example, mapping of the REs of the DMRS and the REs of the DMRS to which the FD OCC having a length of 4 is applied is similar to that of FIG. 20 described above. In the present example, the DMRS of PRB index 0 and the DMRS of PRB index 1 have phase consistency. The receiver can perform joint channel estimation, using the DMRS of PRB index 0 and the DMRS of PRB index 1.

[0236] FIG. 33 shows an example of DMRS configuration type 2. In the present example, the FD OCC having a length of 2 is applied to the DMRS mapped to RE indices 0 and 1 in PRB index 0. The FD OCC having a length of 2 is applied to the DMRS mapped to RE indices 0 and 1 in PRB index 1. The DMRS of PRB index 0 and the DMRS of PRB index 1 are subjected to FDM. In the present example, the DMRS of PRB index 0 and the DMRS of PRB index 1 have phase consistency. The receiver can perform joint channel estimation, using the DMRS of PRB index 0 and the DMRS of PRB index 1.

[0237] According to the present embodiment, channel estimation accuracy can be enhanced using the DMRS across a plurality of PRBs.

<Supplements>

[0238] At least one operation of the embodiments described above may be applied only to the UE that has reported a specific UE capability or that supports the specific UE capability.

[0239] The specific UE capability may indicate at least one of the following:

[0240] Support specific processing/operation/control/information regarding at least one of the embodiments described above.

[0241] Support a larger number of DMRS ports than that in existing specifications for the PDSCH/PUSCH.

[0242] Support a larger number of DMRS ports than that in existing specifications for the PDSCH/PUSCH DMRS, using the TD-OCC/FD-OCC/FDM.

[0243] A maximum number of DMRS ports for the PDSCH/PUSCH.

[0244] Support the FD OCC having a length of 6/4/3.

[0245] Support the FD OCC across a plurality of PRBs.

[0246] Support joint channel estimation across a plurality of PRBs.

[0247] A joint UE capability may be reported for a plurality of DMRS configuration types. Separate UE capabilities may be reported for a plurality of DMRS configuration types.

[0248] A joint UE capability may be reported for the single symbol DMRS and the double symbol DMRS. Separate UE capabilities may be reported for the single symbol DMRS and the double symbol DMRS.

[0249] The specific UE capability may be a capability applied to all of the frequencies (in common regardless of a frequency), may be a capability for each frequency (for example, a cell, a band, a BWP), may be a capability for each frequency range (for example, Frequency Range 1 (FR1), FR2, FR3, FR4, FR5, FR2-1, FR2-2), or may be a capability for each subcarrier spacing (SCS).

[0250] The specific UE capability may be a capability applied to all of the duplex schemes (commonly irrespective of the duplex scheme), or may be a capability for each duplex scheme (for example, time division duplex (TDD), frequency division duplex (FDD)).

[0251] At least one operation of the embodiments described above may be applied when the UE is configured with specific information related to the embodiments described above using higher layer signaling. For example, the specific information may be information indicating enabling of at least one operation of the embodiments described above, any RRC parameter for a specific release (for example, Rel. 18), or the like.

[0252] When the UE does not support at least one operation of the specific UE capabilities or is not configured with the specific information, the UE may apply Rel-15/16/17 operation, for example.

(Supplementary Note)

[0253] Regarding one embodiment of the present disclosure, the following supplementary notes of the invention will be given.

{Supplementary Note 1}

[0254] A terminal including:

[0255] a receiving section that receives a configuration of a demodulation reference signal (DMRS) of a shared channel; and

[0256] a control section that applies a frequency domain (FD)-orthogonal cover code (OCC) having a length of 4 or 3 to the DMRS, based on the configuration.

{Supplementary Note 2}

[0257] The terminal according to supplementary note 1, wherein

[0258] the FD-OCC is applied across a plurality of physical resource blocks.

{Supplementary Note 3}

[0259] The terminal according to supplementary note 1 or 2, wherein

[0260] the FD-OCC is applied to a plurality of consecutive resource elements.

{Supplementary Note 4}

[0261] The terminal according to any one of supplementary notes 1 to 3, wherein

[0262] the control section estimates a channel using the DMRS in a plurality of resource elements across a plurality of physical resource blocks.

(Radio Communication System)

[0263] Hereinafter, a structure of a radio communication system according to one embodiment of the present disclosure will be described. In this radio communication system, the radio communication method according to each embodiment of the present disclosure described above may be used alone or may be used in combination for communication.

[0264] FIG. 34 is a diagram to show an example of a schematic structure of the radio communication system according to one embodiment. The radio communication system 1 may be a system implementing a communication using Long Term Evolution (LTE), 5th generation mobile communication system New Radio (5G NR) and so on the specifications of which have been drafted by Third Generation Partnership Project (3GPP).

[0265] The radio communication system 1 may support dual connectivity (multi-RAT dual connectivity (MR-DC)) between a plurality of Radio Access Technologies (RATs). The MR-DC may include dual connectivity (E-UTRA-NR Dual Connectivity (EN-DC)) between LTE (Evolved Universal Terrestrial Radio Access (E-UTRA)) and NR, dual connectivity (NR-E-UTRA Dual Connectivity (NE-DC)) between NR and LTE, and so on.

[0266] In EN-DC, a base station (eNB) of LTE (E-UTRA) is a master node (MN), and a base station (gNB) of NR is a secondary node (SN). In NE-DC, a base station (gNB) of NR is an MN, and a base station (eNB) of LTE (E-UTRA) is an SN.

[0267] The radio communication system 1 may support dual connectivity between a plurality of base stations in the same RAT (for example, dual connectivity (NR-NR Dual Connectivity (NN-DC)) where both of an MN and an SN are base stations (gNB) of NR).

[0268] The radio communication system 1 may include a base station 11 that forms a macro cell C1 of a relatively wide coverage, and base stations 12 (12a to 12c) that form small cells C2, which are placed within the macro cell C1 and which are narrower than the macro cell C1. The user terminal 20 may be located in at least one cell. The arrangement, the number, and the like of each cell and user terminal 20 are by no means limited to the aspect shown in the diagram. Hereinafter, the base stations 11 and 12 will be collectively referred to as "base stations 10," unless specified otherwise.

[0269] The user terminal 20 may be connected to at least one of the plurality of base stations 10. The user terminal 20 may use at least one of carrier aggregation (CA) and dual connectivity (DC) using a plurality of component carriers (CCs).

[0270] Each CC may be included in at least one of a first frequency band (Frequency Range 1 (FR1)) and a second frequency band (Frequency Range 2 (FR2)). The macro cell C1 may be included in FR1, and the small cells C2 may be included in FR2. For example, FR1 may be a frequency band of 6 GHz or less (sub-6 GHz), and FR2 may be a frequency band which is higher than 24 GHz (above-24 GHz). Note

that frequency bands, definitions and so on of FR1 and FR2 are by no means limited to these, and for example, FR1 may correspond to a frequency band which is higher than FR2.

[0271] The user terminal **20** may communicate using at least one of time division duplex (TDD) and frequency division duplex (FDD) in each CC.

[0272] The plurality of base stations **10** may be connected by a wired connection (for example, optical fiber in compliance with the Common Public Radio Interface (CPRI), the X2 interface and so on) or a wireless connection (for example, an NR communication). For example, if an NR communication is used as a backhaul between the base stations **11** and **12**, the base station **11** corresponding to a higher station may be referred to as an “Integrated Access Backhaul (IAB) donor,” and the base station **12** corresponding to a relay station (relay) may be referred to as an “IAB node.”

[0273] The base station **10** may be connected to a core network **30** through another base station **10** or directly. For example, the core network **30** may include at least one of Evolved Packet Core (EPC), 5G Core Network (5GCN), Next Generation Core (NGC), and so on.

[0274] The user terminal **20** may be a terminal supporting at least one of communication schemes such as LTE, LTE-A, and 5G.

[0275] In the radio communication system **1**, an orthogonal frequency division multiplexing (OFDM)-based wireless access scheme may be used. For example, in at least one of the downlink (DL) and the uplink (UL), Cyclic Prefix OFDM (CP-OFDM), Discrete Fourier Transform Spread OFDM (DFT-s-OFDM), Orthogonal Frequency Division Multiple Access (OFDMA), Single Carrier Frequency Division Multiple Access (SC-FDMA), and so on may be used.

[0276] The wireless access scheme may be referred to as a “waveform.” Note that, in the radio communication system **1**, another wireless access scheme (for example, another single carrier transmission scheme, another multi-carrier transmission scheme) may be used for a wireless access scheme in the UL and the DL.

[0277] In the radio communication system **1**, a downlink shared channel (Physical Downlink Shared Channel (PDSCH)), which is used by each user terminal **20** on a shared basis, a broadcast channel (Physical Broadcast Channel (PBCH)), a downlink control channel (Physical Downlink Control Channel (PDCCH)) and so on, may be used as downlink channels.

[0278] In the radio communication system **1**, an uplink shared channel (Physical Uplink Shared Channel (PUSCH)), which is used by each user terminal **20** on a shared basis, an uplink control channel (Physical Uplink Control Channel (PUCCH)), a random access channel (Physical Random Access Channel (PRACH)) and so on may be used as uplink channels.

[0279] User data, higher layer control information, System Information Blocks (SIBs) and so on are communicated on the PDSCH. User data, higher layer control information and so on may be communicated on the PUSCH. The Master Information Blocks (MIBs) may be communicated on the PBCH.

[0280] Lower layer control information may be communicated on the PDCCH. For example, the lower layer control information may include downlink control information (DCI) including scheduling information of at least one of the PDSCH and the PUSCH.

[0281] Note that DCI for scheduling the PDSCH may be referred to as “DL assignment,” “DL DCI,” and so on, and DCI for scheduling the PUSCH may be referred to as “UL grant,” “UL DCI,” and so on. Note that the PDSCH may be interpreted as “DL data,” and the PUSCH may be interpreted as “UL data.”

[0282] For detection of the PDCCH, a control resource set (CORESET) and a search space may be used. The CORESET corresponds to a resource to search DCI. The search space corresponds to a search area and a search method of PDCCH candidates. One CORESET may be associated with one or more search spaces. The UE may monitor a CORESET associated with a given search space, based on search space configuration.

[0283] One search space may correspond to a PDCCH candidate corresponding to one or more aggregation levels. One or more search spaces may be referred to as a “search space set.” Note that a “search space,” a “search space set,” a “search space configuration,” a “search space set configuration,” a “CORESET,” a “CORESET configuration” and so on of the present disclosure may be interchangeably interpreted.

[0284] Uplink control information (UCI) including at least one of channel state information (CSI), transmission confirmation information (for example, which may be referred to as Hybrid Automatic Repeat reQuest ACKnowledgement (HARQ-ACK), ACK/NACK, and so on), and scheduling request (SR) may be communicated by means of the PUCCH. By means of the PRACH, random access preambles for establishing connections with cells may be communicated.

[0285] Note that the downlink, the uplink, and so on in the present disclosure may be expressed without a term of “link.” In addition, various channels may be expressed without adding “Physical” to the head.

[0286] In the radio communication system **1**, a synchronization signal (SS), a downlink reference signal (DL-RS), and so on may be communicated. In the radio communication system **1**, a cell-specific reference signal (CRS), a channel state information-reference signal (CSI-RS), a demodulation reference signal (DMRS), a positioning reference signal (PRS), a phase tracking reference signal (PTRS), and so on may be communicated as the DL-RS.

[0287] For example, the synchronization signal may be at least one of a primary synchronization signal (PSS) and a secondary synchronization signal (SSS). A signal block including an SS (PSS, SSS) and a PBCH (and a DMRS for a PBCH) may be referred to as an “SS/PBCH block,” an “SS Block (SSB),” and so on. Note that an SS, an SSB, and so on may also be referred to as a “reference signal.”

[0288] In the radio communication system **1**, a sounding reference signal (SRS), a demodulation reference signal (DMRS), and so on may be communicated as an uplink reference signal (UL-RS). Note that DMRS may be referred to as a “user terminal specific reference signal (UE-specific Reference Signal).”

(Base Station)

[0289] FIG. **35** is a diagram to show an example of a structure of the base station according to one embodiment. The base station **10** includes a control section **110**, a transmitting/receiving section **120**, transmitting/receiving antennas **130** and a communication path interface (transmission line interface) **140**. Note that the base station **10** may include

one or more control sections **110**, one or more transmitting/receiving sections **120**, one or more transmitting/receiving antennas **130**, and one or more communication path interfaces **140**.

[0290] Note that, the present example primarily shows functional blocks that pertain to characteristic parts of the present embodiment, and it is assumed that the base station **10** may include other functional blocks that are necessary for radio communication as well. Part of the processes of each section described below may be omitted.

[0291] The control section **110** controls the whole of the base station **10**. The control section **110** can be constituted with a controller, a control circuit, or the like described based on general understanding of the technical field to which the present disclosure pertains.

[0292] The control section **110** may control generation of signals, scheduling (for example, resource allocation, mapping), and so on. The control section **110** may control transmission and reception, measurement and so on using the transmitting/receiving section **120**, the transmitting/receiving antennas **130**, and the communication path interface **140**. The control section **110** may generate data, control information, a sequence and so on to transmit as a signal, and forward the generated items to the transmitting/receiving section **120**. The control section **110** may perform call processing (setting up, releasing) for communication channels, manage the state of the base station **10**, and manage the radio resources.

[0293] The transmitting/receiving section **120** may include a baseband section **121**, a Radio Frequency (RF) section **122**, and a measurement section **123**. The baseband section **121** may include a transmission processing section **1211** and a reception processing section **1212**. The transmitting/receiving section **120** can be constituted with a transmitter/receiver, an RF circuit, a baseband circuit, a filter, a phase shifter, a measurement circuit, a transmitting/receiving circuit, or the like described based on general understanding of the technical field to which the present disclosure pertains.

[0294] The transmitting/receiving section **120** may be structured as a transmitting/receiving section in one entity, or may be constituted with a transmitting section and a receiving section. The transmitting section may be constituted with the transmission processing section **1211**, and the RF section **122**. The receiving section may be constituted with the reception processing section **1212**, the RF section **122**, and the measurement section **123**.

[0295] The transmitting/receiving antennas **130** can be constituted with antennas, for example, an array antenna, or the like described based on general understanding of the technical field to which the present disclosure pertains.

[0296] The transmitting/receiving section **120** may transmit the above-described downlink channel, synchronization signal, downlink reference signal, and so on. The transmitting/receiving section **120** may receive the above-described uplink channel, uplink reference signal, and so on.

[0297] The transmitting/receiving section **120** may form at least one of a transmit beam and a receive beam by using digital beam forming (for example, precoding), analog beam forming (for example, phase rotation), and so on.

[0298] The transmitting/receiving section **120** (transmission processing section **1211**) may perform the processing of the Packet Data Convergence Protocol (PDCP) layer, the processing of the Radio Link Control (RLC) layer (for

example, RLC retransmission control), the processing of the Medium Access Control (MAC) layer (for example, HARQ retransmission control), and so on, for example, on data and control information and so on acquired from the control section **110**, and may generate bit string to transmit.

[0299] The transmitting/receiving section **120** (transmission processing section **1211**) may perform transmission processing such as channel coding (which may include error correction coding), modulation, mapping, filtering, discrete Fourier transform (DFT) processing (as necessary), inverse fast Fourier transform (IFFT) processing, precoding, digital-to-analog conversion, and so on, on the bit string to transmit, and output a baseband signal.

[0300] The transmitting/receiving section **120** (RF section **122**) may perform modulation to a radio frequency band, filtering, amplification, and so on, on the baseband signal, and transmit the signal of the radio frequency band through the transmitting/receiving antennas **130**.

[0301] On the other hand, the transmitting/receiving section **120** (RF section **122**) may perform amplification, filtering, demodulation to a baseband signal, and so on, on the signal of the radio frequency band received by the transmitting/receiving antennas **130**.

[0302] The transmitting/receiving section **120** (reception processing section **1212**) may apply reception processing such as analog-digital conversion, fast Fourier transform (FFT) processing, inverse discrete Fourier transform (IDFT) processing (as necessary), filtering, de-mapping, demodulation, decoding (which may include error correction decoding), MAC layer processing, the processing of the RLC layer and the processing of the PDCP layer, and so on, on the acquired baseband signal, and acquire user data, and so on.

[0303] The transmitting/receiving section **120** (measurement section **123**) may perform the measurement related to the received signal. For example, the measurement section **123** may perform Radio Resource Management (RRM) measurement, Channel State Information (CSI) measurement, and so on, based on the received signal. The measurement section **123** may measure a received power (for example, Reference Signal Received Power (RSRP)), a received quality (for example, Reference Signal Received Quality (RSRQ)), a Signal to Interference plus Noise Ratio (SINR), a Signal to Noise Ratio (SNR), a signal strength (for example, Received Signal Strength Indicator (RSSI)), channel information (for example, CSI), and so on. The measurement results may be output to the control section **110**.

[0304] The communication path interface **140** may perform transmission/reception (backhaul signaling) of a signal with an apparatus included in the core network **30** or other base stations **10**, and so on, and acquire or transmit user data (user plane data), control plane data, and so on for the user terminal **20**.

[0305] Note that the transmitting section and the receiving section of the base station **10** in the present disclosure may be constituted with at least one of the transmitting/receiving section **120**, the transmitting/receiving antennas **130**, and the communication path interface **140**.

[0306] The transmitting/receiving section **120** may transmit a configuration of a demodulation reference signal (DMRS) of a shared channel. The control section **110** may apply a frequency domain (FD)-orthogonal cover code (OCC) having a length of 4 or 3 to the DMRS, based on the configuration.

(User Terminal)

[0307] FIG. 36 is a diagram to show an example of a structure of the user terminal according to one embodiment. The user terminal 20 includes a control section 210, a transmitting/receiving section 220, and transmitting/receiving antennas 230. Note that the user terminal 20 may include one or more control sections 210, one or more transmitting/receiving sections 220, and one or more transmitting/receiving antennas 230.

[0308] Note that, the present example primarily shows functional blocks that pertain to characteristic parts of the present embodiment, and it is assumed that the user terminal 20 may include other functional blocks that are necessary for radio communication as well. Part of the processes of each section described below may be omitted.

[0309] The control section 210 controls the whole of the user terminal 20. The control section 210 can be constituted with a controller, a control circuit, or the like described based on general understanding of the technical field to which the present disclosure pertains.

[0310] The control section 210 may control generation of signals, mapping, and so on. The control section 210 may control transmission/reception, measurement and so on using the transmitting/receiving section 220, and the transmitting/receiving antennas 230. The control section 210 generates data, control information, a sequence and so on to transmit as a signal, and may forward the generated items to the transmitting/receiving section 220.

[0311] The transmitting/receiving section 220 may include a baseband section 221, an RF section 222, and a measurement section 223. The baseband section 221 may include a transmission processing section 2211 and a reception processing section 2212. The transmitting/receiving section 220 can be constituted with a transmitter/receiver, an RF circuit, a baseband circuit, a filter, a phase shifter, a measurement circuit, a transmitting/receiving circuit, or the like described based on general understanding of the technical field to which the present disclosure pertains.

[0312] The transmitting/receiving section 220 may be structured as a transmitting/receiving section in one entity, or may be constituted with a transmitting section and a receiving section. The transmitting section may be constituted with the transmission processing section 2211, and the RF section 222. The receiving section may be constituted with the reception processing section 2212, the RF section 222, and the measurement section 223.

[0313] The transmitting/receiving antennas 230 can be constituted with antennas, for example, an array antenna, or the like described based on general understanding of the technical field to which the present disclosure pertains.

[0314] The transmitting/receiving section 220 may receive the above-described downlink channel, synchronization signal, downlink reference signal, and so on. The transmitting/receiving section 220 may transmit the above-described uplink channel, uplink reference signal, and so on.

[0315] The transmitting/receiving section 220 may form at least one of a transmit beam and a receive beam by using digital beam forming (for example, precoding), analog beam forming (for example, phase rotation), and so on.

[0316] The transmitting/receiving section 220 (transmission processing section 2211) may perform the processing of the PDCP layer, the processing of the RLC layer (for example, RLC retransmission control), the processing of the MAC layer (for example, HARQ retransmission control),

and so on, for example, on data and control information and so on acquired from the control section 210, and may generate bit string to transmit.

[0317] The transmitting/receiving section 220 (transmission processing section 2211) may perform transmission processing such as channel coding (which may include error correction coding), modulation, mapping, filtering, DFT processing (as necessary), IFFT processing, precoding, digital-to-analog conversion, and so on, on the bit string to transmit, and output a baseband signal.

[0318] Note that, whether to apply DFT processing or not may be based on the configuration of the transform precoding. The transmitting/receiving section 220 (transmission processing section 2211) may perform, for a given channel (for example, PUSCH), the DFT processing as the above-described transmission processing to transmit the channel by using a DFT-s-OFDM waveform if transform precoding is enabled, and otherwise, does not need to perform the DFT processing as the above-described transmission processing.

[0319] The transmitting/receiving section 220 (RF section 222) may perform modulation to a radio frequency band, filtering, amplification, and so on, on the baseband signal, and transmit the signal of the radio frequency band through the transmitting/receiving antennas 230.

[0320] On the other hand, the transmitting/receiving section 220 (RF section 222) may perform amplification, filtering, demodulation to a baseband signal, and so on, on the signal of the radio frequency band received by the transmitting/receiving antennas 230.

[0321] The transmitting/receiving section 220 (reception processing section 2212) may apply reception processing such as analog-digital conversion, FFT processing, IDFT processing (as necessary), filtering, de-mapping, demodulation, decoding (which may include error correction decoding), MAC layer processing, the processing of the RLC layer and the processing of the PDCP layer, and so on, on the acquired baseband signal, and acquire user data, and so on.

[0322] The transmitting/receiving section 220 (measurement section 223) may perform the measurement related to the received signal. For example, the measurement section 223 may perform RRM measurement, CSI measurement, and so on, based on the received signal. The measurement section 223 may measure a received power (for example, RSRP), a received quality (for example, RSRQ, SINR, SNR), a signal strength (for example, RSSI), channel information (for example, CSI), and so on. The measurement results may be output to the control section 210.

[0323] Note that the transmitting section and the receiving section of the user terminal 20 in the present disclosure may be constituted with at least one of the transmitting/receiving section 220 and the transmitting/receiving antennas 230.

[0324] The transmitting/receiving section 220 may receive a configuration of a demodulation reference signal (DMRS) of a shared channel (for example, a PDSCH/PUSCH). The control section 210 may apply a frequency domain (FD)-orthogonal cover code (OCC) having a length of 4 or 3 to the DMRS, based on the configuration.

[0325] The FD-OCC may be applied across a plurality of physical resource blocks.

[0326] The FD-OCC may be applied to a plurality of consecutive resource elements.

[0327] The control section 210 may estimate a channel using the DMRS in a plurality of resource elements across a plurality of physical resource blocks.

(Hardware Structure)

[0328] Note that the block diagrams that have been used to describe the above embodiments show blocks in functional units. These functional blocks (components) may be implemented in arbitrary combinations of at least one of hardware and software. Also, the method for implementing each functional block is not particularly limited. That is, each functional block may be realized by one piece of apparatus that is physically or logically coupled, or may be realized by directly or indirectly connecting two or more physically or logically separate pieces of apparatus (for example, via wire, wireless, or the like) and using these plurality of pieces of apparatus. The functional blocks may be implemented by combining softwares into the apparatus described above or the plurality of apparatuses described above.

[0329] Here, functions include judgment, determination, decision, calculation, computation, processing, derivation, investigation, search, confirmation, reception, transmission, output, access, resolution, selection, designation, establishment, comparison, assumption, expectation, considering, broadcasting, notifying, communicating, forwarding, configuring, reconfiguring, allocating (mapping), assigning, and the like, but function are by no means limited to these. For example, functional block (components) to implement a function of transmission may be referred to as a “transmitting section (transmitting unit),” a “transmitter,” and the like. The method for implementing each component is not particularly limited as described above.

[0330] For example, a base station, a user terminal, and so on according to one embodiment of the present disclosure may function as a computer that executes the processes of the radio communication method of the present disclosure. FIG. 37 is a diagram to show an example of a hardware structure of the base station and the user terminal according to one embodiment. Physically, the above-described base station 10 and user terminal 20 may each be formed as a computer apparatus that includes a processor 1001, a memory 1002, a storage 1003, a communication apparatus 1004, an input apparatus 1005, an output apparatus 1006, a bus 1007, and so on.

[0331] Note that in the present disclosure, the words such as an apparatus, a circuit, a device, a section, a unit, and so on can be interchangeably interpreted. The hardware structure of the base station 10 and the user terminal 20 may be configured to include one or more of apparatuses shown in the drawings, or may be configured not to include part of apparatuses.

[0332] For example, although only one processor 1001 is shown, a plurality of processors may be provided. Furthermore, processes may be implemented with one processor or may be implemented at the same time, in sequence, or in different manners with two or more processors. Note that the processor 1001 may be implemented with one or more chips.

[0333] Each function of the base station 10 and the user terminals 20 is implemented, for example, by allowing given software (programs) to be read on hardware such as the processor 1001 and the memory 1002, and by allowing the processor 1001 to perform calculations to control communication via the communication apparatus 1004 and control at least one of reading and writing of data in the memory 1002 and the storage 1003.

[0334] The processor 1001 controls the whole computer by, for example, running an operating system. The processor 1001 may be configured with a central processing unit

(CPU), which includes interfaces with peripheral apparatus, control apparatus, computing apparatus, a register, and so on. For example, at least part of the above-described control section 110 (210), the transmitting/receiving section 120 (220), and so on may be implemented by the processor 1001.

[0335] Furthermore, the processor 1001 reads programs (program codes), software modules, data, and so on from at least one of the storage 1003 and the communication apparatus 1004, into the memory 1002, and executes various processes according to these. As for the programs, programs to allow computers to execute at least part of the operations of the above-described embodiments are used. For example, the control section 110 (210) may be implemented by control programs that are stored in the memory 1002 and that operate on the processor 1001, and other functional blocks may be implemented likewise.

[0336] The memory 1002 is a computer-readable recording medium, and may be constituted with, for example, at least one of a Read Only Memory (ROM), an Erasable Programmable ROM (EPROM), an Electrically EPROM (EEPROM), a Random Access Memory (RAM), and other appropriate storage media. The memory 1002 may be referred to as a “register,” a “cache,” a “main memory (primary storage apparatus)” and so on. The memory 1002 can store executable programs (program codes), software modules, and the like for implementing the radio communication method according to one embodiment of the present disclosure.

[0337] The storage 1003 is a computer-readable recording medium, and may be constituted with, for example, at least one of a flexible disk, a floppy (registered trademark) disk, a magneto-optical disk (for example, a compact disc (Compact Disc ROM (CD-ROM) and so on), a digital versatile disc, a Blu-ray (registered trademark) disk), a removable disk, a hard disk drive, a smart card, a flash memory device (for example, a card, a stick, and a key drive), a magnetic stripe, a database, a server, and other appropriate storage media. The storage 1003 may be referred to as “secondary storage apparatus.”

[0338] The communication apparatus 1004 is hardware (transmitting/receiving device) for allowing inter-computer communication via at least one of wired and wireless networks, and may be referred to as, for example, a “network device,” a “network controller,” a “network card,” a “communication module,” and so on. The communication apparatus 1004 may be configured to include a high frequency switch, a duplexer, a filter, a frequency synthesizer, and so on in order to realize, for example, at least one of frequency division duplex (FDD) and time division duplex (TDD). For example, the above-described transmitting/receiving section 120 (220), the transmitting/receiving antennas 130 (230), and so on may be implemented by the communication apparatus 1004. In the transmitting/receiving section 120 (220), the transmitting section 120a (220a) and the receiving section 120b (220b) can be implemented while being separated physically or logically.

[0339] The input apparatus 1005 is an input device that receives input from the outside (for example, a keyboard, a mouse, a microphone, a switch, a button, a sensor, and so on). The output apparatus 1006 is an output device that allows sending output to the outside (for example, a display, a speaker, a Light Emitting Diode (LED) lamp, and so on).

Note that the input apparatus **1005** and the output apparatus **1006** may be provided in an integrated structure (for example, a touch panel).

[0340] Furthermore, these types of apparatus, including the processor **1001**, the memory **1002**, and others, are connected by a bus **1007** for communicating information. The bus **1007** may be formed with a single bus, or may be formed with buses that vary between pieces of apparatus.

[0341] Also, the base station **10** and the user terminals **20** may be structured to include hardware such as a microprocessor, a digital signal processor (DSP), an Application Specific Integrated Circuit (ASIC), a Programmable Logic Device (PLD), a Field Programmable Gate Array (FPGA), and so on, and part or all of the functional blocks may be implemented by the hardware. For example, the processor **1001** may be implemented with at least one of these pieces of hardware.

(Variations)

[0342] Note that the terminology described in the present disclosure and the terminology that is needed to understand the present disclosure may be replaced by other terms that convey the same or similar meanings. For example, a “channel,” a “symbol,” and a “signal” (or signaling) may be interchangeably interpreted. Also, “signals” may be “messages.” A reference signal may be abbreviated as an “RS,” and may be referred to as a “pilot,” a “pilot signal,” and so on, depending on which standard applies. Furthermore, a “component carrier (CC)” may be referred to as a “cell,” a “frequency carrier,” a “carrier frequency” and so on.

[0343] A radio frame may be constituted of one or a plurality of periods (frames) in the time domain. Each of one or a plurality of periods (frames) constituting a radio frame may be referred to as a “subframe.” Furthermore, a subframe may be constituted of one or a plurality of slots in the time domain. A subframe may be a fixed time length (for example, 1 ms) independent of numerology.

[0344] Here, numerology may be a communication parameter applied to at least one of transmission and reception of a given signal or channel. For example, numerology may indicate at least one of a subcarrier spacing (SCS), a bandwidth, a symbol length, a cyclic prefix length, a transmission time interval (TTI), the number of symbols per TTI, a radio frame structure, a specific filter processing performed by a transceiver in the frequency domain, a specific windowing processing performed by a transceiver in the time domain, and so on.

[0345] A slot may be constituted of one or a plurality of symbols in the time domain (Orthogonal Frequency Division Multiplexing (OFDM) symbols, Single Carrier Frequency Division Multiple Access (SC-FDMA) symbols, and so on). Furthermore, a slot may be a time unit based on numerology.

[0346] A slot may include a plurality of mini-slots. Each mini-slot may be constituted of one or a plurality of symbols in the time domain. A mini-slot may be referred to as a “sub-slot.” A mini-slot may be constituted of symbols less than the number of slots. A PDSCH (or PUSCH) transmitted in a time unit larger than a mini-slot may be referred to as “PDSCH (PUSCH) mapping type A.” A PDSCH (or PUSCH) transmitted using a mini-slot may be referred to as “PDSCH (PUSCH) mapping type B.”

[0347] A radio frame, a subframe, a slot, a mini-slot, and a symbol all express time units in signal communication. A

radio frame, a subframe, a slot, a mini-slot, and a symbol may each be called by other applicable terms. Note that time units such as a frame, a subframe, a slot, mini-slot, and a symbol in the present disclosure may be interchangeably interpreted.

[0348] For example, one subframe may be referred to as a “TTI,” a plurality of consecutive subframes may be referred to as a “TTI,” or one slot or one mini-slot may be referred to as a “TTI.” That is, at least one of a subframe and a TTI may be a subframe (1 ms) in existing LTE, may be a shorter period than 1 ms (for example, 1 to 13 symbols), or may be a longer period than 1 ms. Note that a unit expressing TTI may be referred to as a “slot,” a “mini-slot,” and so on instead of a “subframe.”

[0349] Here, a TTI refers to the minimum time unit of scheduling in radio communication, for example. For example, in LTE systems, a base station schedules the allocation of radio resources (such as a frequency bandwidth and transmission power that are available for each user terminal) for the user terminal in TTI units. Note that the definition of TTIs is not limited to this.

[0350] TTIs may be transmission time units for channel-encoded data packets (transport blocks), code blocks, or codewords, or may be the unit of processing in scheduling, link adaptation, and so on. Note that, when TTIs are given, the time interval (for example, the number of symbols) to which transport blocks, code blocks, codewords, or the like are actually mapped may be shorter than the TTIs.

[0351] Note that, in the case where one slot or one mini-slot is referred to as a TTI, one or more TTIs (that is, one or more slots or one or more mini-slots) may be the minimum time unit of scheduling. Furthermore, the number of slots (the number of mini-slots) constituting the minimum time unit of the scheduling may be controlled.

[0352] A TTI having a time length of 1 ms may be referred to as a “normal TTI” (TTI in 3GPP Rel. 8 to Rel. 12), a “long TTI,” a “normal subframe,” a “long subframe,” a “slot” and so on. A TTI that is shorter than a normal TTI may be referred to as a “shortened TTI,” a “short TTI,” a “partial or fractional TTI,” a “shortened subframe,” a “short subframe,” a “mini-slot,” a “sub-slot,” a “slot” and so on.

[0353] Note that a long TTI (for example, a normal TTI, a subframe, and so on) may be interpreted as a TTI having a time length exceeding 1 ms, and a short TTI (for example, a shortened TTI and so on) may be interpreted as a TTI having a TTI length shorter than the TTI length of a long TTI and equal to or longer than 1 ms.

[0354] A resource block (RB) is the unit of resource allocation in the time domain and the frequency domain, and may include one or a plurality of consecutive subcarriers in the frequency domain. The number of subcarriers included in an RB may be the same regardless of numerology, and, for example, may be 12. The number of subcarriers included in an RB may be determined based on numerology.

[0355] Also, an RB may include one or a plurality of symbols in the time domain, and may be one slot, one mini-slot, one subframe, or one TTI in length. One TTI, one subframe, and so on each may be constituted of one or a plurality of resource blocks.

[0356] Note that one or a plurality of RBs may be referred to as a “physical resource block (Physical RB (PRB)),” a “sub-carrier group (SCG),” a “resource element group (REG),” a “group/set/pair of PRBs/RBs,” and so on.

[0357] Furthermore, a resource block may be constituted of one or a plurality of resource elements (REs). For example, one RE may correspond to a radio resource field of one subcarrier and one symbol.

[0358] A bandwidth part (BWP) (which may be referred to as a “fractional bandwidth,” and so on) may represent a subset of contiguous common resource blocks (common RBs) for given numerology in a given carrier. Here, a common RB may be specified by an index of the RB based on the common reference point of the carrier. A PRB may be defined by a given BWP and may be numbered in the BWP.

[0359] The BWP may include a UL BWP (BWP for the UL) and a DL BWP (BWP for the DL). One or a plurality of BWPs may be configured in one carrier for a UE.

[0360] At least one of configured BWPs may be active, and a UE does not need to assume to transmit/receive a given signal/channel outside active BWPs. Note that a “cell,” a “carrier,” and so on in the present disclosure may be interpreted as a “BWP”.

[0361] Note that the above-described structures of radio frames, subframes, slots, mini-slots, symbols, and so on are merely examples. For example, structures such as the number of subframes included in a radio frame, the number of slots per subframe or radio frame, the number of mini-slots included in a slot, the numbers of symbols and RBs included in a slot or a mini-slot, the number of subcarriers included in an RB, the number of symbols in a TTI, the symbol length, the cyclic prefix (CP) length, and so on can be variously changed.

[0362] Also, the information, parameters, and so on described in the present disclosure may be represented in absolute values or in relative values with respect to given values, or may be represented in another corresponding information. For example, radio resources may be specified by given indices.

[0363] The names used for parameters and so on in the present disclosure are in no respect limiting. Furthermore, mathematical expressions that use these parameters, and so on may be different from those expressly disclosed in the present disclosure. For example, since various channels (PUCCH, PDCCH, and so on) and information elements can be identified by any suitable names, the various names allocated to these various channels and information elements are in no respect limiting.

[0364] The information, signals, and so on described in the present disclosure may be represented by using any of a variety of different technologies. For example, data, instructions, commands, information, signals, bits, symbols, chips, and so on, all of which may be referenced throughout the herein-contained description, may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or photons, or any combination of these.

[0365] Also, information, signals, and so on can be output in at least one of from higher layers to lower layers and from lower layers to higher layers. Information, signals, and so on may be input and/or output via a plurality of network nodes.

[0366] The information, signals, and so on that are input and/or output may be stored in a specific location (for example, a memory) or may be managed by using a management table. The information, signals, and so on to be input and/or output can be overwritten, updated, or appended. The information, signals, and so on that are

output may be deleted. The information, signals, and so on that are input may be transmitted to another apparatus.

[0367] Reporting of information is by no means limited to the aspects/embodiments described in the present disclosure, and other methods may be used as well. For example, reporting of information in the present disclosure may be implemented by using physical layer signaling (for example, downlink control information (DCI), uplink control information (UCI)), higher layer signaling (for example, Radio Resource Control (RRC) signaling, broadcast information (master information block (MIB), system information blocks (SIBs), and so on), Medium Access Control (MAC) signaling and so on), and other signals or combinations of these.

[0368] Note that physical layer signaling may be referred to as “Layer 1/Layer 2 (L1/L2) control information (L1/L2 control signals),” “L1 control information (L1 control signal),” and so on. Also, RRC signaling may be referred to as an “RRC message,” and can be, for example, an RRC connection setup message, an RRC connection reconfiguration message, and so on. Also, MAC signaling may be reported using, for example, MAC control elements (MAC CEs).

[0369] Also, reporting of given information (for example, reporting of “X holds”) does not necessarily have to be reported explicitly, and can be reported implicitly (by, for example, not reporting this given information or reporting another piece of information).

[0370] Determinations may be made in values represented by one bit (0 or 1), may be made in Boolean values that represent true or false, or may be made by comparing numerical values (for example, comparison against a given value).

[0371] Software, whether referred to as “software,” “firmware,” “middleware,” “microcode,” or “hardware description language,” or called by other terms, should be interpreted broadly to mean instructions, instruction sets, code, code segments, program codes, programs, subprograms, software modules, applications, software applications, software packages, routines, subroutines, objects, executable files, execution threads, procedures, functions, and so on.

[0372] Also, software, commands, information, and so on may be transmitted and received via communication media. For example, when software is transmitted from a website, a server, or other remote sources by using at least one of wired technologies (coaxial cables, optical fiber cables, twisted-pair cables, digital subscriber lines (DSL), and so on) and wireless technologies (infrared radiation, microwaves, and so on), at least one of these wired technologies and wireless technologies are also included in the definition of communication media.

[0373] The terms “system” and “network” used in the present disclosure can be used interchangeably. The “network” may mean an apparatus (for example, a base station) included in the network.

[0374] In the present disclosure, the terms such as “precoding,” a “precoder,” a “weight (precoding weight),” “quasi-co-location (QCL),” a “Transmission Configuration Indication state (TCI state),” a “spatial relation,” a “spatial domain filter,” a “transmission power,” “phase rotation,” an “antenna port,” an “antenna port group,” a “layer,” “the number of layers,” a “rank,” a “resource,” a “resource set,” a “resource group,” a “beam,” a “beam width,” a “beam

angular degree,” an “antenna,” an “antenna element,” a “panel,” and so on can be used interchangeably.

[0375] In the present disclosure, the terms such as a “base station (BS),” a “radio base station,” a “fixed station,” a “NodeB,” an “eNB (eNodeB),” a “gNB (gNodeB),” an “access point,” a “transmission point (TP),” a “reception point (RP),” a “transmission/reception point (TRP),” a “panel,” a “cell,” a “sector,” a “cell group,” a “carrier,” a “component carrier,” and so on can be used interchangeably. The base station may be referred to as the terms such as a “macro cell,” a “small cell,” a “femto cell,” a “pico cell,” and so on.

[0376] A base station can accommodate one or a plurality of (for example, three) cells. When a base station accommodates a plurality of cells, the entire coverage area of the base station can be partitioned into multiple smaller areas, and each smaller area can provide communication services through base station subsystems (for example, indoor small base stations (Remote Radio Heads (RRHs))). The term “cell” or “sector” refers to part of or the entire coverage area of at least one of a base station and a base station subsystem that provides communication services within this coverage.

[0377] In the present disclosure, the base station transmitting information to the terminal may be interchangeably interpreted as the base station indicating control/operation based on the information to the terminal.

[0378] In the present disclosure, the terms “mobile station (MS),” “user terminal,” “user equipment (UE),” and “terminal” may be used interchangeably.

[0379] A mobile station may be referred to as a “subscriber station,” “mobile unit,” “subscriber unit,” “wireless unit,” “remote unit,” “mobile device,” “wireless device,” “wireless communication device,” “remote device,” “mobile subscriber station,” “access terminal,” “mobile terminal,” “wireless terminal,” “remote terminal,” “handset,” “user agent,” “mobile client,” “client,” or some other appropriate terms in some cases.

[0380] At least one of a base station and a mobile station may be referred to as a “transmitting apparatus,” a “receiving apparatus,” a “radio communication apparatus,” and so on. Note that at least one of a base station and a mobile station may be a device mounted on a moving object or a moving object itself, and so on.

[0381] The moving object is a movable object with any moving speed, and naturally a case where the moving object is stopped is also included. Examples of the moving object include a vehicle, a transport vehicle, an automobile, a motorcycle, a bicycle, a connected car, a loading shovel, a bulldozer, a wheel loader, a dump truck, a fork lift, a train, a bus, a trolley, a rickshaw, a ship and other watercraft, an airplane, a rocket, a satellite, a drone, a multicopter, a quadcopter, a balloon, and an object mounted on any of these, but these are not restrictive. The moving object may be a moving object that autonomously travels based on a direction for moving.

[0382] The moving object may be a vehicle (for example, a car, an airplane, and the like), may be a moving object which moves unmanned (for example, a drone, an automatic operation car, and the like), or may be a robot (a manned type or unmanned type). Note that at least one of a base station and a mobile station also includes an apparatus which does not necessarily move during communication operation.

For example, at least one of a base station and a mobile station may be an Internet of Things (IoT) device such as a sensor.

[0383] FIG. 38 is a diagram to show an example of a vehicle according to one embodiment. A vehicle 40 includes a driving section 41, a steering section 42, an accelerator pedal 43, a brake pedal 44, a shift lever 45, right and left front wheels 46, right and left rear wheels 47, an axle 48, an electronic control section 49, various sensors (including a current sensor 50, a rotational speed sensor 51, a pneumatic sensor 52, a vehicle speed sensor 53, an acceleration sensor 54, an accelerator pedal sensor 55, a brake pedal sensor 56, a shift lever sensor 57, and an object detection sensor 58), an information service section 59, and a communication module 60.

[0384] The driving section 41 includes, for example, at least one of an engine, a motor, and a hybrid of an engine and a motor. The steering section 42 at least includes a steering wheel, and is configured to steer at least one of the front wheels 46 and the rear wheels 47, based on operation of the steering wheel operated by a user.

[0385] The electronic control section 49 includes a micro-processor 61, a memory (ROM, RAM) 62, and a communication port (for example, an input/output (IO) port) 63. The electronic control section 49 receives, as input, signals from the various sensors 50 to 58 included in the vehicle. The electronic control section 49 may be referred to as an Electronic Control Unit (ECU).

[0386] Examples of the signals from the various sensors 50 to 58 include a current signal from the current sensor 50 for sensing current of a motor, a rotational speed signal of the front wheels 46/rear wheels 47 acquired by the rotational speed sensor 51, a pneumatic signal of the front wheels 46/rear wheels 47 acquired by the pneumatic sensor 52, a vehicle speed signal acquired by the vehicle speed sensor 53, an acceleration signal acquired by the acceleration sensor 54, a depressing amount signal of the accelerator pedal 43 acquired by the accelerator pedal sensor 55, a depressing amount signal of the brake pedal 44 acquired by the brake pedal sensor 56, an operation signal of the shift lever 45 acquired by the shift lever sensor 57, and a detection signal for detecting an obstruction, a vehicle, a pedestrian, and the like acquired by the object detection sensor 58.

[0387] The information service section 59 includes various devices for providing (outputting) various pieces of information such as drive information, traffic information, and entertainment information, such as a car navigation system, an audio system, a speaker, a display, a television, and a radio, and one or more ECUs that control these devices. The information service section 59 provides various pieces of information/services (for example, multimedia information/multimedia service) for an occupant of the vehicle 40, using information acquired from an external apparatus via the communication module 60 and the like.

[0388] The information service section 59 may include an input device (for example, a keyboard, a mouse, a microphone, a switch, a button, a sensor, a touch panel, and the like) for receiving input from the outside, or may include an output device (for example, a display, a speaker, an LED lamp, a touch panel, and the like) for implementing output to the outside.

[0389] A driving assistance system section 64 includes various devices for providing functions for preventing an accident and reducing a driver’s driving load, such as a

millimeter wave radar, Light Detection and Ranging (LiDAR), a camera, a positioning locator (for example, a Global Navigation Satellite System (GNSS) and the like), map information (for example, a high definition (HD) map, an autonomous vehicle (AV) map, and the like), a gyro system (for example, an inertial measurement apparatus (inertial measurement unit (IMU)), an inertial navigation apparatus (inertial navigation system (INS)), and the like), an artificial intelligence (AI) chip, and an AI processor, and one or more ECUs that control these devices. The driving assistance system section 64 transmits and receives various pieces of information via the communication module 60, and implements a driving assistance function or an autonomous driving function.

[0390] The communication module 60 can communicate with the microprocessor 61 and the constituent elements of the vehicle 40 via the communication port 63. For example, via the communication port 63, the communication module 60 transmits and receives data (information) to and from the driving section 41, the steering section 42, the accelerator pedal 43, the brake pedal 44, the shift lever 45, the right and left front wheels 46, the right and left rear wheels 47, the axle 48, the microprocessor 61 and the memory (ROM, RAM) 62 in the electronic control section 49, and the various sensors 50 to 58, which are included in the vehicle 40.

[0391] The communication module 60 can be controlled by the microprocessor 61 of the electronic control section 49, and is a communication device that can perform communication with an external apparatus. For example, the communication module 60 performs transmission and reception of various pieces of information to and from the external apparatus via radio communication. The communication module 60 may be either inside or outside the electronic control section 49. The external apparatus may be, for example, the base station 10, the user terminal 20, or the like described above. The communication module 60 may be, for example, at least one of the base station 10 and the user terminal 20 described above (may function as at least one of the base station 10 and the user terminal 20).

[0392] The communication module 60 may transmit at least one of signals from the various sensors 50 to 58 described above input to the electronic control section 49, information obtained based on the signals, and information based on an input from the outside (a user) obtained via the information service section 59, to the external apparatus via radio communication. The electronic control section 49, the various sensors 50 to 58, the information service section 59, and the like may be referred to as input sections that receive input. For example, the PUSCH transmitted by the communication module 60 may include information based on the input.

[0393] The communication module 60 receives various pieces of information (traffic information, signal information, inter-vehicle distance information, and the like) transmitted from the external apparatus, and displays the various pieces of information on the information service section 59 included in the vehicle. The information service section 59 may be referred to as an output section that outputs information (for example, outputs information to devices, such as a display and a speaker, based on the PDSCH received by the communication module 60 (or data/information decoded from the PDSCH)).

[0394] The communication module 60 stores the various pieces of information received from the external apparatus in the memory 62 that can be used by the microprocessor 61. Based on the pieces of information stored in the memory 62, the microprocessor 61 may perform control of the driving section 41, the steering section 42, the accelerator pedal 43, the brake pedal 44, the shift lever 45, the right and left front wheels 46, the right and left rear wheels 47, the axle 48, the various sensors 50 to 58, and the like included in the vehicle 40.

[0395] Furthermore, the base station in the present disclosure may be interpreted as a user terminal. For example, each aspect/embodiment of the present disclosure may be applied to the structure that replaces a communication between a base station and a user terminal with a communication between a plurality of user terminals (for example, which may be referred to as “Device-to-Device (D2D),” “Vehicle-to-Everything (V2X),” and the like). In this case, user terminals 20 may have the functions of the base stations 10 described above. The words such as “uplink” and “downlink” may be interpreted as the words corresponding to the terminal-to-terminal communication (for example, “sidelink”). For example, an uplink channel, a downlink channel and so on may be interpreted as a sidelink channel.

[0396] Likewise, the user terminal in the present disclosure may be interpreted as base station. In this case, the base station 10 may have the functions of the user terminal 20 described above.

[0397] Actions which have been described in the present disclosure to be performed by a base station may, in some cases, be performed by upper nodes of the base station. In a network including one or a plurality of network nodes with base stations, it is clear that various operations that are performed to communicate with terminals can be performed by base stations, one or more network nodes (for example, Mobility Management Entities (MMEs), Serving-Gateways (S-GWs), and so on may be possible, but these are not limiting) other than base stations, or combinations of these.

[0398] The aspects/embodiments illustrated in the present disclosure may be used individually or in combinations, which may be switched depending on the mode of implementation. The order of processes, sequences, flowcharts, and so on that have been used to describe the aspects/embodiments in the present disclosure may be re-ordered as long as inconsistencies do not arise. For example, although various methods have been illustrated in the present disclosure with various components of steps in exemplary orders, the specific orders that are illustrated herein are by no means limiting.

[0399] The aspects/embodiments illustrated in the present disclosure may be applied to Long Term Evolution (LTE), LTE-Advanced (LTE-A), LTE-Beyond (LTE-B), SUPER 3G, IMT-Advanced, 4th generation mobile communication system (4G), 5th generation mobile communication system (5G), 6th generation mobile communication system (6G), xth generation mobile communication system (xG (where x is, for example, an integer or a decimal)), Future Radio Access (FRA), New-Radio Access Technology (RAT), New Radio (NR), New radio access (NX), Future generation radio access (FX), Global System for Mobile communications (GSM (registered trademark)), CDMA 2000, Ultra Mobile Broadband (UMB), IEEE 802.11 (Wi-Fi (registered trademark)), IEEE 802.16 (WiMAX (registered trademark)), IEEE 802.20, Ultra-WideBand (UWB), Bluetooth (regis-

tered trademark), systems that use other adequate radio communication methods and next-generation systems that are enhanced, modified, created, or defined based on these. A plurality of systems may be combined (for example, a combination of LTE or LTE-A and 5G, and the like) and applied.

[0400] The phrase “based on” (or “on the basis of”) as used in the present disclosure does not mean “based only on” (or “only on the basis of”), unless otherwise specified. In other words, the phrase “based on” (or “on the basis of”) means both “based only on” and “based at least on” (“only on the basis of” and “at least on the basis of”).

[0401] Reference to elements with designations such as “first,” “second,” and so on as used in the present disclosure does not generally limit the quantity or order of these elements. These designations may be used in the present disclosure only for convenience, as a method for distinguishing between two or more elements. Thus, reference to the first and second elements does not imply that only two elements may be employed, or that the first element must precede the second element in some way.

[0402] The term “judging (determining)” as in the present disclosure herein may encompass a wide variety of actions. For example, “judging (determining)” may be interpreted to mean making “judgments (determinations)” about judging, calculating, computing, processing, deriving, investigating, looking up, search and inquiry (for example, searching a table, a database, or some other data structures), ascertaining, and so on.

[0403] Furthermore, “judging (determining)” may be interpreted to mean making “judgments (determinations)” about receiving (for example, receiving information), transmitting (for example, transmitting information), input, output, accessing (for example, accessing data in a memory), and so on.

[0404] In addition, “judging (determining)” as used herein may be interpreted to mean making “judgments (determinations)” about resolving, selecting, choosing, establishing, comparing, and so on. In other words, “judging (determining)” may be interpreted to mean making “judgments (determinations)” about some action.

[0405] In addition, “judging (determining)” may be interpreted as “assuming,” “expecting,” “considering,” and the like.

[0406] “The maximum transmission power” according to the present disclosure may mean a maximum value of the transmission power, may mean the nominal maximum transmission power (the nominal UE maximum transmit power), or may mean the rated maximum transmission power (the rated UE maximum transmit power).

[0407] The terms “connected” and “coupled,” or any variation of these terms as used in the present disclosure mean all direct or indirect connections or coupling between two or more elements, and may include the presence of one or more intermediate elements between two elements that are “connected” or “coupled” to each other. The coupling or connection between the elements may be physical, logical, or a combination thereof. For example, “connection” may be interpreted as “access.”

[0408] In the present disclosure, when two elements are connected, the two elements may be considered “connected” or “coupled” to each other by using one or more electrical wires, cables and printed electrical connections, and, as some non-limiting and non-inclusive examples, by using

electromagnetic energy having wavelengths in radio frequency regions, microwave regions, (both visible and invisible) optical regions, or the like.

[0409] In the present disclosure, the phrase “A and B are different” may mean that “A and B are different from each other.” Note that the phrase may mean that “A and B are each different from C.” The terms “separate,” “be coupled,” and so on may be interpreted similarly to “different.”

[0410] When terms such as “include,” “including,” and variations of these are used in the present disclosure, these terms are intended to be inclusive, in a manner similar to the way the term “comprising” is used. Furthermore, the term “or” as used in the present disclosure is intended to be not an exclusive disjunction.

[0411] For example, in the present disclosure, when an article such as “a,” “an,” and “the” in the English language is added by translation, the present disclosure may include that a noun after these articles is in a plural form.

[0412] In the present disclosure, “equal to or smaller than,” “smaller than,” “equal to or larger than,” “larger than,” “equal to,” and the like may be interchangeably interpreted. In the present disclosure, words such as “good,” “poor,” “large,” “small,” “high,” “low,” “early,” “late,” “wide,” “narrow,” and the like may be interchangeably interpreted irrespective of positive degree, comparative degree, and superlative degree. In the present disclosure, expressions obtained by adding “i-th” (i is any integer) to words such as “good,” “poor,” “large,” “small,” “high,” “low,” “early,” “late,” “wide,” “narrow,” and the like may be interchangeably interpreted irrespective of positive degree, comparative degree, and superlative degree (for example, “highest” may be interpreted as “i-th highest,” and vice versa).

[0413] In the present disclosure, “of,” “for,” “regarding,” “related to,” “associated with,” and the like may be interchangeably interpreted.

[0414] Now, although the invention according to the present disclosure has been described in detail above, it should be obvious to a person skilled in the art that the invention according to the present disclosure is by no means limited to the embodiments described in the present disclosure. The invention according to the present disclosure can be implemented with various corrections and in various modifications, without departing from the spirit and scope of the invention defined by the recitations of claims. Consequently, the description of the present disclosure is provided only for the purpose of explaining examples, and should by no means be construed to limit the invention according to the present disclosure in any way.

1. A terminal comprising:

- a receiving section that receives a configuration of a demodulation reference signal (DMRS) of a shared channel; and
- a control section that applies a frequency domain (FD)-orthogonal cover code (OCC) having a length of 4 or 3 to the DMRS, based on the configuration.

2. The terminal according to claim 1, wherein

the FD-OCC is applied across a plurality of physical resource blocks.

3. The terminal according to claim 1, wherein

the FD-OCC is applied to a plurality of consecutive resource elements.

4. The terminal according to claim 1, wherein the control section estimates a channel using the DMRS in a plurality of resource elements across a plurality of physical resource blocks.
5. A radio communication method for a terminal, the radio communication method comprising:
 - receiving a configuration of a demodulation reference signal (DMRS) of a shared channel; and
 - applying a frequency domain (FD)-orthogonal cover code (OCC) having a length of 4 or 3 to the DMRS, based on the configuration.
6. A base station comprising:
 - a transmitting section that transmits a configuration of a demodulation reference signal (DMRS) of a shared channel; and
 - a control section that applies a frequency domain (FD)-orthogonal cover code (OCC) having a length of 4 or 3 to the DMRS, based on the configuration.

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